TURBO 800S/HX2.0

Programming Manual

for Machining Center (MC)

Serial No. : PG-20160607

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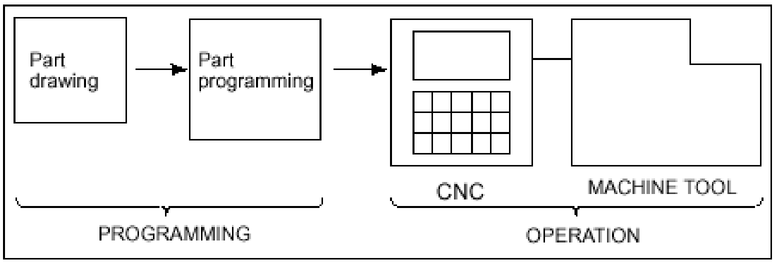
1 General

In general, CNC machine tool consists of CNC control unit, actuator (motor) and body. It includes a lot of hardware and software in the fields of electricity, electronics, computer, and machinery. Generally, axis control of servo motor is done by G code and ON/OFF control of PLC by M code.

Machining Center (MC) means a machine tool (CNC milling with ATC ? Auto Tool Changer ? mounted) that cuts work pieces while automatically replacing tools for consecutive works of several processes.

The cutting work that is carried out by CNC machine tool is: making out a drawing first, making out a program, and then, doing the work.



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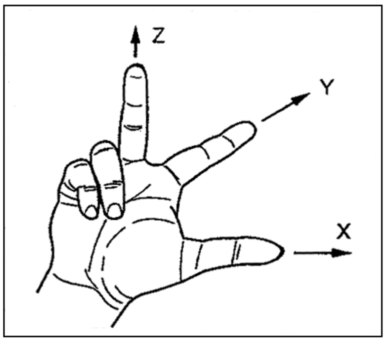
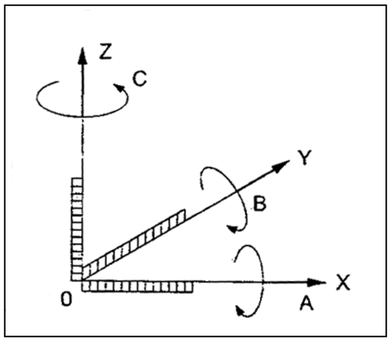
1.1 Coordinate

There are two coordinates: one is machine coordinate that is based on the first reference position of a machine; and the other is work piece coordinate that is based on the reference position of a program. Right-hand orthogonal coordinate is usually used. As straight lined feed axis, X, Y, and Z are used, and as an additional axis (rotating axis) to each straight lined feed axis, A, B, and C are used (A rotates, each, around X axis, B rotates, around Y and C, around Z).





The coordinates for MC are set based on the following rules:
- The direction in parallel with the spindle is in parallel with Z axis.
- Seeing MC in front of it, the left and the right are in parallel with X axis.
- Y axis expresses Z and X with the right hand coordinate, which are vertical to X and Z.

1.1.1 Machine Coordinate

Each CNC machine tool has its own reference position and based on this reference position, the machine coordinate is defined. In principle, the reference position is based on the machine origin of the spindle, but in case of a tool mounted, it is difficult to return to the reference position so a special point is used as the reference position.

Setting of the machine coordinate is completed by manually executing reference position return after making power on.

The machine coordinate system becomes a standard for setting Stored Stroke Limit of G22/G23, Over Travel, and the second/the third/the fourth reference position. The machine coordinate value of the reference position is X0, Y0, Z0.

1.1.2 Work-piece Coordinate

In the work-piece coordinate system, the reference point for cutting a work piece is set as the reference position. This point for cutting conforms to the reference position of a program drawing that is used for making NC program. The reference position should set in order for a worker to conveniently make out NC program, referring to the drawing and to apply the NC program to cutting as it is.

In general, the reference position of the work-piece coordinate system is set, using a method of setting the work-piece coordinate system, and by use of G54 ~ G59, the work-piece coordinate system is selected for cutting.

In case a parameter maintaining the work-piece coordinate system (located on the left top of the coordinate screen) is set, the lastly set coordinate value is applied as it is even though the initial power is ON. And, by the initial command mode parameter (PI 146), G90 mode (absolute command) or G91 mode (incremental command) can be set.

In the program, it is possible to set the work-piece coordinate system like G92 X _ Y _ Z _ . The reference position of the work-piece coordinate system becomes the reference point for absolute command, and based on the reference point on the program, the absolute coordinate [G90] value becomes X0, Y0, Z0.

1.1.3 Local Coordinate

In case of making a program with the work-piece coordinate system, we set a random point as the reference point based on the work-piece coordinate system. At this time, we call it local coordinate system.

The local coordinate system can make a new coordinate system based on all the work-piece coordinate systems G54~G59, and the reference position of each local coordinate system is X_ Y_ Z_ that is designated by each work-piece coordinate system. If a new work-piece coordinate is set, the local coordinate system gets cleared to be 0.

1.1.4 Relative Coordinate

Relative coordinate system is used for setting a random point as the reference position. It is usually used for setting the work-piece coordinate system, simply moving handles, and setting coordinate systems.

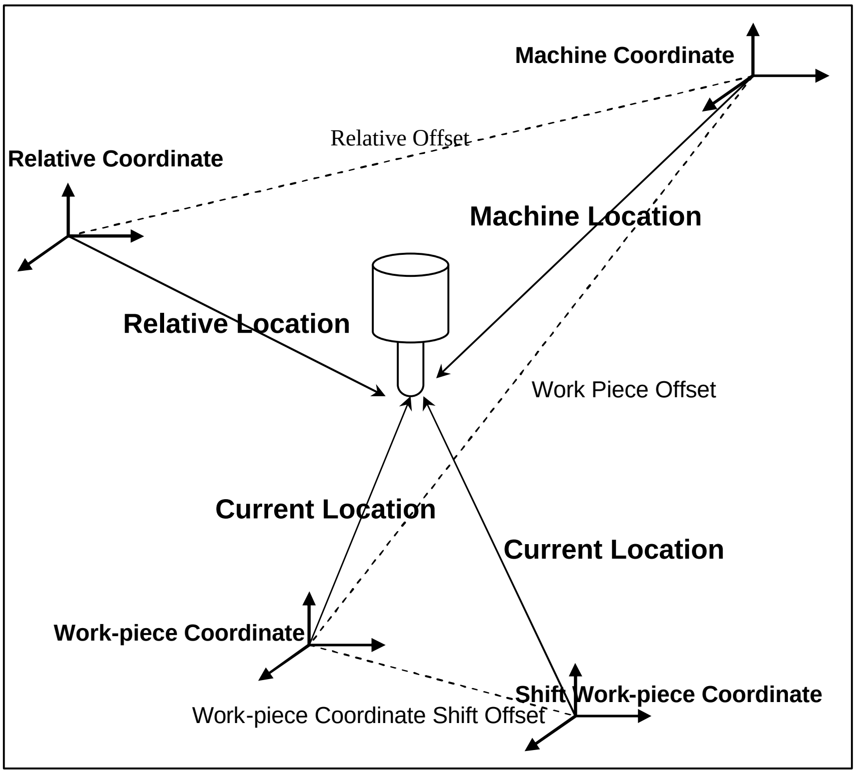
1.1.5 Distance To Go

Execute the program at auto mode [AUTO, MDI mode] and command axis feed, and a value appears on the residual coordinate systems. The value means the remaining distance to go of the block now in run. It is used to check any error with the program path in case of cutting a test sample after setting a work-piece.

1.1.6 Offset and Location by Coordinate

Machine coordinate system, work-piece coordinate system and relative coordinate system are related with each other, having deviation of offset. The relative coordinate system has the deviation from the machine coordinate system as much as the relative offset, and the work-piece coordinate system has the deviation from the machine coordinate system as much as the work-piece offset. The work-piece coordinate system can move and at this time, it gets to have deviation as much as the shift offset of work-piece.

Each coordinate system indicates the coordinate of a tool with location. On the machine coordinate system, it is called as machine location, on the relative coordinate system, it is called as relative location, and on the work-piece coordinate system, it is called as current location.



1.2 Setting Conditions for CNC Work

1.2.1 Cutting velocity V [m/min]

Cutting velocity V means the maximum relative velocity between a tool and a structure. V has serious influence on the lifetime of tool and it is closely related with roughness of the cutting face and material removal rate. M/min is used as a basic variable in the cutting phenomenon.

$$V = \frac{p\ DN}{1000}$$

D : Diameter of tool [mm]

N : Revolution speed of the spindle [rpm]

1.2.2 Speed of Revolution N [rpm]

$$N = \frac{1000\quad V}{p\ D}$$

V : Cutting velocity [m/min]

D : Diameter of tool [mm]

1.2.3 Feedrate F [mm/min, mm/rev]

Feedrate F means the relative movement size between tool and structure in cutting. It is indicated as feedrate per minute (G94) or feedrate per revolution (G95). The feedrate per minute (F) means the feedrate of axis per one minute and the unit for it is [mm/min]. The feedrate is commanded in the form of G94 F200. The feedrate per revolution (f) is decided by the feed amount per thread, which means feedrate of the axis per spindle ' s one time of rotation. The unit for it is [mm/rev] and it is commanded in the form of G95 F0.2.

Feedrate of table (F) = Feedrate per rotation × Speed of revolution (N)
= Feed per thread(f) × No. of threads (Z) × Speed of revolution (N)

F : Feedrate of table [mm/min]

f : Feed per thread [mm/tooth]
Z : No. of threads [teeth/rev]
N : Speed of revolution [rpm]

If the feedrate is given as the moving distance per revolution [mm/rev] in the table of cutting conditions, it should be converted into moving distance per minute [mm/min].

- In case of drill and reamer counter sink

F [mm/min] = N [rpm] × f [mm/rev]

- In case of milling cutter

F [mm/min] = N [rpm] × No. of cutter threads [teeth/rev] × f [mm/teeth]

- In case of tapping and screw cutting

F [mm/min] = N [rpm] × Pitch of screw

1.2.4 Material Removal Rate Q [cm^3 / min]

- In case of drill

Q = Area of drill × No. of revolutions × Feedrate

The area of drill whose diameter is d is $\pi d^2/4$ [mm^2]

$$Q(Cm^3) = \left(\frac{\text{Area}}{100}\right)\times(Nrpm)\times\left(\frac{Fmm/rev}{10}\right) = \frac{\text{Area} \times F(mm/\min)}{1000}$$

- In case of milling cutter

Q = Width of cutter × Depth of cutting × Feedrate per cutter thread × No. of cutter threads

1.2.5 Actual Cutting Time T [sec]

For calculation of length, surplus at the time of starting the cutting and of finishing it should be included into the cutting length.

$$T = \frac{L}{F} \times 60$$

L : length of cutting processing [mm]

F : Feedrate per minute [mm/min]

2 Program

Program is a method to give various commands for the location of a tool's feed path and for machine's several actions when a machine works on cutting. In this system, it is made out in text file under AUTO mode and it is directly made out on the edition window under MDI mode.

2.1 Program File

The file having the contents of the program is called, “ Program File ” , which is saved in the form of “ program name.extension ” . The program name can be made with a combination of numbers or letters. The sub-program should be named with four-digit number only. The name of sub-program from 9000 to 9029 is used as macro program in the system so users can ’ t use it as a general sub-program. The extension is “ NC ” .

Basically the program file is retrieved from and saved in NC folder which is under the folder in which the system execution file exists. It also can be retrieved from and saved in an arbitrary folder or the folder on other PC of network. There is no limitation to the capacity of the program, but it is restricted by the capacity of the saving medium of CNC.

> Note :
> The macro programs of 9000 ? 9029 should be in /Nc/Macro folder.

2.2 Factors of Program

The program is made in the language that CNC can read. It is composed of blocks that indicate a series of action commands. Block is composed of a group of words and the word is composed of address and data.

It is made out with the letters based on ASCII code. In case the letters that are not defined in the CNC system such as ~, !, $, ^, & are commanded, an alarm, F_82001 (which means, "There is an undefined letter.") happens.

2.2.1 Address

In address, there are basic address and special address. The basic address is defined by one of A to Z, which informs the meaning of a command.

Address	Description	Example
D	Tool diameter compensation value	G41 D1
H	Tool length compensation value	G43 H1
F	Command of velocity	F100.
G	Command of get ready	G00
I, J, K	Command of locating at the center for circular interpolation	G02 X10. I20.
M	Command of assist	M00
N	Statement label	N10
O	Program name	O1234
S	Command of spindle	S1000
T	Command of tool	T1010
X, Y, Z	Command of location	G00 X10.

In the special address, there are comments like %, ;, (,), and optional block skip statement, /.

2.2.2 Data

Data informs a necessary value for address command. All the data command should be given within 12 letters except for signs. Otherwise, an alarm, F_82002 (The number exceeds the maximum buffer) happens. If more than one decimal is commanded in the data that are commanded in the type of real number, an alarm, F_82004 (which means, "The decimal is more than one.") happens.

2.2.3 Word

Word is the basic unit of functional commands, which consists of address and data. Data gets located next to address.

▯II Example of application

N100

G00

X100.

If more than two words, both of which have the same contents, are commanded in one block, the first word is ignored and the second word gets carried out.

▯II Example of application

G00 X100. X200. (X100.is ignored and X200. is executed.)

2.2.4 Block

Block is a basic command unit to operate the machine. It consists of one word or a combination of several words. One block conforms to one line of the program. The maximum number of letters for one block is 300, including space. If the number of the letters for one block is over 300, an alarm, F_82111 (which means, "The letters for one block should be 300 or less.") happens.

N○○	G○○	X ○.○	Y ○.○	Z ○.○	M○○	S○○○	T○○	F○○
Command of statement label	Command of get ready	Command of coordinate			Command of assist	Command of spindle	Command of tool	Command of speed

In case the program is run, the order of each word doesn't matter in the same block. Generally, it is okay to input in the order as shown on the above.

Once one block is completely executed, it is progressed to the next block.

2.3 Progress of Program

The program consists of a combination of blocks. According to the contents of the blocks, the functions are different. In the first block, a program file name like O 1234 is specified and by use of the function of comment, explanation of the program is made. For the program file name having O, it is good to input, having one space after O. If O and R are input without space, it is misunderstood as OR operator, and accordingly, an alarm happens. If other command is given in the block starting with O, it gets ignored. At the end of the block, there should be M02 or M30, an auxiliary command to close the program. Without the command of M02 or M30, an alarm, F_82016 (which means, "The program has been closed without M02 or M30.") happens.

2.3.1 Comment

%, O, ;, ( ) are the commands that are handled as comments. All the contents in the block after the commands of %, O, ; are handled as comment. The command of ( ) handles some contents in the middle of a block as comment.

2.3.2 Statement Label

This is a command using N address and number at the head of a block. It is used for designating a block range for canned cycles or for moving performance of a program to a particular block. It is possible to designate a label regardless of order, but for convenient interpretation of the program, it is good to designate in order from a higher block to a lower one.

If the same statement label is given to several blocks, an alarm, F_82018 (which means, "There is the same block in progress.") happens. In case only "N" (N__) is designated without number next to N, it is understood as a statement label.

If the total number of statement labels designated in a program is over 1000, an alarm, F_82019 (which means, "The number of statement labels exceeds the maximum number of buffers.") happens. In order to carry out the program without being influenced by the limited number of statement labels, PI 134 (#3124), a parameter of checking whether there is a statement label or not, is used. If the value of the parameter is set at 0, it retrieves statement labels and the command of statement label can't exceed the limited number of 1000, but if the value of parameter is set at 1, it doesn't retrieve statement labels and there is no limitation to the allowed number of statement labels to be commanded.

2.3.3 Optional Block Skip

This is used with the command of / at the head of a block. It selects whether to perform the current block or not through an external signal. The command of / should be located in the forefront of the block and otherwise, an alarm, F_8201 (which means, "Giving a command is possible only at the head of the block"), happens. If there is / command at the head of the block and the switch is ON, it skips the current block and moves to the next one.

Optional Block Skip can handle one signal with / command, but in addition, it can work for several signals too. In this system, 10 external signals can be handled and the command is indicated as /0 ~ /10. Here, /0 works for the same signal as /.

2.4 Main Program and Sub-program

The program is divided into main and sub-program. The main program initially starts processing from the status of stop. The sub-program is executed by paging the main program, and if the execution is closed, it returns to the main program. The execution program can be composed of the main program only or the main program and several sub-programs.

2.4.1 Sub-program

Just like the case that a shape to process is repeated or a pattern repeats a certain action, it is possible to program the shapes that are often used in a program and then, use them in other program. Sub-program paging is to make repeated parts into a program and page them whenever necessary. The program that is paged at this time is a sub-program. The name of sub-program should be 4-digit numbers only.

> **Notes:**
> The name of sub-program should be 4-digit numbers only. 9000-9029 are the names used in the system so users can't use them.

2.4.2 Multiplicity of Sub-Program Paging

The main program can page a sub-program and the sub-program can page another sub-program. If to page a sub-program once is one-fold, it is possible to page 9 times. If a sub-program is paged more than 9 times, an alarm, F_82022 (which means, "It exceeds the maximum chances of paging a sub-program.") happens.

2.4.3 Sub-program Paging and Return

<Table >

M98	Sub-program Paging
M99	Sub-program Close
P _	Name of sub-program in case of M98 (_ is 4-digit number) and statement label of the block to return in case of M98
Q _	Statement label of the block in which the sub-program starts (Do from the first block once it is omitted)
R _	Statement label of the last block for sub-program. (Progressed until M99 appears if omitted)
L _	Frequency of repeating sub-program paging

The sub-program should be in the same folder as the main program (storage space). If there is no sub-program to page in the same folder, an alarm, F_82109 (which means, "There is not a program file that is selected."). The program of 9000 ? 9029 should be in the /Nc/Macro folder. .

In case the grammar is not accurate at the time of paging a sub-program, an alarm, F_82021 (which means, "The paging grammar for the sub-program is not correct.") happens. For the last block of the paged sub-program, M99 should be used independently. If M99 is not commanded in the sub-program under the condition that R_ command is not used, an alarm, F_82024 (which means, "There is no M99 in the sub-program.") happens. And if the grammar is not correct in case of M99 command given, an alarm, F_82025 (which means, "M99 grammar is not correct.) happens. In case of paging a sub-program consecutively over one-fold, if the program that was paged previously is paged again, the program gets to fall into infinite loop. As shown on the above, if a sub-program that has been paged already is paged again, an alarm, F_82023 (which means, "It has been already paged.") happens.

Under the MDI mode, it is impossible to page a sub-program. In case of paging it, an alarm, F_82113 (which means, "It is impossible to page a sub-program under MDI mode.")

happens.

▯ II Example of application

```
(Main program)
O Paging sub-program
G54 G00 X0. Y0. Z0.
X20. Y40.
M98 P10 Q1 R2 L2
X70. Y20.
M98 P10
G91 G28 X0. Y0. M05
M30

(Sub-program)
O 0010
N1 G90 G00 Z5.
G01 Z-10. F20.
Z-25. F100
N2 G00 Z50.
M99
```

2.4.4 Main Program Repeat

<Table>

M99	Main program repeat
P _	Statement label of starting block at the time of repetition
L _	Frequency of repetition

With the command of M99 in the main program, it carries out the function of repetition. If the command of P_ is not given, it repeats from the first block in the main program, and if the command of L_ is not given, it repeats without limitation.

Example of application

```
O Main program repeat
G54 G00 X0. Y0. Z0.
N1 G91 G01 Z5. F500.
X100.
Y100.
X-100.
Y-100.
M99 P1 L3
M30
```

3 G Code

G Code defines a type of tool ' s moving and working, and carries out operation of machine and program. There are One Shot command and Modal command. One Shot command is effective in the block in which G code is commanded, and the Model command is continuously effective in the next block until the G Code in the same group is commanded. The groups of 0 and 23 are One Shot command and the others are for Modal command.

Item	Description
One Shot command	Effective only in the block in which G code is given
Modal command	Effective until G code in the same group is given

G Code, a preparatory function, is defined as shown the below G Code table. If G code that is not specified here is given, an alarm, F_82030 (which means, "The G code is not available.") happens. When G code in the same group is commanded in the same block, the G code that is commanded later is effective and it is possible to command G code of each different group in the same block. If G code in no. 1 group is commanded during the canned cycle mode, the cycle mode is cancelled automatically (G80).

3.1 Table of G Code

G code that sets G code (which becomes default when initialized) with parameter

G code that becomes default when initialized

G Code	Group	Function	Remark
G00	1	Positioning	4.1
G01		Linear interpolation	4.3
G02		CW Circular interpolation	4.4
G03		CCW Circular interpolation	
G04	0	Dwell	5.4
G09		Exact Stop(Inposition)	5.3.1
G10		Compensation value setting of tool length and tool diameter	13.1.3
G10.3	21	High-speed machining mode	15.1
G11.3		High-speed machining mode off	
G15	17	Polar coordinate command cancellation	8.3
G16		Polar coordinate command	
G17	16	XY Plane select	7.4
G18		ZX Plane select	
G19		YZ Plane select	
G20	6	Inch data input	8.4
G21		Metric data input	
G22	9	Stored Stroke Limit function ON	15.5.3
G23		Stored Stroke Limit function OFF	
G27	0	Reference point return check	6.4
G28		Return to Reference point	6.1
G29		Return from Reference point	6.3
G30		2nd,3rd,4th reference point return	6.2
G31	23	Skip Function 1	4.9
G31.1		Skip Function 1	
G31.2		Skip Function 2	
G31.3		Skip Function 3	
G31.4		Skip Function 4	
G33	1	Thread cutting	4.8
G37	0	Automatic tool compensation 1	4.10
G37.1		Automatic tool compensation 1	
G37.2		Automatic tool compensation 2	
G37.3		Automatic tool compensation 3	
G37.4		Automatic tool compensation 4	

G39		Corner off circular interpolation	13.2.1
G40	7	Cutter Diameter compensation cancel	13.2.1
G41		Cutter Diameter compensation Left	13.2.1/13.2.2
G42		Cutter Diameter compensation Right	13.2.1/13.2.2
G43	13	Tool length compensation + direction	13.1.1
G44		Tool length compensation - direction	
G45	0	Tool offset increase	13.3
G46		Tool offset decrease	
G47		Tool offset double increase	
G48		Tool offset double decrease	
G49	13	Tool length compensation cancel	13.1.1
G50	11	Scaling, Mirror image cancel	15.2 / 15.3
G51		Scaling, Mirror image	
G52	0	Local coordinate setting	7.3
G53		Machine coordinate setting	7.1.1
G54	14	Work-piece coordinate 1 select	7.2.2
G55		Work-piece coordinate 2 select	
G56		Work-piece coordinate 3 select	
G57		Work-piece coordinate 4 select	
G58		Work-piece coordinate 5 select	
G59		Work-piece coordinate 6 select	
G60	0	Single direction positioning	4.2
G61	15	Exact stop mode	5.3.2
G62		Automatic corner override mode	5.3.5
G63		Tapping mode	5.3.4
G64		Cutting mode	5.3.3
G65	0	Macro calling	14.1.1
G66	12	Custom Macro modal calling	14.1.2/14.1.8
G67		Custom Macro modal calling cancel	14.1.2
G68	18	Coordinate rotation	15.4
G69		Coordinate rotation cancel	
G73	10	High-speed peck drilling cycle	12.3.4
G74		Reverse tapping cycle	12.3.6
G76		Fine boring cycle	12.3.9
G80		Canceling canned cycle	12.2.1
G81		Drilling cycle	12.3.1
G82		Dwell drilling cycle, Counter boring cycle	12.3.2
G83		Peck drilling cycle	12.3.3
G84		Tapping cycle	12.3.5
G84.2		Rigid tapping cycle	
G84.3		Counter rigid tapping cycle	12.3.6
G85		Boring cycle	12.3.7
G86		Boring stop cycle	12.3.8
G87		Back boring cycle	12.3.10
G88		Manual boring cycle	12.3.11
G89		Dwell boring cycle	12.3.12

G90	3	Absolute command	8.1 / 12.1.3
G91		Incremental command	8.2 / 12.1.3
G92	0	Setting for workpiece coordinate	7.2.1 / 9.3
G94	5	Feed per minute	5.2.1
G95		Feed per revolution	5.2.2
G96	2	Constant surface speed control On	9.1
G97		Constant surface speed control Off	9.2
G98	19	Return to initial position	12.1.4
G99		Return to R position	
G107	22	Cylindrical coordinate interpolation	4.7
G112	20	Polar coordinate interpolation mode	4.6
G113		Polar coordinate interpolation mode off	

3.2 Initialization of G code modal

All the groups except for One-Shot command group (0 and 23) and group no. 22 (setting circular interpolation mode) have initial modal G code. The initial modal G code is effective without commanding G code in the group.

In case of the groups ? no. 66 (G20 / G21), no. 9(G22 / G23), and no. 16(G17 / G18 / G19), modal G code is initialized only when the power is on, and the other modal G codes are initialized when the power is ON and RESET. G code that is initialized is painted gray and (G00, G01), (G17, G18, G19), (G20, G21), (G22, G23), (G50, G51), (G68, G69), (G90, G91) can change G code to be initialized in parameter.

< Parameter ▸ Program ▸ Default setting >

			Default Setting
PI 0144	0		Modal G Code (0:G00 1:G01)
PI 0145	0		Modal Plane (0:XY(G17) 1:ZX(G18) 2:YZ(G19))
PI 0146	0		Modal Absolute/Increment (0:Absolute(G90) 1:Increment(G91))
PI 0147	0		Modal Unit of Input (0:Metric(G21) 1:Inch(G20))
PI 0148	0		Modal Prohibition Area Inspection(0:Perform(G22) 1:Cancel(G23)
PI 0150	0		Modal Coordinate System Rotation (0:Cancel(G69) 1:Apply(G68)
PI 0117	0.000	deg	Coordinate System Rotation Angle

< Parameter ▸ Program ▸ Schedule setting>

			Scale Setting
PI 0149	0		Modal Scale Setting (0:Cancel (G50) 1:Apply (G51))

4 Interpolation Functions

Interpolation function is the basic function for cutting, which gives a command according to the way that a tool moves in the section from start to end.

4.1 Rapid Traverse Positioning (G00)

[G90 / G91] G00 X _ Y _ Z _

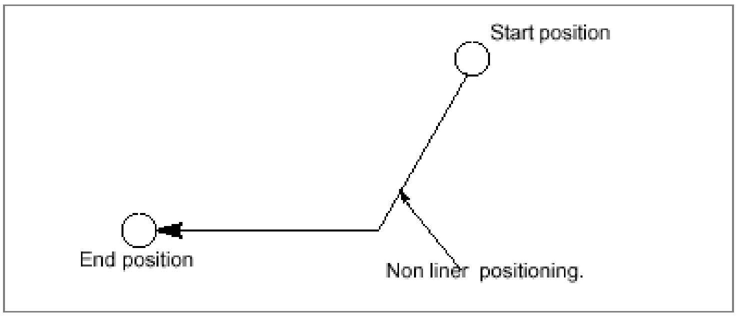
G90 / G91	Absolute / Incremental Command
G00	Traverse Positioning
X _ Y _ Z _	Forwarding Location

This is a function to forward to the location commanded on X, Y, Z (A, B, C are possible too in case there is an additional axis) at rapid feedrate set with parameter PM 2759-2790 (#22759-22790) for each axis. After completion of In-Position investigation, it moves to next block. It is possible to change the feedrate by use of the switch, Rapid Feed Override (Refer to 5.1).

< Relevant parameter>

번호	내용
PM 2759-2790 (#22759-22790)	급송 이송 속도 설정
PM 2928-2959 (#22928-22959)	In Position 범위

Forwarding is done independently by each axis. A short axis reaches the destination first so it doesn't make a straight line.



G00 is a modal command. If there is no other interpolation command (group 1), it continuously carries out positioning for the axis command.

Ⅱ Example of application

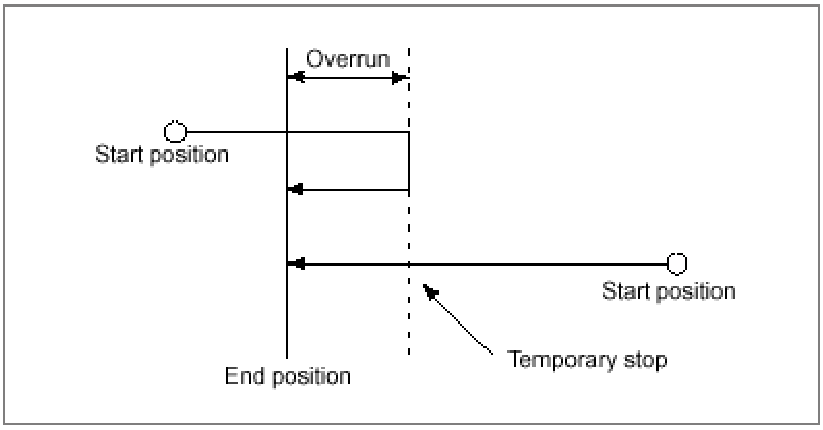
```
O Positioning
G90 G54 G00 X0. Y0. Z0.
X100. Y50.
Y100.
Z100.
G91 X-100.
Y-100.
Z-100.
M30
```

4.2 Single Direction Positioning (G60)

G60 [G00 / G81-G89] X _ Y _ Z _

G60 Single Direction Positioning
G00 / G81-89 Positioning / Drill Cycle
X _ Y _ Z _ Forwarding Location

This function is executed at the end of rapid feed in order to run highly accurate position control (the backlash of a machine compensated). It stops at the position that is far away as much as the overrun set for the axis that is designated in G60 block, and then, moves to the next commanded position to remove influence of backlash. The amount of overrun is designated at parameter PI 61-69(#3061-3069) for each axis. (Location: Parameter ▸ Program ▸ Single Direction Positioning overrun forwarding amount).



It is applied only for the commands of positioning and canned cycle, but not for Z axis in the drill cycle. In case of overrun forwarding at Mirror Image, the direction is not affected by Mirror Image.

Ⅱ Example of application

O Single Direction Positioning

```
G54 G00 X0. Y0. Z0
G90 G60 X50.
G60 Y50.
G60 Z50.
G91 G60 X50. Y-50. Z-50.
M30
```

4.3 Linear Interpolation (G01)

[G90 / G91] G01 X _ Y _ Z _ F _

G90 / G91	Absolute / Incremental Command
G01	Linear Interpolation
X _ Y _ Z _	Forwarding Location
F _	Feedrate

This function is to linearly move each axis at the same time to the designated location at the given speed for cutting that you want.

Feedrate is given in F_, and as for the feed method, there are feed per minute (G94, mm/min) and feed per revolution (G95, mm/rev). Usually, feed per minute is used. Here, the freedrate per revolution is defined as the amount to move during one revolution of the spindle. You can change the federate, using the switch of federate override. F that is given once, modal, is continuously effective in the next block (refer to 5.2).

If the feedrate by F code is not designated in G01 block or the previous block, it moves at the feed set with parameter PM 2871 (the lowest speed limit for cutting feed).

When the feed amount of X is Lx, the feed amount of Y is Ly and the feed amount of Z is Lz, the feedrate of each axis is calculated as follows:

$$L = \sqrt{Lx^2 + Ly^2 + Lz^2}$$

$$Fx = \frac{Lx}{L} \times F$$

$$Fy = \frac{Ly}{L} \times F$$

$$Fz = \frac{Lz}{L} \times F$$

G01, modal command, is continuously effective to carry out linear interpolation for the axis command if there is no other interpolation command (group 1).

▯ll Example of application

O Linear interpolation

```
G54 G00 X0. Y0. Z0.
G90 G01 X50. F100.
Y-40.
Z-30.
G91 X50. Y40Z30.
M30
```

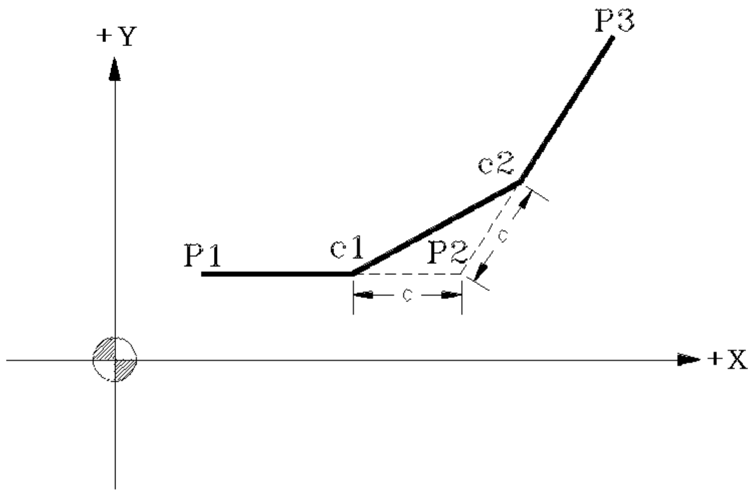
4.3.1 Chamfering Arbitrary Angle (Taping, Chamfering)

By use of [,C] command between two blocks having arbitrary angle, you can command chamfering simply. [,C] command is used at the end of the block in which chamfering starts.

G01 X _ Y _ ,C _ F _

G01	Linear Interpolation Command
X _ Z _	Forwarding Location
,C _	Chamfering

< Tool ' s path by chamfering at arbitrary angle>



▯ II Example of application

G00 X10 Y30;	(Move to P1)
G01 X30, C5 F300 ;	(Move to P1 ▸ P2(c1) ▸P2(c2))
X50 Y50	(Move to P2(c2) ▸ P3)

Note :

① Chamfering and rounding arbitrary angle are calculated by the command of axis feed for the designated block and the next block so next to the block in which the command of chamfering and rounding is given, there should be a block related with the axis (move T code command within 5 blocks).

② The length of chamfering and rounding [C, R] is a radius command, which should be smaller than the length of segment P1-P2 and segment P2-P3.

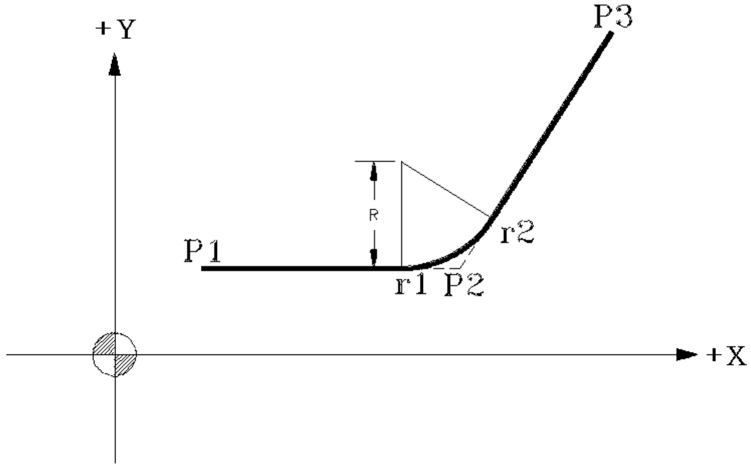
4.3.2 Rounding Arbitrary Angle (Corner R, Rounding)

By use of [,R] command between two blocks having arbitrary angle, it is possible to command rounding. [,R] command is used at the end of the block in which rounding starts.

G01 X _ Y _ ,R _ F_

G01	Linear Interpolation Command
X _ Y _	Forwarding Location
,R _	Rounding

< Tool ' s path by rounding>



▯II Example of application

G00 X10 Y30 ;	(Move to P1)
G01 X30, R15 F300 ;	(Move to P1 ‣ P2(r1) ‣ P2(r2))
X50 Z50	(Move to P2(r2) ‣ P3)

Note：

① Chamfering and rounding arbitrary angle are calculated by the command of axis feed in the designated block and the next block so there should be a block related with the axis feed next to the block in which chamfering and rounding are commanded (move T code command within 5 blocks).

② The length of chamfering and rounding is a diameter command, which should be smaller than the length of segment P1-P2 and segment P2-P3.

4.4 Circular Interpolation (G02/G03)

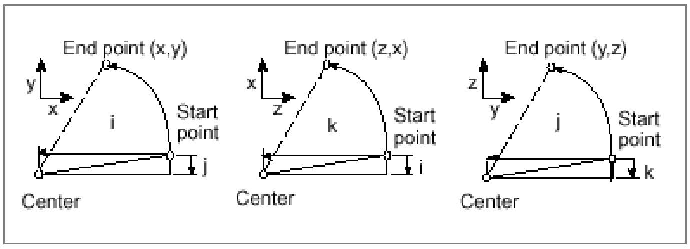
[G17] {G02 / G03} X _ Y _ {I _ J _ / R _} F _ [G18] {G02 / G03} X _ Z _ {I _ K _ / R _} F _ [G19] {G02 / G03} Y _ Z _ {J _ K _ / R _} F _

G17 / G18 / G19	Plane Command
G02 / G03	Circular Interpolation in the Direction of CW/CCW
X _ Y _ Z _	Forwarding Location
I _ J _ K _	Vector from the Start of a Circle to Its Center (direction and distance) Use it instead of R command
R _	Radius of circle
F _	Feedrate

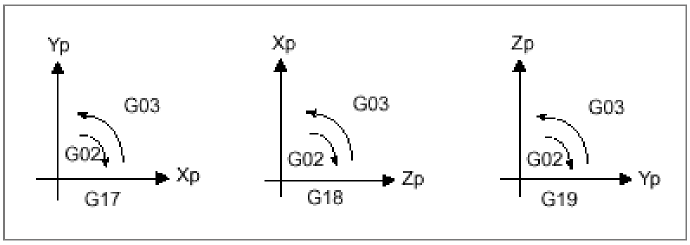
This function is to rotate and forward to the designated location at the designated speed based on the designated center point. The speed here means the velocity of the line in the direction of a tangent. To the position of X _ Y _ / X _ Z _ / Y _ Z _ on the work-piece coordinate system based on I _ J _ / I _ K _ / J _ K _ or the center point by R_, it rotates and forwards at F_ speed.

Feedrate can be changed by the switch of feedrate override. The F which is given once, modal, is continuously effective in the next block (refer to 5.2).

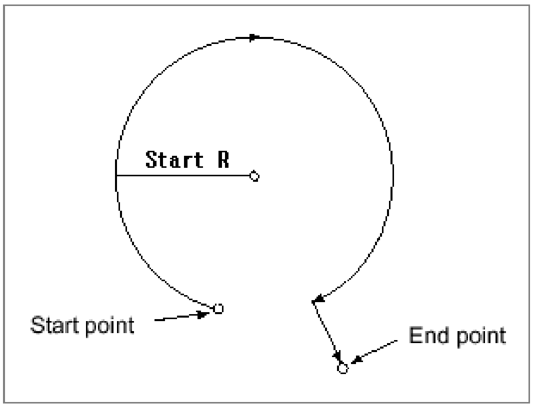
The nominal value of the center point by I _ J _ / I _ K _ / J _ K _ is the incremental distance from the starting point to the center point regardless of absolute/incremental command. In case of I0, J0, K0 command, they can be omitted. In case the starting point and the ending point are same, 360 ° circle can be commanded by I _ J _ K _.



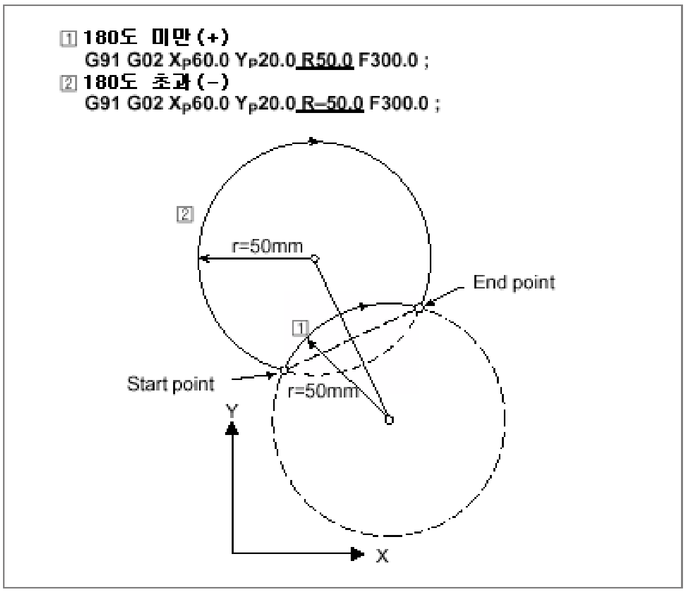
G17 is defined as XY plane, G18 by ZX plane and G19 by YZ plane. Based on the center point against each plane, G02 rotates clockwise (CW) while G03, counterclockwise (CCW).



The allowable error for the radius from the starting point to the center point and the radius from the ending point to the center point can be defined at the parameter PI 151(#3151). If the allowable error is exceeded, an alarm, F_82112 (which means, "The center point of the circular arc is not appropriate.") happens. In case the starting radius of a circular arc and the ending radius is different because the allowable error for parameter PI 151 is set big, it rotates at the starting radius and it forwards linearly for remaining.



In case of designating the center point by radius R_ instead of I _ J _ K _ , there can be two types of center point, either of which is decided according to +, - of R value. In case the difference between half of the distance between the starting point and the ending point, and R nominal value is larger than the allowable error of parameter PI 151, an alarm, F_82100 (which means, "It is impossible to find the center point of circular arc.") happens. In case I _ J _ K _ and R _ are commanded at the same time, I _ J _ K _ gets ignored and by R _, the center point is applied. It is impossible to command 360 ° whole circle with R_.



II Example of application

O Circular interpolation

G54 G00 X0. Y0. Z0.

(XY plane)

G90 G17 G02 X50. Y50. I50. F100.

G03 X0. Y0. R-50.

G91 G03 X100. Y100. J100.

G02 X-100. Y-100. R-100.

(ZX plane)

G90 G18 G02 Z50. X50. K50. F200.

G03 Z0. X0. R-50.

G91 G03 Z100. X100. I100.

G02 Z-100. X-100. R-100.

```
(YZ plane)
G90 G19 G02 Y50. Z50. J50. F300.
G03 Y0. Z0. R50.
G91 G03 Y100. Z100. K100.
G02 Y-100. Z-100. R-100.
(Whole circle)
G17 G02 I50.
G03 J50.
G02 I50. J50.
(I is ignored and R is applied)
G02 X-100. I30. R50.
M30
```

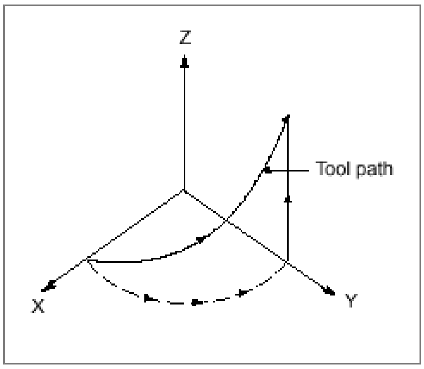
4.5 Helical Interpolation (G02/G03)

[G17] {G02 / G03} X _ Y _ {I _ J _ / R _} Z _ F _
[G18] {G02 / G03} X _ Z _ {I _ K _ / R _} Y _ F _
[G19] {G02 / G03} Y _ Z _ {J _ K _ / R _} X _ F _

G17/G18/G19	Plane Command
G02 / G03	Command of Circular Interpolation in the Direction of CW/CCW
X _ Y _ Z _	Forwarding Location
I _ J _ K _	Center Point
R _	Radius
F _	Feedrate

Helical interpolation nominates other axis which is not available on the circular interpolation plane during circular interpolation, synchronizes with the circular interpolation, and carries out linear interpolation, which is used for cutting cylindrical cam or screw.

Helical Interpolation ' s selection of plane	
G17 Plane	Circular interpolation for X and Y, Linear interpolation for Z
G18 Plane	Circular interpolation for Z and X, Linear interpolation for Y
G19 Plane	Circular interpolation for Y and Z, Linear interpolation for X



Feedrate is nominated as F_, which can be changed by use of the switch of feedrate override. The given F, as modal, is continuously effective for the next block (refer to 5.2).

In the helical interpolation, tool diameter compensation is applied to circular interpolation only, not to the blocks in which offset command or tool length compensation is commanded.

■ Example of application

```
O Helical interpolation
G54 G00 X0. Y0. Z0.
(XY plane)
G90 G17 G02 X50. Y50. I50. Z50. F100.
G03 X0. Y0. Z0. R-50.
G91 G03 X100. Y100. J100. Z100.
G02 X-100. Y-100. Z-100. R-100.
(ZX plane)
G90 G18 G02 Z50. X50. K50. Y50. F200.
G03 Z0. X0. Y0. R-50.
G91 G03 Z100. X100. I100. Y100.
G02 Z-100. X-100. Y-100. R-100.
(YZ plane)
G90 G19 G02 Y50. Z50. J50. X50. F300.
G03 Y0. Z0. X0. R-50.
G91 G03 Y100. Z100. X100. K100.
G02 Y-100. Z-100. X-100. R-100.
```

(Whole circle)

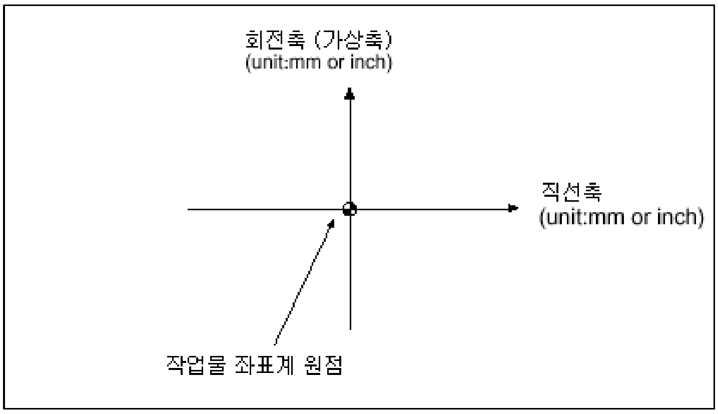
```
G90 G17 G02 I50. Z50.
G03 J50. Z0.
G02 I100. J100. Z100.
M30
```

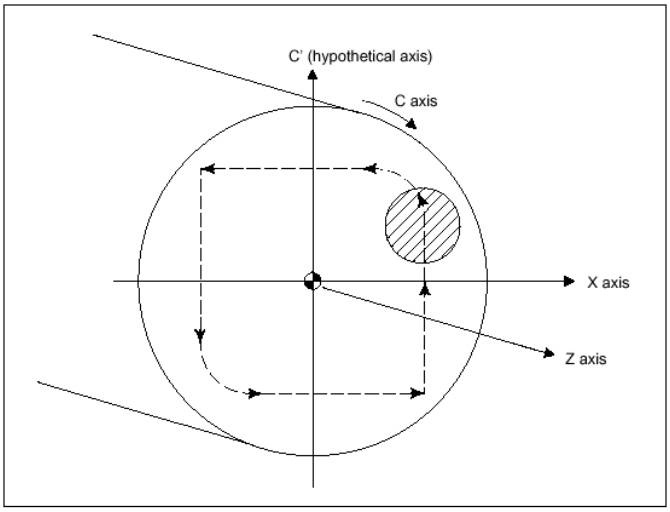
4.6 Polar Coordinate Interpolation (G112/G113)

G112 G01, G02, G03 ... G113

G112	Polar Coordinate Interpolation Start
G01, G02, G03 ...	Processing Interpolation command
G113	Polar Coordinate Interpolation Close

Polar coordinate interpolation is: hypothetically sets coordinate system that consists oflinear axis, axis of rotation,and the plane vertical to the axis of rotation, and does programming; and by these two axes, machining is done on the vertical planeIt is suitable for face milling or cam shaft machining.





If G112 is commanded, polar coordinate interpolation mode, as modal, begins, and if G113 is given, the mode is cancelled. In the polar coordinate interpolation mode, only the linear and the circular interpolation can be commanded. If there is any other command than those, an alarm, F_82204 (which means, "It can't be commanded for polar coordinate interpolation.") happens.

The linear axis is used for X, Y, and Z, and the axis of rotation, for A, B, and C. Having these two axes on XY plane, the interpolation is performed. Here, X is used as linear axis and Y, as axis of rotation. At the time of circular interpolation command, I, J. K command are possible only with I, J.

Selection of linear axis can be set at parameter PM 4623(#24623), and selection of the axis of rotation, at parameter PM 4624(#24624).

Tool diameter compensation that has been progressed before the polar coordinate interpolation command is given is cancelled as soon as the polar coordinate interpolation command is given, and it is progressed again in the polar coordinate interpolation mode. When the polar coordinate interpolation is cancelled, compensation is cancelled again and the previous status of compensation is progressed.

For feedrate, the tangent speed on the polar coordinate interpolation plane is commanded as F under the status of feedrate per minute (G94).

Within the block of polar coordinate interpolation mode, it is impossible to restart the program.

Example of application

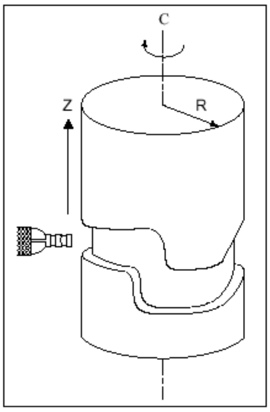
```
O Polar coordinate interpolation
G54 G90 G00 X60. C0. Z50.
G112
G42 G01 X20. F100.
C10.
G03 X10. C20. R10.
G01 X-20.
C-10.
G03 X-10. C-20. I10. J0.
G01 X20.
C0.
G40 X60.
G113
M30
```

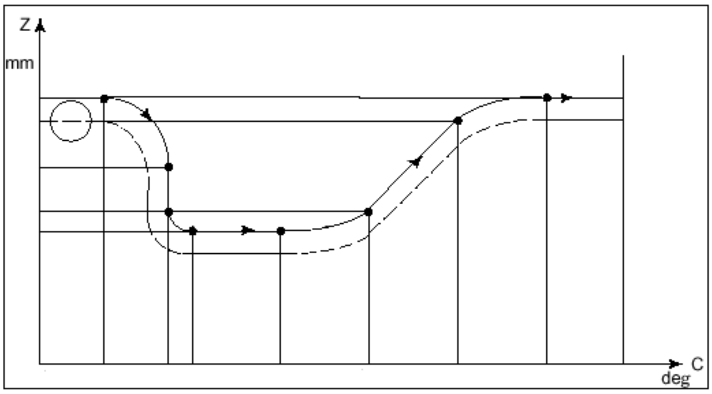
4.7 Cylindrical Interpolation (G107)

G107 {A _ / B _ / C _}
G01, G02, G03 ...
G107 {A 0 / B 0 / C 0}

G107	Cylindrical Interpolation Command
A _ / B _ / C _	Cylindrical Interpolation Start and Axis of Rotation and Radius Setting
G01, G02, G03 ...	Machining Interpolation
A 0 / B 0 / C 0	Cylindrical Interpolation Cancel

Cylindrical interpolation is for easily machining the circumference of a cylinder by the radius of the axis of rotation. It is usually used for machining cylindrical cam. The instructed angle for the axis of rotation is converted into distance, which is converted into angle again after calculating linear and circular interpolation. The axis that is commanded together with G107 becomes the axis of rotation, and the value that is commanded together with the axis of rotation is applied as radius of a cylinder. In case the radius is set at 0, cylindrical interpolation is cancelled.





If the mode is set by the cylindrical interpolation command, it is applied as modal until cancellation is commanded. While the cylindrical interpolation mode is applied, only the linear and the circular interpolation can be commanded. In case of any other command available than linear and circular interpolation, an alarm, F_82208 (which means, "It can't be commanded during the cylindrical interpolation.") happens.

```
G107 C30.0
G18 G01 Z10.0 C20.
G02 C40. R20.
G00 Z20. <Alarm>
G107 C0 <Cancel>
```

You can decide which axis on the plane should be used as the axis of rotation when there is the circular interpolation command during cylindrical interpolation at the time of inspecting a tool's path. Parameter PI 155(#3155) sets it and according to the nominated plane (G17, G18, G 19), it is applied.

Parameter PI 155 (#3155) :Setting the axis of rotation for cylindrical interpolation
This parameter is to set the axis to map the axis of rotation at the time of circular interpolation in the mode of cylindrical interpolation. X, Y, Z can be selected and by mapping the axis of rotation to the selected axis, circular interpolation is carried out. For example, in case the axis of rotation is mapped into X and it is XY plane (G17), the width becomes the axis of rotation and the length becomes Y. If the plane is ZX (G18), the width becomes X and the lengthl becomes the axis of rotation. However, if the plane is YZ (G19), it is impossible to carry out circular interpolation with the commanded axis of rotation.

In actual machining, regardless of the plane, the horizontal axis becomes the axis of rotation and the vertical axis becomes linear axis. The linear axis is decided by the axis of rotation, and A corresponds with A, B with Y, and C with Z.

```
<Select X as the axis of rotation>
G17 C_ Y_
G02(G03) C_ Y_ R_
G18 Z_ C_
G02(G03) Z_ C_ R_
```

During the circular interpolation command, I, J, K can ' t be used.   Only R can be used.
Tool diameter compensation that was progressed before cylindrical interpolation is commanded is cancelled at the time of cylindrical interpolation command, and compensation is progressed again under the mode of cylindrical interpolation. When the cylindrical interpolation is cancelled, compensation is cancelled again and the previous status of compensation is progressed again. In the block of cylindrical interpolation mode, it is impossible to restart the program.

Ⅱ Example of application

```
O Cylindrical interpolation
G54 G00 G90 Z100.0 C0
G01 G91 G18 Z0 C0
G107 C57.299
G90 G01 Z120. F250
C30.
G02 Z90. C60. R30.
G01 Z70.
G03 Z60. C70. R10.
G01 C150.
G03 Z70. C190. R75.
G01 Z110. C230.
G02 Z120. C270. R75.
G01 C360.
Z100.
```

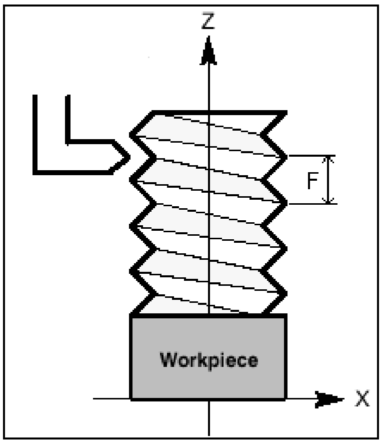
```
G107  C0
M30
```

4.8 Constant Lead Threading (G33)

G33 X _ Y _ Z _ F _

G33	Constant Lead Threading Command
X _ Y _ Z _	Forwarding Location
F _	Threading Lead

In milling, linear machining of constant lead threading is possible. At this time, the number of spindle ' s revolutions and the federate of tool should be synchronized.



Feedrate is not influenced by the switch of feedrate override. If the button of Feed Hold is pressed during threading, it is continued to work. After completion of threading, the function of Feed Hold is applied.

Example of application

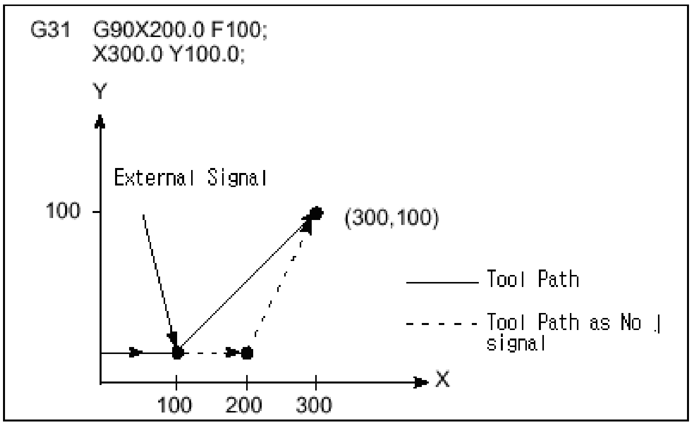
O Constant lead threading
G90 G54 G00 X0. Y0. Z0.
Z100.
G33 Z0. F5.
M30

4.9 Skip Function (G31/G31.1/G31.2/G31.3/G31.4)

{G31 / G31.1 / G31.2 / G31.3 / G31.4} X _ Y _ Z _ F _

G31 / G31.1 / G31.2 / G31.3 / G31.4	G Code by external signal
X _ Y _ Z _	Forwarding Location
F _	Feedrate

The method of command and the type of axis feed are applied same as those for linear interpolation (G01). Once a defined external signal is input, feed is stopped and it is progressed in the next block. At this time, the location of the machine (at the point that is stopped) is input to the system variable N 319 ~ 350 (#6319 ~ 6350) by axis. This function is used for measuring a size of work-piece or for knowing a particular location during machining.



Feedrate is not influenced by the switch of feedrate override.

Ⅱ Example of application

```
O Skip
G90 G54 G00 X0. Y0. Z0.
G31 X100. F500.          (Stop by Input of External Signal 1)
G31.2 Y100.              (Stop by Input of External Signal 2)
```

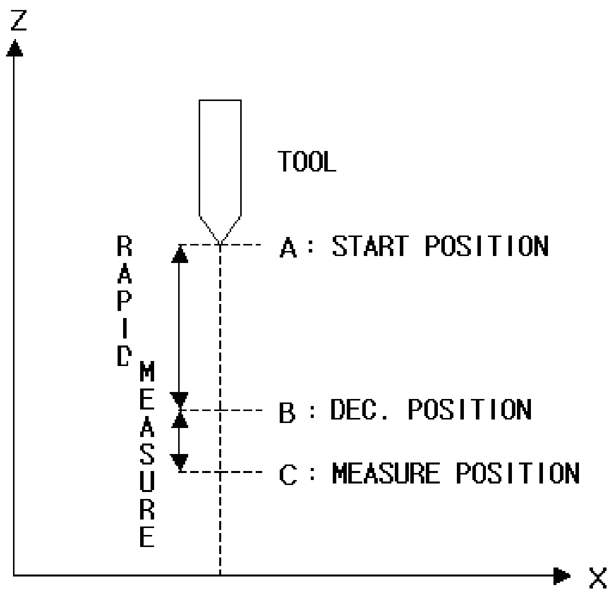
```
G31.3 X0.          (Stop by Input of External Signal 3)
G31.4 Y0.          (Stop by Input of External Signal 4)
M30
```

4.10 Auto Tool Length Measure (G37/G37.1/G37.2/G37.3/G37.4)

{G37 / G37.1 / G37.2 / G37.3 / G37.4} {X_ /Y_ /Z_}

G37	Auto tool length measure 1
G37.1	Auto tool length measuring 1
G37.2	Auto tool length measuring 2
G37.3	Auto tool length measuring 3
G37.4	Auto tool length measuring 4
X _ / Y _ / Z _	Measuring Location

It is used when a tool measuring equipment is installed on a particular position of the machine. It automatically measures compensation value of a tool and compensates. After moving a tool to a particular position with NC program, it calculates new tool compensation value from the difference between actually nominated position and the position where the auto tool measure signal happens (SKIP signal).



(1) Auto tool measure signal (SKIP signal)

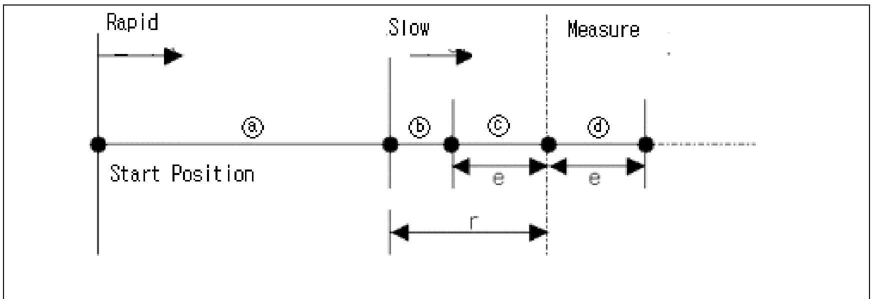
If SKIP signal becomes high, the feed command for the relevant block is closed and tool measure is carried out. After SKIP signal ' s happening, axis feed stops within 0 ~ 2x(IPO Sampling Time) msec, and the location where SKIP signal happens is put into the system variable SN 319~350 (#6319~ 6350). (Initial screen -> System management -> Status information).

G code corresponding to each SKIP signal is as shown on the below, and following G code, one of X, Y, or Z is nominated.

SKIP Signal	G Code
SKIP1	G37.1(G37)
SKIP2	G37.2
SKIP3	G37.3
SKIP4	G37.4

(2) Type of axis feed during auto tool measure

Once the command of auto tool measure is given, it rapidly moves a tool at first (ⓐ section). If a tool comes into the section of speed reduction set by parameter PI 141(#3141), the tool is moved at a low speed set by the parameter PI 143 (#3143) (ⓑ section). If the tool reaches the measuring point and the measuring signal (SKIP signal) is ON, moving of a tool is stopped.



r : Dec. Start Length
e : Automatic Measure Area

If the measuring signal is sensed in any other sections than ⓒ, ⓓ, an alarm, F_84022 (which means, "Alarm of tool measuring has happened.") happens. If the measuring signal is not sensed until it goes out of ⓓ section, an alarm, F_84022 (which means, "Alarm of tool measuring has happened.") happens.

(3) Calculation of auto tool compensation value

New compensation value is produced out by adding the difference (between the coordinate values of a tool at the time of reaching the measuring location and the nominated measuring location) to the currently used compensation value.

Compensation value = Current compensation value + (Location where measuring signal becomes ON – Nominated measuring point)

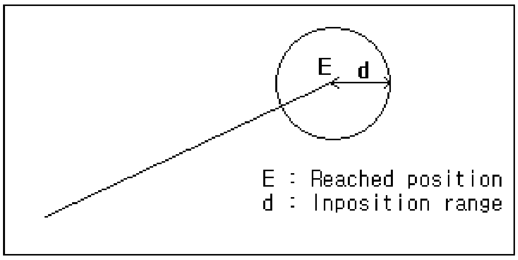
5 Feed Function

Feed functions mean the functions that a tool carries out while feeding under the situation of machining. In the feed functions, there are feedrate, cutting speed control (exact stop mode, cutting mode, tapping mode, automatic corner override mode), acceleration/deceleration, and dwell.

5.1 Rapid Traverse

Rapid traverse is that a tool moves under the positioning (G00) command. Rapid traverse goes to the next block after positioning of the designated block is completed and in-position investigation is completed. Rapid traverse speed is defined by axis on the parameter PM 2759-2790(#22759-22790).

In-position means that the reached position is within the allowable range from the targeted position. The range for each axis is defined on the parameter PM 2928-2959(#22928-22959).



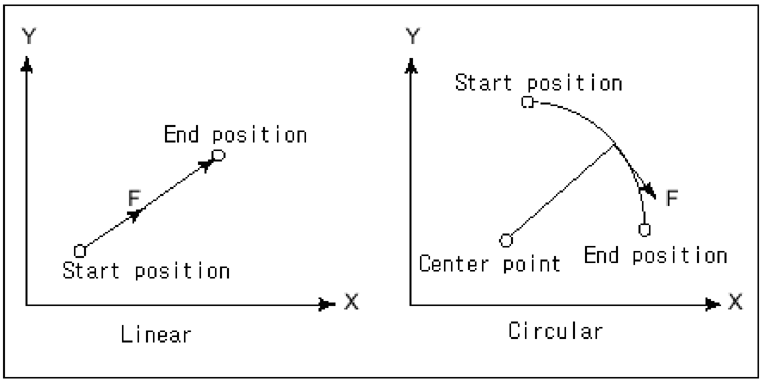
Rapid traverse speed can be adjusted by the override switch and the percentage of speed for each override switch can be set on the parameter, PM 2791-2822(#22791-22822). If it is set at 0 on the override parameter table, traverse is not carried out.

The rapid traverse speed under the dry-run mode is different, depending on the values set on the parameter PM 2828(#22828). If it is set at 0, it traverses at the manual speed selected by the switch, and if it is set at 1, it traverses at the speed of the rapid traverse speed parameter.

5.2 Cutting Feed

Cutting feed is that a tool moves by the speed given by F command under linear interpolation (G01) and circular interpolation (G2, G03) commands. Feed per minute (G94) and feed per revolution (G95) are available.

Cutting feed carries out control while maintaining the tangential speed to be the commanded speed.



Cutting feedrate can be adjusted by the override switch and the percentage of speed per position of each override switch can be set on the parameter PM 2891-2922(#22891-22922). If it is set at 0 on the override parameter table, cutting feed is not carried out.

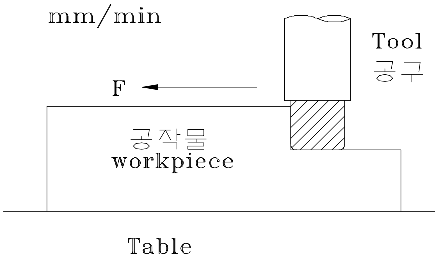
If F command is not given at all or if F0 command is given, it feeds at the lower limit for feedrate set on the parameter PM2871(#22871). If the feed that exceeds the upper limit for feedrate defined on the parameter PM2870(#22870) is given, an alarm, F_84020 (which means, " The command of cutting speed has exceed the maximum value. ") happens.

5.2.1 Feed per Minute (G94)

G94 F _

G94	Feed per Minute Mode
F _	Feedrate (mm/min or inch/min)

After this function is commanded, the speed of F command is applied as feed per minute.
This function, as modal command (group 5), is maintained before G95 (feed per revolution) is given. At the time of power on or RESET, the initialization of modal is set by this command.



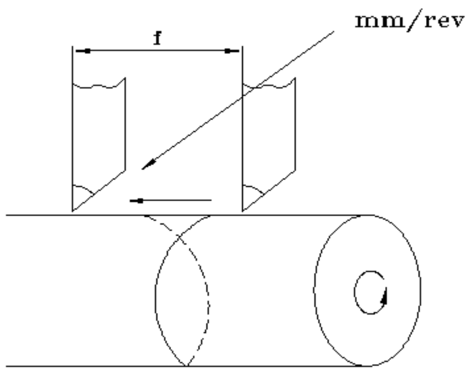
▌Ⅱ Example of application
G54 G00 X0. Y0. Z0.
G94 G01 X100. F500.
M30

5.2.2 Feed per Revolution (G95)

G95 **F _**

G95	Feed per Revolution Mode
F _	Feedrate (mm/rev or inch/rev)

Feedrate per revolution is defined as the amount of feed for one revolution of the spindle. After this function is commanded, the speed of F command is applied as feed per revolution. This function, as modal command (group 5), is maintained until G94 (feed per minute) is commanded.



▯ II Example of application

G54 G00 X0. Y0. Z0.
M03 S100
G95 G01 X100. F5.
M30

5.3 Cutting Feedrate Control)

The method of cutting feed control can be differently applied according to the commands like Exact Stop or Exact Stop Mode, Cutting Mode, Tapping Mode, and Automatic Corner Override.

5.3.1 Exact Stop (G09)

G09

G09	Command of Exact Stop

Everything has inertia so it is difficult to momentarily stop a certain thing or change its direction to a desirable position with power. In general, in the cutting mode, in consideration of this matter, at the time of accelerating or decelerating feed of an axis, the current block is decelerated in advance and then, the next block is accelerated again.

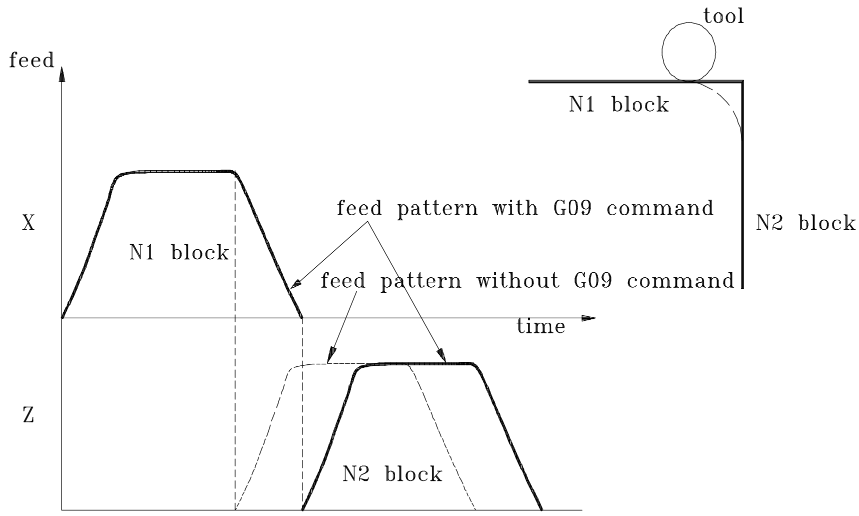
```
N1 G00 X50. Y0.
N2 G01 Y50. F2000.
N3 X0.
```

Running the above feed program, Y is decelerated in advance before it reaches the coordinate of Y50, and X is accelerated in advance. Therefore, at the corner, a phenomenon of rounding happens.

With the command of Exact Stop, NC checks whether the motor (at the time of decelerating) is in the range of In-Position set by the parameter, and then, performs feed of the next axis. Accordingly, by using the function of Exact Stop, it can prevent rounding at the corner.

The range of in-position for axis is set by parameter PM 2928~2959(#22928~22959).

⏭ Difference in cutting feedrate between the Exact Stop command in use and the Exact Stop command not in use



The above figure shows changes in the cutting feedrate by time. In case there is no Exact Stop command (G 09), feed of Z starts before feed of X is over so the corners get to have a phenomenon of rounding.

Note:

① This function is one-shot command, which is effective only in the instructed block.
② If this function is used, a phenomenon of minute stop happens at the node of a curved surface so the status of the machining surface is not good, the tool is seriously worn out and it takes more time for machining.

II Example of application

G54 G00 X0. Y0. Z0.
G09 G01 X100. F500.
Y100.
X0.
M30

5.3.2 Exact Stop Mode (G61)

G61

G61 Command of Exact Stop Mode

Exact Stop mode (G61)carries out cutting feed in the same way as Exact Stop command (G09). The difference between these two is that G09 is one-shot command and G61 is modal command (group 15).

G61 is continuously applied to cutting feed until G62 (automatic corner override mode), G63 (tapping mode), and G64 (cutting mode) are commanded.

Refer to 5.3.1 for details of Exact Stop. .

Name	G cod	Function
Exact Stop (One Shot)	G09	Effective only in the designated block Decelerated at the ending point of the designated block ⏩ Check in-position ⏩ Execute the next block
Exact Stop Mode	G61	Effective until G62, G63, G64 are commanded Decelerated at the ending point of the designated block ⏩ Check in-position ⏩ Execute the next block
Cutting Mode	G64	Effective until G61, G63, G64 are instructed No deceleration at the ending point of the designated block ⏩ Execute the next block
Tapping Mode	G63	Effective until G61, G62, G64 are instructed Feed override is fixed at 100% until other G code in the same group of the designated block appears
Automatic Corner Override Mode	G62	Effective until G61, G63, G64 are instructed It automatically reduces feed in case of machining inside of the corners. (Refer to Parameter→Program →Automatic override setting)

⏭ Exact Stop G09 / G61, Exact Stop Cancel G64 [Cutting Mode]

① The amount of deviation of the round that happens at the time of using G64 cutting mode is very small so it doesn't give big influence on precision of a product.

② If the function of Exact Stop is used, a phenomenon of minute stop happens at the node of curved surfaces, the condition of the machining surface is not good, the tool gets seriously worn out and it takes more time for machining.

③ Exact Stop has two commands of One Shot and Modal, which is the only case in G codes. In case of Positioning Mode [Rapid Positioning: G00, Single Direction Positioning: G60], it works same as Exact Stop even though Exact Stop is not executed.

④ G64 [Cutting Mode] is to cancel the command of Modal Exact Stop of G61.

▯ II Example of application

```
G54 G00 X0. Y0. Z0.
G61 G01 X100. F500.
Y100.
G64 X0.
Y0.
M30
```

5.3.3 Cutting Mode (G64)

G64

G64 Cutting Mode (Exact Stop Mode Cancel)

This function is applied general cutting feed.

Everything has inertia so it is difficult to momentarily stop a certain thing or change its direction to a desirable position with power. In general, in the cutting mode, in consideration of this matter, at the time of accelerating or decelerating feed of an axis, the current block is decelerated in advance and then, the next block is accelerated again.

```
N1 G00 X50. Y0.
N2 G01 Y50. F2000.
N3 X0.
```

Running the above feed program, Y is decelerated in advance before it reaches the coordinate of Y50, and X is accelerated. Therefore, at the corner, a phenomenon of rounding happens.

This function, as modal command (group 15), is continuously applied to cutting feed until G61 (Exact Command Mode), G62 (Automatic Corner Override Mode) and G63 (Tapping Mode) are commanded. At the time of power ON or RESET, the initialization of modal is set by this command.

Example of application

```
G54 G00 X0. Y0. Z0.
G64 G01 X100. F500.
Y100.
G61 X0.
Y0.
M30
```

5.3.4 Tapping Mode (G63)

G63

G63 Tapping Mode

This function carries out cutting feed same as G64 (cutting mode), but the difference is that cutting feedrate override is fixed at 100% and the function of Feed Hold is not applied (refer to 5.3.3).

This function, as modal command (group 15), is continuously applied to cutting feed until G61 (Exact Command Mode), G62 (Automatic Corner Override Mode) and G64 (Cutting Mode) are commanded.

Ⅱ Example of application

```
G54 G00 X0. Y0. Z0.
G63 G01 X100. F500.
Y100.
M30
```

5.3.5 Automatic Corner Override Mode (G62)

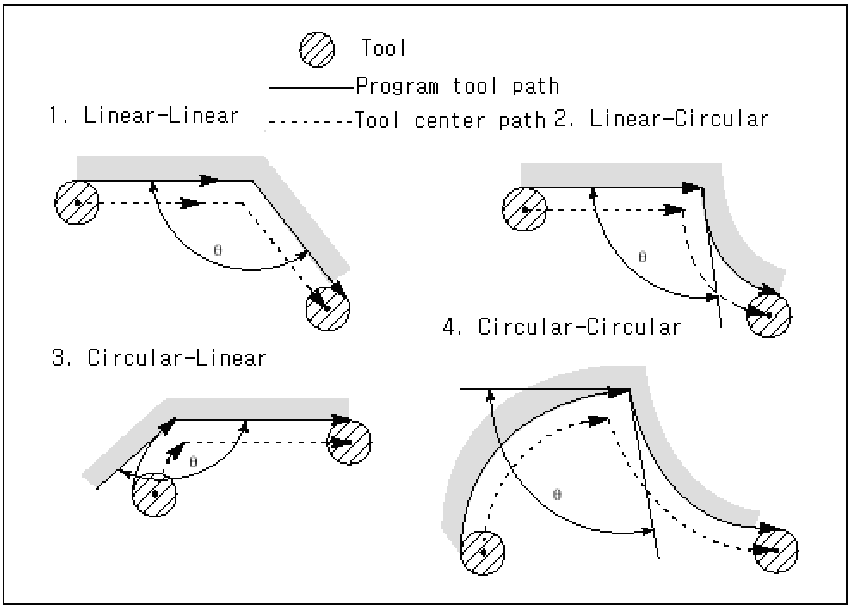
(1) General Automatic Corner Override

G62

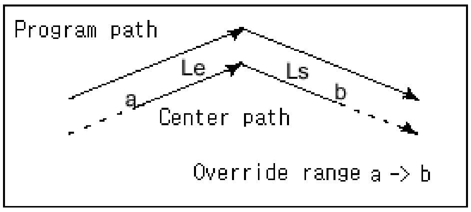
G62	Automatic Corner Override Mode

In case of cutting the internal corner during tool diameter compensation, the tool gets to have load a lot if the cutting is done at the nominated cutting feedrate so the machining surface can be rough. This function automatically decelerates the feedrate of a tool on this part to reduce load on the tool. It is not applied in the positioning feed section.

There are 4 types of corner that can happen during feed, according to the combinations of linear interpolation and circular interpolation. The zone for the corner angle Θ to which automatic corner override is applied is 2 ≤ θ ≤ Θp ≤ 178, and Θp is defined with parameter PI 130(#3130). In case the parameter value is 0 ~ 2, 178 is applied.



If the corner to which automatic corner override is applied happens, override is applied to all the starting section and the ending section of the corner. The starting section is set with PI 124(#3124) and the ending section with PI 125(#3125). If 0 is set at each section, override is not applied to the section.



The amount of override in the corner section that this function is applied is set with PI 126(#3126). The actual feedrate in this section is as follows: If the parameter value is set at 0, feed is not performed.

Feedrate in the deceleration section = Nominated feedrate x Amount of automatic corner override x Amount of feedrate override

At the time of starting and ending the tool diameter compensation, this function is not applied.

This function, as modal command (group 15), is continuously applied to cutting feed until G61 (Exact Command Mode), G63 (Tapping Mode) and G64 (Cutting Mode) are instructed.

Example of application

```
G54 G00 X0. Y0. Z0.
G41 D1
G62 G01 X100. F500.
G03 X130. Y30. R30.
G01 X0.
Y100.
M30
```

(2) Always Automatic Corner Override

This function is same as automatic corner override in terms of the execution method and the details of application. Only the difference is that it is applied to the whole cutting feed, not only to tool diameter compensation.

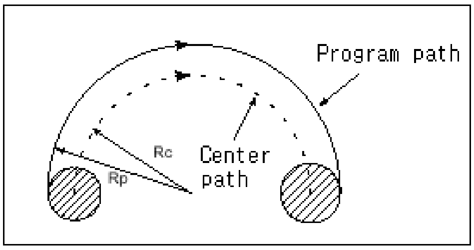
This function is applied by setting PI 131(#3131). If it is set at 0, it means automatic corner override and if it is set at 1, it means always automatic corner override.

(3) Automatic Corner Feedrate

This function is same as automatic corner override. In the section that override is applied to, actual feedrate is not changed by override and instead, it is instructed by parameter PI 153(#3153). This function works only if the parameter value is 0 or higher. In case it is 0, automatic corner override works. If the feedrate set with the parameter is larger than the actually commanded rate, it feeds at the commanded rate.

5.3.6 Internal Circular Cutting Feedrate Change)

Just like the internal corner section to which automatic corner override is applied during tool diameter compensation, the internal circle falls in the same situation. In this internal circular section, deceleration is automatically applied without a particular command. The feedrate in the circular section that is decelerated automatically is as follows:



Feedrate in the deceleration section = Nominated feedrate x Rc/Rp x Amount of automatic corner override x Amount of feedrate override

If Rc/Rp is close to 0, the speed is almost in the status of stop so the minimum rate is set on the parameter PI 132(#3132). If the feedrate is smaller than this minimum value, the value on the parameter is applied. If the parameter value is 0, Rc/Rp is unconditionally applied.

Under the situation that automatic corner override function is applied, if there is a circular arc on the corner, this function and the automatic corner override function are applied at the same time.

5.4 Dwell (G04)

G04	Dwell
X _	sec Unit Time / No. of revolutions
P _	mili-sec Unit Time (1/1000 sec) / No. of revolutions

This function is to stop the axis feed of the program during the given time. Even though this function is commanded, revolution of the spindle and supply of cutting oil are maintained.

If the cutting feedrate is feed per revolution (G95) and parameter PI 120(#3120) is 1, the instructed X_ or P _ becomes the number of revolutions. That is, after the spindle revolves as many as the commanded number of revolutions, the next block is progressed. The instructed X_ or P _ after that becomes stop time.

Ⅱ Example of application

```
O Dwell
G54 G00 X0. Y0. Z0.
G94 G01 X10 F500.
G04 X3 (Time dwell)
X20.
G95 M03 S60. F5.
Y10.
#100 = #3120
#3120 = 1
G04 P1 (Revolution number dwell)
#3120 = #100

M30
```

5.5 Automatic Acceleration / Deceleration

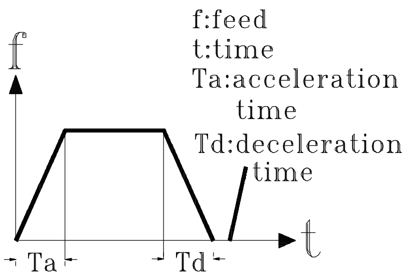
If we try to momentarily move a thing (which is in the status of stop) or to momentarily stop a moving thing, the thing gets to have a lot of load and inertia so it is not easy to stop it on the exact position.

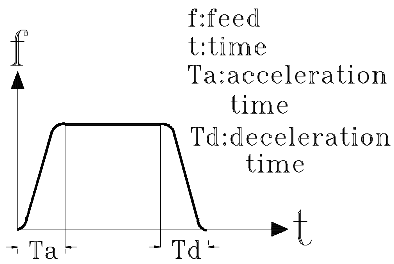
In case of moving a table to the location for machining at rapid feedrate, it is difficult to stop it at exact point.

In order to solve this problem, it is necessary to accelerate for moving and to decelerate for stop in front of the stop point. This function of automatic acceleration/deceleration makes positioning of high accuracy possible. It smoothly executes start and stop of axis feed during rapid traverse and cutting feed.

(1) Acceleration/deceleration against rapid feedrate

- Positioning G00
- Rapid Traverse
- Manual Feed JOG
- Manual Handle Feed
- Rapid feed acceleration/deceleration time are set with parameter PM 561~592(#20561 ~ 20592).





(2) Acceleration/deceleration against cutting feedrate

- Cutting Feed G01, G02, G03
- Automatic acceleration/deceleration type during cutting feed is set with parameter PM 598(#20598).
- Acceleration/deceleration time during cutting feed are

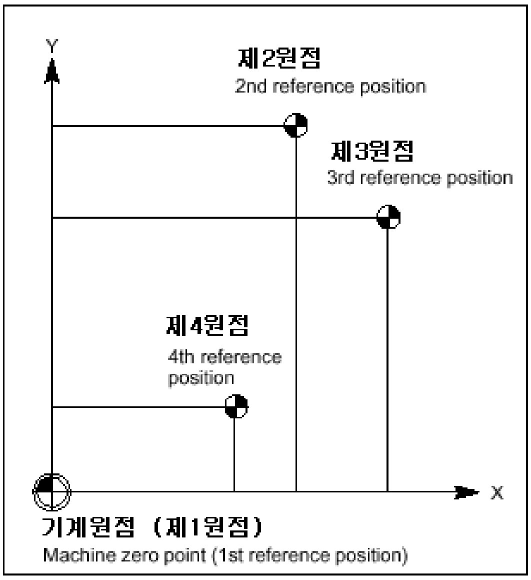
set with parameter PM 599(#20599).

If this value is set high, the machine is moved smoothly, but machining error can be increased.

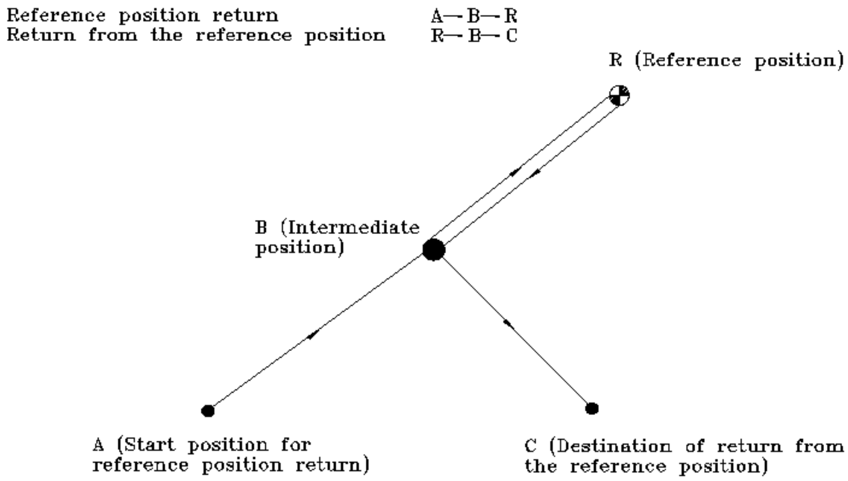
Note:
It is better not to change automatic acceleration/deceleration time if it is not specially required.

6 Reference Position

Reference position means the fixed standard position in machines. There are several kinds of reference position as shown on the below:



As the types of feed related with reference position, there are Reference Position Return and Return from the Reference Position.



What is reference position (the first reference position)?
: This is an arbitrary point on a machine and is a point that a tool can be moved to easily. With simple operation, it is possible to return to this point and a machine manufacturer sets this reference position.

It is the position that becomes a standard for operation of a program and machine. Therefore, machine coordinate system and workpiece coordinate system are automatically made only after making power on and performing reference position return.

Only the staff in charge of the manufacturer is allowed to change the position.

6.1 **Reference Position Return** (G28)
6.2 **2nd, 3rd, 4th Reference Position Return** (G30)
6.3 **Return from Reference Position** (G29)
6.4 **Reference Position Return Check** (G27)

6.1 Reference Position Return (G28)

[G90 / G91] G28 X _ Y _ Z _

G90 / G91 Absolute / Incremental Command

G28 Reference Position Return command

X _ Y _ Z _ Positioning the Central Point against the Axis for Reference Position Return

This function is to automatically return the instructed axis to the first reference position at rapid feedrate through the central point of work-piece coordinate system, X_ Y_ Z_ .

Note:

① The axis which is not commanded doesn't return to the reference position.

② For safety, compensation of diameter and length is cancelled in advance.

③ The nominated central point is used as the central point under G29(return from reference position).

▌Ⅱ Example of application 2

O Reference position return

G91 G28 X0. Y0. Z0. (Use the current point as the central point and return to reference position)

G90 G54 X10. Y20. Z30. (Feed to X10. Y20. Z30 on the G54 work-piece coordinate system)

▌Ⅱ Example of application 2

O Reference position return

G91 G28 Z0. (Use the current point as the central point and return only Z to the reference position)

G28 X0. Y0. (Use the current point as the central point and return X and Y to the reference position)

```
G90 G54 X10. Y20. Z30.
```

6.2 2nd, 3rd, 4th Reference Position Return (G30)

[G90 / G91] G30 {P2 / P3 / P4} X _ Y _ Z _

G90 / G91	Absolute /Incremental Command
G30	Command of 2nd, 3rd, 4th Reference Position Return
P2 / P3 / P4	Select 2nd, 3rd, or 4th Reference Position Type
X _ Y _ Z _	Location of the central point against the axis for reference position return

This function is to automatically return the commanded axis to the 2nd, 3rd and 4th reference position defined by the parameter at rapid feedrate through the central point of the work-piece coordinate system, X_ Y_ Z_ . The second reference position is commanded by P2 and P2 can be omitted. The third reference position is commanded by P3 and the fourth, by P4.

< Relevant parameter>

No.	Description
PM 2097-2128(#22097-22128)	Locate each axis to the 2nd reference position (machine coordinate system)
PM 2129-2160(#22129-22160)	Locate each axis to the 3rd reference position (machine coordinate system)
PM 2161-2192(#22161-22192)	Locate each axis to the 4th reference position (machine coordinate system)

Note:

① The axis that is not commanded doesn't return to the reference position.

② For safety, compensation of diameter and length is cancelled in advance.

③ The nominated central point is used as the central point under G29(return from reference position).

Ⅱ Example of application

O 2nd, 3rd, 4th reference position return

G90 G54 X0. Y0. Z0.	(Set G54 coordinate system. Move to X0. Y0. Z0.)
G30 P2 X30. Y50. Z10.	(Having X30. Y50. Z10 on the G54 coordinate system as the central point and returning to the second reference position)
G30 P3 X0.	(Having X0. on the G54 coordinate system as the central point and returning only X the to the third reference position)
G30 P4 Y0.	(Having Y0. on the G54 coordinate system as the central point and returning only Y to the 4th reference position)
M30	

6.3 Return from Reference Position (G29)

[G90 / G91] G29 X _ Y _ Z _

G90 / G91	Absolute / Incremental Command
G29	Command of Return from Reference Position
X _ Y _ Z _	Forwarding Location

This is usually used after reference position return or $2^{nd}/3^{rd}/4^{th}$ reference position return. Through the central point that has been nominated previously, it rapidly feeds to the forwarding location on the work-piece coordinate system. In case there is no central point because G28 (Reference Position Return) or G30 ($2^{nd}/3^{rd}/4^{th}$ Reference Position Return) has not been commanded before commanding this function, the first reference position becomes the central point.

Note:

① For safety, compensation of diameter and length is cancelled in advance.

② Feed to the central point is carried out for all the axes that are instructed previously, and feed to the position by G29 command is carried out for the relevant instructed axis.

Ⅱ Example of application

O Return from Reference Position

G90 G54 X0. Y0. Z0.	(Set G54 coordinate system. Move to X0. Y0. Z0.)
G28 X30. Y50. Z10.	(Return to the first reference position through X30.Y50.Z10 on the G54 coordinate system)
G29 X0. Y0. Z0.	(Rapidly feed to the forwarding location that is instructed (X0.Y0.Z0. on the G54 coordinate) through the central point that has been previously instructed (X30.Y50.Z10 on the G54 coordinate)
M30	

6.4 Reference Position Return Check (G27)

[G90 / G91] G27 X _ Y _ Z _

G90 / G91	Absolute / Incremental Command
G27	Command of Reference Position Return Check
X _ Y _ Z _	Forwarding Location

If you rapidly feed to X_ Y_ Z_ instructed on the workpiece coordinate system and reach the reference position, the lamp of reference position return completed becomes on. If it is not the reference position, an alarm, F_82209 (which means "It is not the reference position.") happens.

▯ ⏭

① For safety, compensation of diameter and length is cancelled in advance.

② For the location that you reach with G27, it temporarily cancels offset even during compensation and rapidly feeds to the first reference position.

③ In case G27 is commanded under the status of Machine Lock On, it is impossible to check whether it reaches the reference position or not.

④ In case of using inch coordinate system, the input unit is smaller than the minimum motion inch eventhough it is away from the reference position on the mm coordinate system by 1μ in program so the lamp of reference position return may light on.

▯ II Example of application

O Reference Position Return Check (In case X offset is ? 100 and Y offset is ? 100)
G90 G54 X0. Y0. Z0.
G27 X100. Y100. (The lamp of the first reference position return completed
lights on)

G27 X10.　　　　　　(Alarm, F_82209 (which means, “It is not the first reference position.”) happens

M30

7 Coordinate System

In order to move a tool to the location that you want, you should set its position with CNC. The location of a tool is expressed as the coordinate value on the coordinate system and then, set with CNC. You can designate the coordinate values, using the following three coordinate systems.

(1) Machine Coordinate System

(2) Work-piece Coordinate System

(3) Local Coordinate System

7.1 Machine Coordinate System

7.1.1 Machine Coordinate System Selection (G53)

G90 G53 X _ Y _ Z _

G90 Absolute Command
G53 Machine Coordinate System Selection
X_ Y_ Z_ Forwarding Location

In order to use machine coordinate system, G53 is used. G53 is one shot G code so it is effective only in the instructed block. Once this function is run, tool diameter compensation, tool length offset and tool offset are temporarily cancelled. And incremental coordinate command of G91 is temporarily ignored.

The tool moves at rapid traverse if the machine coordinate system instructs a position. This function is effective only after performing manual reference position return over one time by use of automatic reference position return (using G28) or operating key after power on.

⏭

① One-shot command that can be used in the form of G90 G53 X_ Y_ Z_ on the basis of the first reference position, this function doesn ' t apply tool diameter compensation, tool length offset, and tool offset.

② Setting for machine coordinate is done by running manual or automatic reference position return after power on.

③ This is the coordinate system fixed on the 2nd, 3rd, and 4th reference position, and it becomes a reference for setting G22/G23 Stored Stroke Limit, Over Travel, and the 2nd/3rd/4th reference position. The machine coordinate value on the reference position is X0, Y0, Z0.

④ It is used to know the distance between the current location of a tool and the first reference position.

II Example of tool position offset

Explanation

N01 G40 G80 ;

```
N02 G53 G90 X-140 Y-120 Z0 ; (Move to X-140 Y-120 Z0 on the machine
                               coordinate system)
N03 G92 X0 Y0 Z150 ;           (Reset by changing the work-piece
    coordinate
                               system)
N04 G30 G91 Z0 ;
N05 G54 G00 G90 X0 Y0 ;
       ⬇
N60 M30 ;
```

7.2 Work-piece Coordinate System

7.2.1 Work-piece Coordinate System Setting (G92)

G90 G92 X _ Y _ Z _

G90	Absolute Command
G92	Work-piece Coordinate System Setting
X_ Y_ Z_	The current location of the work-piece coordinate system to set

You can simply and accurately make a program by setting the reference point of drawing or work-piece and instructing the machining location from the reference point. However, CNC can't recognize where the reference point of a work-piece is so you need to inform CNC of it. In case of moving a tool to a certain position (first reference position, second reference position) with absolute command, there are many cases of designating the coordinate value from the reference position of a program. This is a G code that sets the reference position of a program at this time.

In order to obtain the reference point, you should perform reference position return or second reference position return and then, set the work-piece coordinate system by use of G92 code.

⏭

① In case of using G92, [Work-piece Coordinate System Setting], reference position return should be performed first and then, the coordinate system should be set. Therefore, unnecessary axis feed is executed.

② The coordinate system having the reference position of program as the reference point is work-piece coordinate system. Once the coordinate system is set, the absolute command that is given after that becomes a location on this work-piece coordinate system.

⏸ G92 example 1 **Explanation**

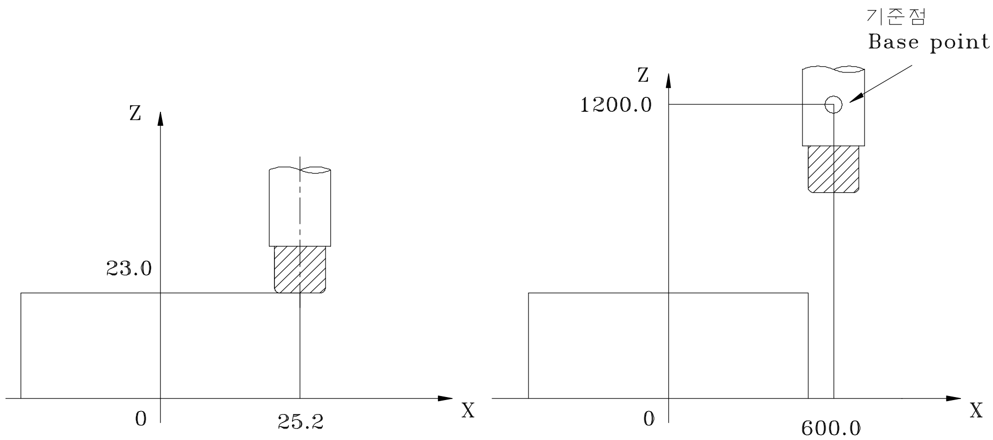
N01 G40 G80 ;

N02 G28 G91 X0 Y0 Z0 ; (Return to the first reference position (way-point is the current location value) [G91 mode])

N03 G92 G90 X140 Y120 Z150 ; (This is setting for work-piece coordinate system in which the current coordinate value of machine position X0 Y0 Z0 is changed to X140 Y120 Z150. The distance from the first reference position of machine to the reference position of work-piece coordinate system is programmed by absolute coordinate system command [G90].)

N04 G30 G91 Z0 ;
N05 G00 G90 X0 Y0 ;
⬇
N60 M30 ;

Ⅱ Example 2



G92 X25.2 Z23 ; : The tip of a tool is the starting point.
G92 X600 Z1200 ; : The reference point in a toolhole is the starting point.

7.2.2 Work-piece Coordinate System Selection 1~6 (G54~G59)

```
G54  X _ Y _ Z _
G55  X _ Y _ Z _
G56  X _ Y _ Z _
G57  X _ Y _ Z _
G58  X _ Y _ Z _
G59  X _ Y _ Z _
```

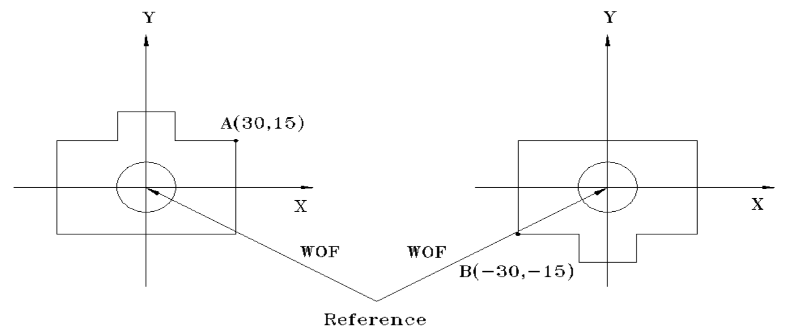
G54 Work-piece Coordinate System 1 Selection
G55 Work-piece Coordinate System 2 Selection
G56 Work-piece Coordinate System 3 Selection
G57 Work-piece Coordinate System 4 Selection
G58 Work-piece Coordinate System 5 Selection
G59 Work-piece Coordinate System 6 Selection
X_ Y_ Z_ Location of work-piece coordinate system

① On this coordinate system, you can conveniently make out a program based on a drawing and apply the made NC program as it is to decide any particular position of a work piece for machining as the reference position. You need to perform reference position return after power on in order to correctly use this coordinate system.

② In case of using G54~G59, you don't need to set a coordinate system with G92.

③ If the value of "work-piece coordinate system (0/1)" on the left top of the coordinate system screen (Initial screen->Set->Coordinate system) is set at "0: maintain", the previously used coordinate becomes effective when power is ON. The coordinate system previously set with G54~G59, G92 or G52 is maintained as it is. If the value of "work-piece coordinate system (0/1)" is set at "1: Cancel", the previously used coordinate system gets cleared all when it is RESET or power is ON.

④ With the right-hand coordinate system applied, from the absolute coordinate reference position, right of X, back of Y and top of Z are "+", and the opposites are "-



“ .

Example 1 of work-piece coordinate system

Explanation WOF : Work-piece reference position offset

G90 G54 G00 X30 Y15 ; (A point that has used G54 work-piece coordinate system)

G55 X-30 Y-15 ; (B point that has used G55 work-piece coordinate system)

Example 2 of work-piece coordinate system

Explanation

G40 G80 ;

G28 G91 X0 Y0 Z0 ; (Return to the first reference position where point is the current location value [G91 mode])

G54 G00 G90 X0 Y0 Z0 ; (Rapidly feed to the reference position, using G54 work-piece coordinate system. That is, rapid feed to the reference position of G54 workpiece coordinate system)

⬇

M30 ;

7.2.3 (Work Zero Point Offset Setting (G54~G59)

Offset setting is to set the distance between the first reference position of a machine to the reference position of a work-piece.

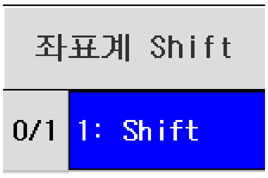
X, Y, Z offset setting for work-piece coordinate system

① Press Mode Select key for Zero Return ⇒ Press Axis Select key for Z ⇒ Press + Axis Feed key for reference position return of Z. And then, perform the reference position return for X and Y too in the same way.
② Fix 10 end mills on the milling chuck and install the milling chuck on the spindle.
③ Main menu ⇒ F2 setting ⇒ F2 offset setting. Move to offset address that you want and input the values of diameter and length compensation of a tool that you use.
④ Main menu ⇒ F2 setting ⇒ F1 coordinate system. Read the screen of coordinate system setting by selecting a coordinate system in order.
⑤ Press Mode Select key JOG ⇒ Press Spindle Override key for 100% ⇒ Press Spindle Start key to revolve the spindle.
⑥ Using the manual mode of Mode Select key (JOG, X1, X10, X100), touch the end mill on the top/the front/the side of a work piece with a tool, select one of work-piece coordinate system G54~G59, and then, input the location value of the machine indicated on the screen.
⑦ In case of inputting the location value of machine through the input window, input the coordinate values for Z as indicated, and for X and Y, input in consideration of the diameter of a tool.
⑧ If you input "I ±5" on the input window, the number (radius of tool) input on the current machine location is increased/decreased to be automatically input. If you input "I", the current machine location is input.
⑨ If setting is completed, each axis is moved in order not to be interfered by a work piece.
⑩ The end of the tool's central point is set to be same as the reference position of the work-piece coordinate system.

7.2.4 Work-piece Coordinate Shift

In case the work-piece coordinate system that is assumed at the time of making a program is different from the coordinate system that is actually set, it is necessary to shift the set coordinate system. Work-piece coordinate shift is set with the parameter, "Coordinate system shift" on the coordinate system screen.

< Initial screen? F2 setting screen? F1 coordinate system screen >



For example, in case the height of a work-piece is different, you can perform machining by use of the function of Shift available on the coordinate system screen, maintaining the tool offset value, after setting this difference in the Shift item for Z.

> **Note:** Suggestions related with setting the amount of shift
> ① Work-piece coordinate shift becomes immediately effective as soon as the amount of shift is set.
> ② If the work-piece coordinate changes under the circumstance that the amount of shift is set, the amount of shift is applied. The work-piece coordinate is changed by G92 coordinate setting and G54~G59 coordinate systems.
> ③ Work-piece coordinate shift is effective when "Coordinate Shift 0/1" parameter is 1. Setting for this parameter is done on the screen of Parameter.
> (Initial screen -> F2 setting -> F1 coordinate system)

7.3 Local Coordinate System (G52)

G52 G90 X _ Y _ Z _
G52 X0 Y0 X0

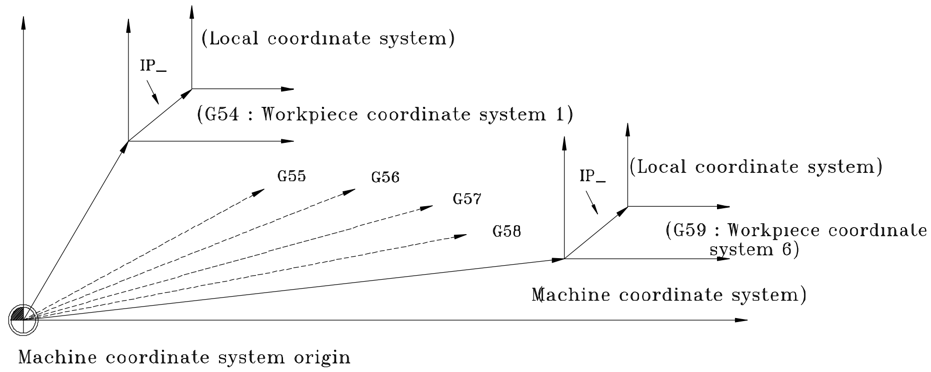
G90	Absolute Command
G52	Local Coordinate System Setting
G52	Local Coordinate System Cancel
X_ Y_ Z_	Reference position of the local coordinate system to set on the current work-piece coordinate system

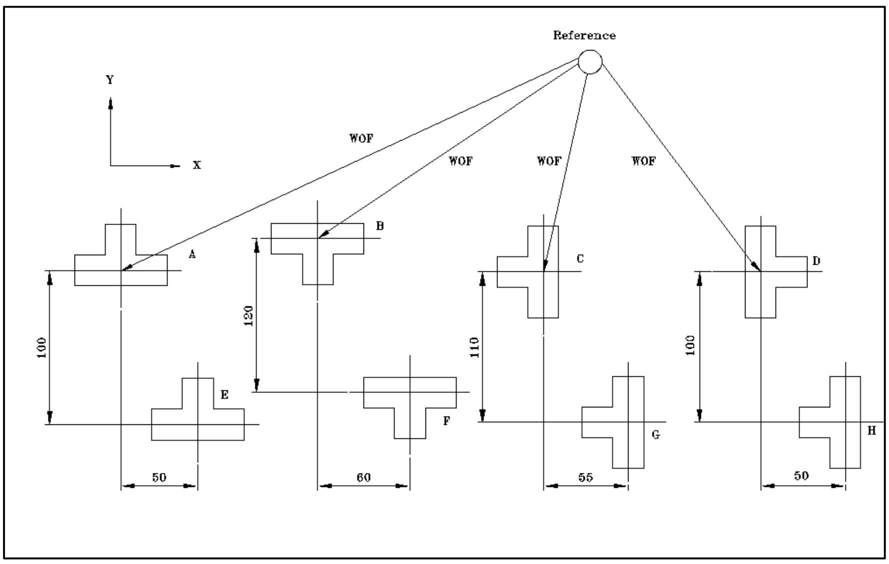
The command of local coordinate cancel is given just like G52 X0 Y0 Z0.

The local coordinate system is that based on the set work-piece coordinate system, an arbitrary point is set as the reference point to use at the time of making a program with work-piece coordinate system.

Local coordinate system can make a new coordinate system based on all the work-piece coordinate system G54~G59, and the reference position of each local coordinate system is X_ Y_ Z_ that is designated by each work-piece coordinate system.

The local coordinate system becomes cleared to be 0 once a new work-piece coordinate system is set.





Ⅱ Example of local coordinate system

⏭

G90 G54 ; (Work-piece A : Work-piece coordinate system 1)

G52 X50 Y-100 ; (Work-piece E: Local coordinate system in Work-piece coordinate system 1)

G55 ; (Work-piece B : Work-piece coordinate system 2)

G52 X60 Y-120 ; (Work-piece F : Local coordinate system in Work-piece

coordinate system 2)

G56 ; (Work-piece C : Work-piece coordinate system 3)

G52 X55 Y-110 ; (Work-piece G : Local coordinate system in Work-piece

coordinate system 3)

G57 ; (Work-piece D : Work-piece coordinate system 4)

G52 X50 Y-100 ; (Work-piece H : Local coordinate system in Work-piece

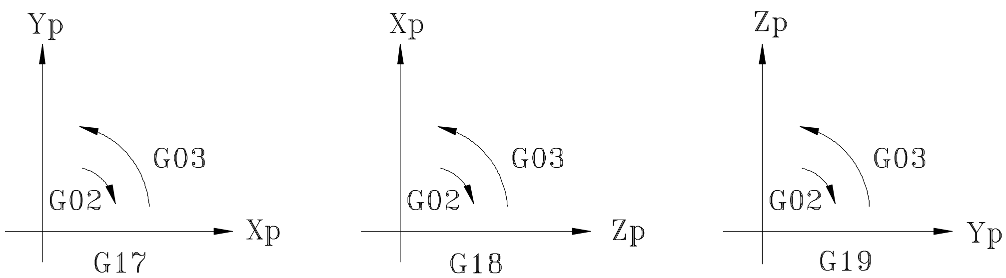
coordinate system 4)

7.4 Plane Selection (G17, G18, G19)

```
G17 [G02 / G03]   X _ Y _ [I _ J _ / R _ ] F _
G18 [G02 / G03]   X _ Z _ [I _ K _ / R _ ] F _
G19 [G02 / G03]   Y _ Z _ [J _ K _ / R _ ] F _
```

G17	X-Y plane selection
G18	Z-X plane selection
G19	Y-Z plane selection
G02 / G03	Circular interpolation CW / Circular interpolation CCW
X _ Y _ Z _	Forwarding location
I _ J _ K _	Central location of a circular arc (indicated in vector)
R _	Radius
F _	Feedrate

Circular interpolation (G02/G03) or tool diameter compensation is possible only in one plane so plane should be selected by use of G code



The plane that is set as default at the time of being initialized is set with parameter PI 145 (#3145).

8 Coordinate Value and Dimension

As the ways to command coordinates, there are Absolute Command, Incremental Command, and Polar Coordinates Command. In this chapter, details of each command are examined and conversion of inch/metric will be explained.

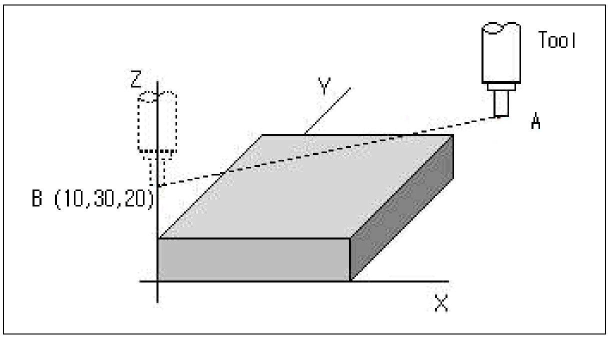
8.1 Absolute Command (G90)

G90 G00 X _ Y _ Z _

G90	Absolute Command
G00	Rapid Traverse (Positioning)
X _ Y _ Z _	Forwarding Location

Absolute Command is to command the location of the terminal with the location of the absolute coordinate system (location based on the reference position of work-piece coordinate system).

Ⅱ Example of Absolute Command (G90)



G90 X10. Y30. Z20. (Command to move from A point to B)

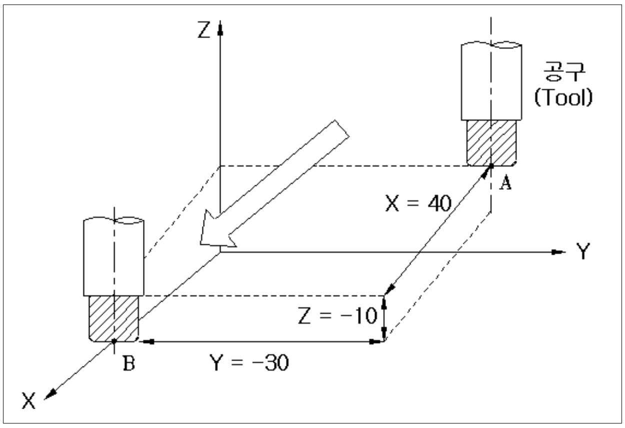
8.2 Incremental Command (G91)

G91 G00 X _ Y _ Z _

G91	Incremental Command
G00	Rapid Traverse (Positioning)
X _ Y _ Z _	Forwarding Location

Incremental Command is to command the distance to the next location from the current location (the amount of movement). It indicates the coordinate values of the current location in the amount of increment based on the previous location. The signal is decided, depending on the direction of the terminal based on the point of time.

Example 1 of Incremental Command (G91)



G91 X40. Y-30. Z-10.(Command to move from A point to B)

8.3 Polar Coordinates Command (G15, G16)

```
G15
G16   X _ Y _
```

G15	Polar Coordinates Command Cancel
G16	Polar Coordinates Command
X _	Circular Radius of Polar Coordinates Command
Y _	Angle Command for Polar Coordinates Location

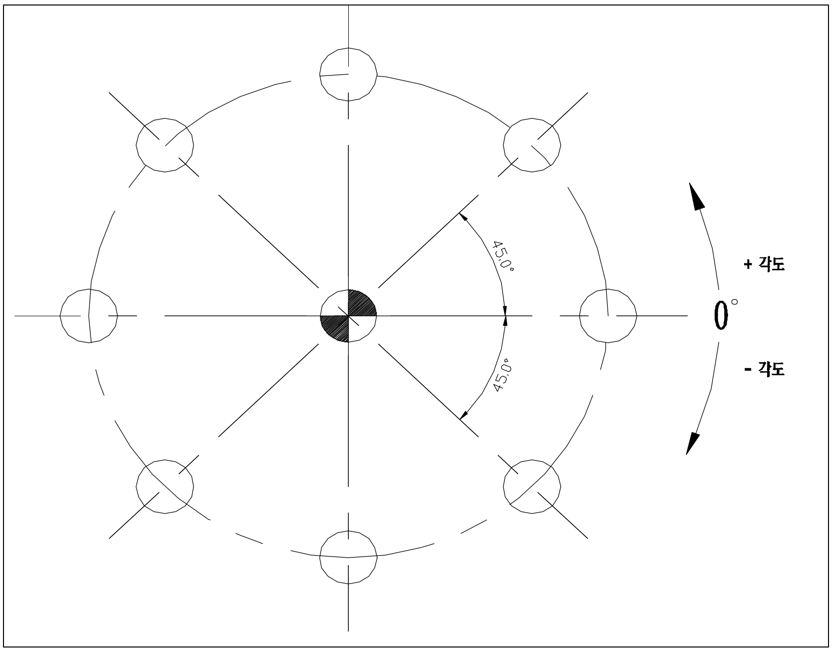
The standard for angle is that the direction of 3 o ’ clock is 0° .

Counterclockwise (CCW) : “ + “ , Clockwise (CW) : “ ? “

Dimension of the shapes on a drawing is indicated as linear values based on the reference point, but as the location values on a cylinder, only the radius and angle of a circular arc are sometimes given so a worker should manually calculate the coordinate values by use of the trigonometric function to make a program.

The polar coordinates command is a function to solve this problem. That is, it automatically calculates the coordinates on a cylinder by commanding the radius and angle of a circular arc. It is to calculate the coordinate values so it is used, together with the canned cycles.

< Circular radius and angle of polar coordinate command >



⏭

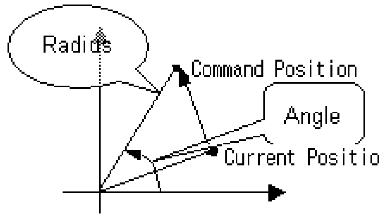
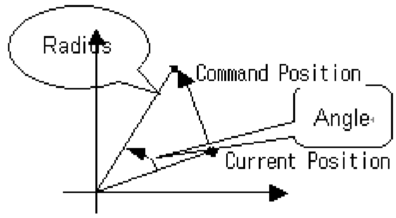
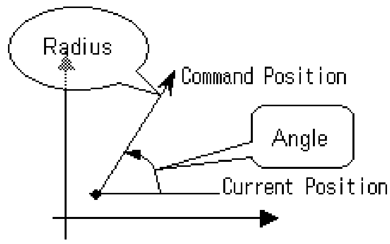
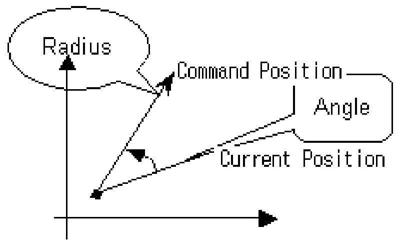
① Polar Coordinates Command and Plane Function Selection

G_Code	Radius	Angle
G17	X	Y
G18	Z	X
G19	Y	Z

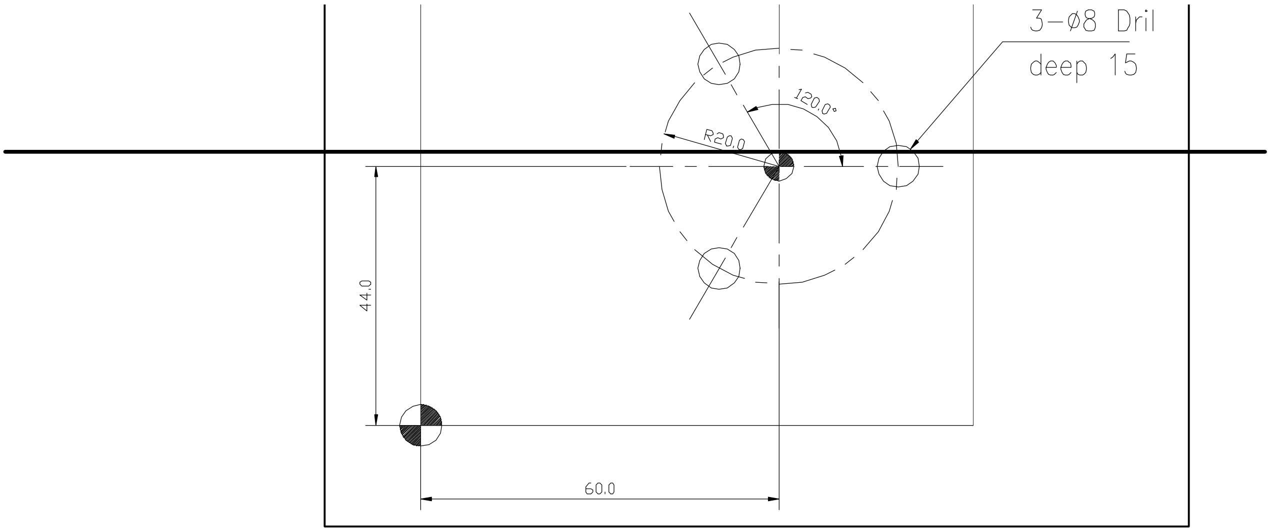
" + directional angle command " is counterclockwise based on X.

② In case Circular Radius command is Absolute Command G90:
: The reference position of work-piece coordinate system or of local coordinate is the center of the polar coordinate. If necessary, it gives polar coordinate command after moving the reference position of work-piece coordinate system or local coordinate system.

③ In case Circular Radius command is Incremental Command G91:
: The current location becomes the reference position of the polar coordinate command. For normal machining, it gives a polar coordinate command after moving the tool to the center of polar coordinate.

In case the command for radius is Absolute Command (G90	In case the command for angle is Absolute Command (G90)	
	In case the command for angle is Incremental Command (G91)	
In case the command for radius is Incremental Command (G91)	In case the command for angle is Absolute Command (G90)	
	In case the command for angle is Incremental Command (G91)	

Example of Polar Coordinates Command



⏭ Circular Arc

⏭ In case the command for circular radius is absolute (G90):

N1 G40 G80 ; (Tool diameter compensation cancel, Canned cycle cancel)

N2 G28 G91 X0 Y0 Z0 ; (Return to reference position of machine)

N3 G92 G90 X150 Y150 Z150 ; (Work-piece coordinate system setting)

⬇

N50 G52 X60 Y44 ; (Local coordinate system setting)

N51 G16 G17 ; (Polar coordinate command on X-Y plane)

N52 G81 G90 G99 X20 Y0 Z-15 R3 F100 ; (Drill machining at ①)

(X20 : Polar coordinate radius, Y0 : Angle)

N53 Y120 ; (Drill machining at ②)

N54 Y240 ; (Drill machining at ③)

N55 G15 G00 Z50 M09 ; (Polar coordinate command cancel)

⏭ In case the command for circular radius is incremental (G91):

⬇

N50 G90 X60 Y44 Z10 ; (Move the tool to the center of polar coordinate)

N51 G16 G17 ; (Polar coordinate command on X-Y plane)

N52 G81 G91 G99 X20 Y0 Z-22 R-7 F100 ; (Drill machining at ①)

N53 Y120 ; (Drill machining at ②)

N54 Y120 (Drill machining at ③)

N55 G15 G00 Z50 M09 ; (Polar coordinate command cancel)

⏭ [Example. Bolt Hole cycle : Start angle of 30 degree, Radian: 120 degrees, Holes: 3]

In case both radius and angle are absolute

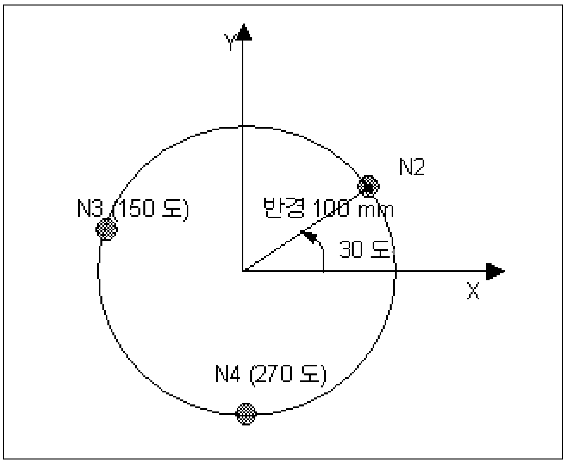
N1 G17 G90 G16

N2 G81 X100. Y30. Z-20. R-5. F200.

N3 Y150.

N4 Y270.

N5 G15 G80



⏭ In case radius is absolute and angle is incremental

N1 G17 G16

N2 G81 G90 X100. Y30. Z-20. R-5. F200. (Radius: 100mm, Revolution angle: 30 degree point)

N3 G91 Y120.

N4 Y120.

N5 G15 G80

Note:

The following commands are not influenced by the commands for radius and angle even though there is a command for axis.

- Dwell (G04 X_)
- Change in offset value by use of G10 (G10)
- Setting values of local coordinate system (G52)
- Setting values of work-piece coordinate system (G92)
- Setting values of machine coordinate system (G53)
- Setting values of the sections where feed is forbidden (G22 X_Y_Z_I_J_K_)
- Center for coordinate system revolution (G68 X_Y_)
- Center for scaling (G51 X_Y_)
- In the polar coordinate command, circular interpolation and helical interpolation (G02/G03) commands are possible.

8.4 Inch Unit / Metric Unit (G20, G21)

G20 G21

G20	Mode of Inch unit input
G21	Mode of mm unit input

This G code is to command the input coordinate values in mm or inch. When the whole dimensions of a drawing are indicated in inch or mm (metric) coordinates, you can simply make a program by converting the unit of machine moving into inch or metric. It is a modal command that controls the coordinate values in a lower block including the block in which G20/G21 are input.

< Minimum setting unit >

G Code	Unit	Minium setting
G20	Inch	0.0001 inch
G21	Metric	0.001 mm

Once this G code, modal command, is executed, the input unit (coordinate and feedrate) of a part program after that follows the designated input unit, and even though the input unit system of G20/G21 is changed, the tool offset values in CNC and the machine location still follow the mm unit system. Consequently, there is no other change than input unit of NC program.

The input unit system that is set at the time of power ON is set with the parameter PI 147 (#3147).

Conversion of the unit system should be commanded for an independent block before setting a coordinate system at the head of the program. It is not good to convert inch or metric in the same program so in case a part of a drawing is indicated in inch or metric, it should be converted.

Note:

① The unit conversion by G20/G21 is conversion of the input unit system. The unit system indicated on the screen is not influenced by G20/G21.

② The unit system indicated on the screen is set with parameter PA0.
(Metric/ Inch setting)

③ The unit system – inch/metric - for axis data indicated on the screen (current location, location of machine) is designated by parameter PA 374~405. The axis for which the parameter is set at 1 is controlled by the unit system set by PA 0. The axis for which the parameter is set at 0 gets to use metric regardless of the unit system set by PA0.

9 Spindle Function

9.1 Constant Surface Speed Control (G96)

G96 S _

G96	Constant Surface Speed Control
S_	Cutting Speed [m / min]
	S is not rpm command, but the value of cutting speed.
	Surface speed is 0 before M03 or M04 is given.

If the diameter of a material is changed in the process of cutting, the cutting speed becomes different. If the cutting speed becomes different, the machining surface (roughness) becomes different and the cutting resistance on bite becomes different too, which causes the tool seriously worn out. Constant Surface Speed Control (G96) is to solve this problem.

In the mode of Constant Surface Speed Control (G96), CNC by itself calculates the number of revolutions in order to maintain the fixed cutting speed regardless of changes in the diameter of a material, and according to the diameter, the number of revolutions changes. If you command the relative surface speed of a tool and a work-piece, using address S, it calculates the number of revolutions in order for the designated surface speed to be maintained according to changes in location of a tool, and sends corresponding voltage to the control part of the spindle in order for the spindle to revolve for right speed.

Constant Surface Speed Control makes the cutting speed maintained, the roughness of the surface even, and the lifetime of a tool lengthened.

Constant Surface Speed Control is cancelled by G97 command.

In general, it is used a lot for CNC lathe. In the field of milling, it, with C axis added, is used for boring or machining the sections having right angle.

Relationship

❶ Cutting Velocity V [m/min]

$$V = \frac{p\ DN}{1000}$$

(D : Diameter of tool [mm], N : Revolution speed of spindle [rpm])

❷ Speed of revolution [Rough Cutting : N rpm]

$$N = \frac{1000\quad V}{p\ D}$$

(D : Diameter of tool [mm], V : Cutting Velocity [m/min])

9.2 Constant Surface Speed Control Cancel (G97)

G97 S _

S _ Designate the revolution speed of spindle [rpm]

It is a modal G code that is automatically set once the initial power is ON.

It is to directly command the number of spindle ' s revolutions in rpm. Give 5-digit number to address S for it.

With G97, constant surface speed control is cancelled and the spindle revolves at the designated speed regardless of changes in the location of the tool.

Example

G97 S800 (Cancel constant surface speed control and directly command the number of revolutions at 800.)

9.3 Clamp at maximum Spindle Speed (G92)

G92 S _

G92 Clamp at maximum spindle speed [rpm]

This function is to directly designate the maximum revolution speed of the spindle in rpm. G96(Constant Surface Speed Control) restricts the designated revolution speed based on the set maximum speed of the spindle. The revolution speed of the spindle that is larger than the maximum is controlled to revolve at the speed set by G92. (G97 mode has no relation with this)

① 주축의 최고속도 지정은 G92 기능과 같은 블록에 지령해야 합니다.
② 주속 일정 제어시 주축 최고속도와 최저속도는 파라미터 PM 3360(#23360) 과 PM 3361(#23361)으로도 설정 할 수 있습니다.
G92 로 주축 최고 속도를 지령하면 그 값은 자동으로 해당 파라미터에 입력 됩니다.

Example

G92 S500 (Designate the maximum revolution speed of the spindle at 500 rpm at the time of Constant Surface Speed Control G96 S2000 (Constant Surface Speed Control at 2000 mm/min)
M03 (The revolution speed of the spindle is restricted to 500 rpm)

10 Tool Function

Replacement of a tool is commanded in the form of T□□△△(□□: Too number, △△: Offset number). For example, the command of T0305 means that “3” is tool number and “5” is offset number.

10.1 Tool Selection

T _

T _ Tool selection

It is possible to select a tool that you want, using two-digit numbers next to T. You should command T code once in a block. If T code is commanded over two times in a block, the last T code is effective.

In case the feed command and the tool command are given in the same block, feed of axis and selection of a tool are made at the same time or a tool is selected after feed of an axis, which depends on the specifications set by the machine manufacturer (MTB, Machine Tool Builder).

10.1.1 Macro Program Paging by T code (tool replacement)

In case of replacing a tool by use of T code in ATC device, a particular type of paging macro program is used. This macro program is that after the action of tool replacement is programmed into 9000.NC~9009.NC, a particular program corresponding to them is paged. In order to use this method, parameter PI 106 (#3106) should be set at 1.

The number of macro program (9000.nc ? 9009.nc) that is paged by T code command is set with PI 105(#3105). If the parameter value is 0, the macro program of no. 9000 is paged as default.

PI 105(#3105)	No. of macro program that is paged by T code command
PI 106(#3106)	Availability or non-availability of macro program at the time of T code command given

10.1.2 Macro Program Paging by M Code (tool replacement)

This is for replacement of tool by M code after designating a tool by T code. In order to use this method, parameter PI 106(#3106) should be set at 0 in order for macro program not to be paged by T code command. And M code that pages macro program (9020~9029.nc) should be set on the parameter PI 95~104 (#3095~3104) (except for M00, M02, M30).

By paging the macro program (in which the action of tool replacement is already programmed) with M code, tool replacement is made.

PI 95 ~ 104 (#3095~3104)	Setting M code for paging macro program

PI 0095	6		9020.nc Calling M Code
PI 0096	10		9021.nc Calling M Code
PI 0097	55		9022.nc Calling M Code
PI 0098	56		9023.nc Calling M Code
PI 0099	57		9024.nc Calling M Code
PI 0100	58		9025.nc Calling M Code
PI 0101	59		9026.nc Calling M Code
PI 0102	0		9027.nc Calling M Code
PI 0103	0		9028.nc Calling M Code
PI 0104	0		9029.nc Calling M Code

11 M Code

When operating a machine tool, you should control several machine behaviors like revolution of spindle motor, injection of cutting oil, replacement of tool, etc. To control these machine behaviors is auxiliary function, M code. For this command, you should designate consecutive numbers next to address M [M00~M999].

If you send consecutive numbers for address M to NC machine tool, this signal is used for ON/OFF control in the machine.

11.1 General M Code

The machine manufacturer (MTB) decides what M code is used for what function. In case the command of motion and M code are used in the same block, MTB also decides whether to start both motion and M code at the same time or to start M code after the motion is completed. Generally, one block is allowed to have maximum 10 of M code

< General M code >

M code	Function	Description
M00	Program Stop	Automatic operation stops if M00 is commanded. Like single block stop, the modal information so far is effective and automatic operation is continued if the button of Cycle Start is pressed.
M01	Optional Stop	This is effective when Optional Stop switches on. It is same as M00. If the switch is not on, this command is ignored.
M02	End of Program	This is to inform end of the program. After the behaviors of the block are completed, the spindle and the cutting oil stop. The cursor goes to the head of the program. It is same as M30. All the commands become RESET.
M03	Clockwise Revolution of Spindle [CW]	Clockwise revolution of spindle. The speed change of gear and the revolution number of spindle should be adjusted in advance before using this command.
M04	Counterclockwise Revolution of Spindle [CCW]	Counterclockwise revolution of spindle. The speed change of gear and the revolution number of spindle should be adjusted in advance before using this command.
M05	Spindle Stop	Spindle stops. It is used for changing the direction of revolution or changing the speed of gear.
M06	Tool Replacement	For tool replacement. It is also used for paging a particular macro program, depending on the type of tool replacement device (ATC).

M08	Cutting Oil On	Cutting oil motor gets operated. Before using this command, Auto switch of cutting oil (in the operation part of the machine) should be ON. If it is OFF, the program is not progressed.
M09	Cutting Oil Off	Cutting oil motor stops.
M30	End of Tape	As soon as the program is closed, the program returns to the head. [Cursor returns to the head of the program] All the commands become RESET.
M98	Sub- program Paging	It is to page sub-program while the auto program is working. .
M99	End of sub-program	It is to end sub-program. Used too for continuously repeating the main program

11.2 Sub-program Paging by M code

In this system, a user can designate M code that he/she wants in order to page a macro program. M code to page a macro program is set with parameter, PI 95 ~ 104 (#3095~3104). The macro programs that can be executed by M code are 9020.nc ~ 9029.nc. It is possible to command local factors (#1 ~ #33) that are used in the macro program. It is also possible to use a new variable in it.

- Macro program paging by M code set with parameter, PI 95~104 (#3095~3104)

G65 P_<Designate factor>

Instead of the above command, the below command can be given.

M __ <Designate factor>

G65 Macro program paging
P _ Name(number) of the program to page
M __ M code for paging macro program that is set with parameter (PI 95~104)

- M FIN and M code command signals are not sent (same as M98, M99).
- Possible to use maximum 10 among M01 ~ M97 (9020 ~ 9029)
- If this MXX is commanded while a sub-program is being paged by G code, M code or T code, it is handled as general M code.
- Possible to page a sub-program by use of M code set with the parameter.

M98 P__

The above command can be given like the below:

M__

M98	M code to page a sub-program
P__	Name(number) of a sub-program to page
M__	M code to page a sub-program set with the parameter

- If this M__ is commanded while a sub-program is being paged by G code, M code or T code, it is handled as general M code.

12 Canned Cycles

Canned Cycles is made to command several machine behaviors (which are used a lot) in one block. For this, the behaviors like drill, tap, boring, etc. are set into patterns and 12 kinds of modal G code from G73 to G89 are available for the behaviors. The Canned Cycles can be cancelled by G80.

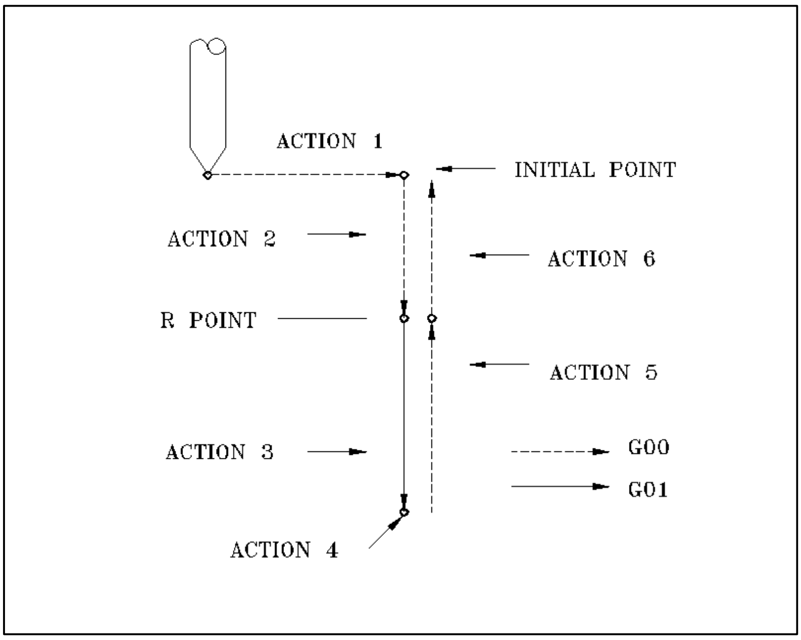
Using the Canned Cycles, you can handle in one block the commands that several blocks need. Therefore, you can easily simply make a program and efficiently use the memory.

12.1 General of Canned Cycles

12.1.1 Actions of Canned Cycles

Generally, canned cycles are divided into 6 actions as shown in the below and according to the action methods, several kinds of canned cycle function are decided:

Action 1 : Positioning X and Y (to be selected according to the plane)
Action 2 : Rapid traverse to R point
Action 3 : Boring
Action 4 : Action at the lowest position of a hole
Action 5 : Actions taken until R point
Action 6 : Rapid traverse to the initial point



: Cutting feed ⟶ - - - - ⟶ Rapid traverse

① Initial point is the location of Z on the absolute coordinate when the mode of Canned Cycle Cancel is changed to Canned Cycles.
② Generally, canned cycles do positioning with X and Y, and machining with Z, but according to the plane selection of the program, X or Y can be used for machining.

12.1.2 Command Type for Canned Cycles (G73 ~ G89)

```
{G73~G89} [G90/ G91] [G98/ G99] X_ Y_ Z_ R_ Q_ P_ F_ K_
```

G17 ~ G18	Work Plane Selection
G73 ~ G89	Canned Cycle
G90 / G91	Absolute Command / Incremental Command
G98 / G99	Return to Initial Point / Return to R Point
X _ Y _	Data of Hole Location
Z _ R _ Q _ P _	Canned Cycle Data
F _	Feedrate
K _	Frequency of Repetition

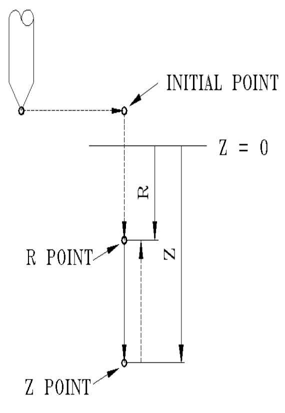
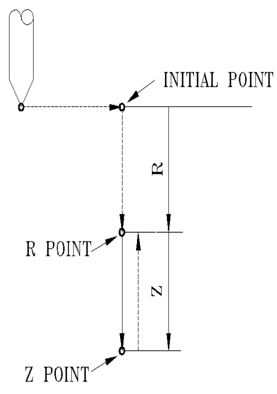
< Data of canned cycles based on X-Y plane (G17)>

Command	Address	Description of Address
Select G17 ~ G19	G	This is a function of plane selection. It decides the axis for boring.
Kind of canned cycles	G	Refer to the action list of canned cycles.
Select G90 ~ G91	G	Absolute or Relative Command is selected, which is omitted in case a command is given already.
Select G98 ~ G99	G	Select either Return to Initial Point or Return to R Point
Location of hole	X, Y	Designate the location for boring with Absolute or Incremental Command. Feed of tool is done by G00 Rapid Traverse.
	Z	It commands the final depth of boring, and performs cutting feed G01 from R point to Z. In case it is absolute, based on the reference position of Z on the work-piece coordinate system, the depth of cutting is commanded, and in case it is incremental, the depth of cutting is commanded based on R.
Drilling Data	R	Refer to <12.1.4 Return to Initial Point (G98) and Return to R (G99)>
	Q	In the commands of G73 and G83, it means the amount of drilling for each case, and in the commands of G76 and G87, it means the amount of shift. It should be incremental.
	P	It commands the time of dwell on the hole bottom.
	F	It commands feedrate of boring.
Frequency of Repetition	K	It commands frequency of repetition of canned cycles. If K is omitted, it is regarded as K1 is commanded, and if K0 is given, only the data of canned cycles, in the current block, are remembered. Once the command of hole location is given, the function of cycles is execute. Only in the block with K command given, the function of repetition is carried out.

12.1.3 Absolute Command and Incremental Command of Canned Cycles (G90 / G91)

In the canned cycles, the reference location for R point and Z terminal are different, depending on Absolute Command and Incremental Command. **In case of Absolute Command,** the reference point for R and Z is Z0 of the work-piece coordinate system. **In case of Incremental Command,** the reference point for R is Initial Point and that for Z is R point.

< Absolute Command and Incremental Command>

G90 [Absolute Command]	G91 [Incremental Command]
The nominated values for X_Y_Z_ are absolute.	The nominated values for X_Y_Z_ are incremental. Z point is distance from R point.
	

Remarks

In the canned cycles, cutting machining is performed in the tool length direction of Z axis, but in case of using angle holder like 5-side tool to work on the side of a work-piece, you should change plane selection and command.

G Code	Drilling Data	
	Hole position	Hole Depth
G17	X, Y	Z

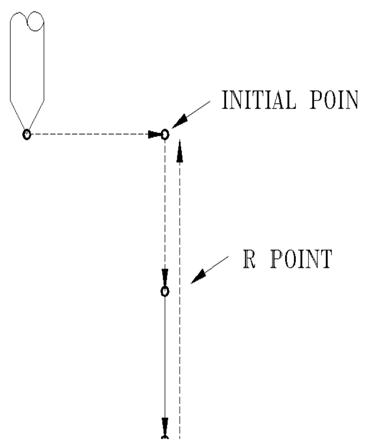
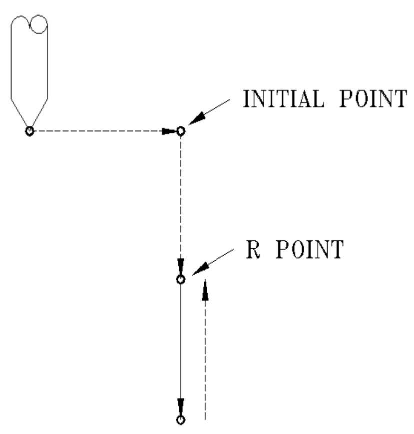
G18	Z, X	Y
G19	Y, Z	X

12.1.4 Return to Initial Point (G98) and Return to R Point (G99)

This is a G code to decide the point that Z axis of the tool should be returned after completion of boring. It is used together with G code for canned cycles. If G98 is commanded, the location of Z returns to Initial Point and if G99 is commanded, it returns to R Point.

Even though boring is done under G99 mode, the Initial Point is not changed. The starting location becomes Initial Point if the return location of the previous case is Initial Point, and it becomes R Point if the return location of the previous case is R.

< Return to Initial Point (G98) and Return to R Point (G99) >

G98 [Return to Initial Point]	G99 [Return to R Point]
This is to command return to Initial Point after end of the cycles. Movement from R to Initial Point is always rapid traverse.	It is to command return to R Point until the action of the cycle ends.
	

Remarks

① G98 (Return to Initial Point)
: Once boring is completed, the tool returns to R Point and then, to Initial Point at rapid traverse.

② G99 (Return to R Point)
: Once boring is completed, the tool returns to R Point and the R Point becomes the start point of the next machining.

③ Reference location of R Point under G90
: It is Z0 on the work-piece coordinate system, and return to R Point after machining is

performed at rapid traverse even without G00 command.

④ Reference location of R Point under G91

: It commands the distance from Initial Point. Return to R Point after machining is performed at rapid traverse even without G00 command.

12.2 Actions of Canned Cycles

MCT	Lathe	Function	Machining	Z point	Return
G80	G80	Canned Cycle Cancel			
G81		Drilling Cycle /Spot Drilling Cycle	Cut Feed		Rapid
G82		Drilling Cycle /Count Boring Cycle	Cut Feed	Dwell	Rapid
G83	G83/ G87	Peck Drilling Cycle	Intermittently Cut Feed		Rapid
G73		High-Speed Peck Drilling Cycle	Intermittently Cut Feed		Rapid
G84	G84/ G88	Tapping Cycle	Cut Feed	Spindle CCW	Cut Feed
G74		Counter Tapping Cycle	Cut Feed	Spindle CW	Cut Feed
G84.2		Rigid Tapping Cycle	Cut Feed	Spindle CCW	Cut Feed
G84.3		Counter Tapping Cycle	Cut Feed	Spindle CW	Cut Feed
G85		Boring Cycle	Cut Feed		Cut Feed
G86	G86/ G89	Boring Cycle	Cut Feed	Spindle Stop	Rapid
G76		Fine Boring Cycle	Cut Feed	주축이 정회전 위치에 정지 후 Shift	Rapid Shift
G87		Back Boring Cycle	주축이 정회전 위치에 정지 후 Shift 급속 이송	Shift Spindle CW	Cut Feed Dwell
G88		Boring Cycle	Cut Feed	Dwell Spindle Stop IPR Stop Manual CYCLE_START	Rapid
G89		Boring Cycle	Cut Feed	Dwell	Cut Feed

12.2.1 Canned Cycle Cancel (G80)

G80

G80 Canned Cycle Cancel

G80 makes canned cycles do modal functions after canceling the current data of canned cycles including R and Z. In the Incremental Command, R becomes 0 and Z becomes 0, and any other boring data than these are all cancelled.

For cancellation of canned cycles, G00. G01, G02, G03, G33 in the G01 group can be given.

📖 Example of canned cycles

① G00 X_ M03 ;

② G81 X_ Y_ Z_ R_ F_ K_ ; : This is first so necessary values for Z_, R_, F_ should be designated. Boring by G81 is repeated K times.

③ Y_ ; : It is same as boring mode and boring data designated at ② so it is possible to omit G81, Z_, R_, F_. The location of a hole moves by Y_ and boring by G81 is performed only one time.

④ G82 X_ P_ K_ ; : Against ③, only X axis moves. Boring by G82 is repeated K_ times with the boring data designated in ② (Z_, R_, F_) and ④ (P_).

⑤ G80 X_ Y_ ; : Boring is not performed. All the boring data except for F_ are cleared.

⑥ G85 X_ Z_ R_ P_ ; : It is necessary to designate Z_, R_ again because they are cancelled at ⑤. X_ is same as F_ designated at ② so it can be omitted. P_ is not necessary for this block, but it is remembered.

⑦ X_ Z_ ; : Boring which is different from ⑥ by Z_ is performed after the location is moved by X-.

⑧ G89 X_ Y_ ; : Boring by G89 is performed with the data of Z_ designated at ⑦, R_, P_ designated at ⑥ and F_ designated at ②.

⑨ G01 X_ Y_ ; : Boring mode and boring data (except for F) are all deleted.

12.2.2 Special Notes for Command of Canned Cycles

① It is necessary to revolve the spindle with M code before giving the command of Canned Cycles.

② For the block including G code (G00 group) in the mode of Canned Cycles, Canned Cycles is not executed and only G code is executed. However, boring mode and boring data are changed to be those designated in the relevant block.

③ In the canned cycles, English letters of L_, J_ are modal commands. Therefore, they are continuously effective until the command is changed. With only the L_, J_ commanded, only the data are changed and boring is not performed.

④ During the mode of Canned Cycles, if any of X_, Y_, Z_, R_ is designated for the block, boring is performed for the block. In addition, in the block that doesn't include any of X_, Y_, Z_, R_, only the machining data are changed and boring is not performed. Even though X_ is commanded, boring is not performed in case of G04 X_ ; .

⑤ Please command boring data Q_, P_ in the block in which boring is executed. That is, give the command in the block having any command of X_, Y_, Z_ data. Even though these data (Q_, P_) are instructed in the block in which boring is not executed, they are not remembered as modal data.

⑥ G74, G84, G86 mode

ⓐ Under the circumstance that canned cycles G74, G84, G86 (that control revolution of the spindle) are used, if boring (the distance from hole location X_, Y_, initial level to R level is short) is continued, there can be a case that the spindle doesn't reach the normal revolution before it goes into cutting action ? Z of the hole. In this case, you need to insert dwell by G04 between actions of boring to have some time.

ⓑ Unlike that other canned cycles are temporarily stopped immediately in the modes of G74 and G84, if Temporary Stop is commanded during cutting feed, the lamp of Temporary Stop lights on, but the machine continuously work on R Point or Initial Point until the action ends.

ⓒ During the machining actions of G74 and G84, it automatically becomes the status of cutting velocity adjust cancel.

⑦ If the command of Canned Cycles and M code are given in the same block, M code is sent to the initial positioning action 1. In case the command of frequency l(K) is given, M code is sent out first once and after that, it is not.

⑧ In the mode of Canned Cycles, the command of tool location offset G45 ~ G48 is ignored.

⑨ In the mode of Canned Cycles, if the tool length offset G43, G44 or G49 is given, the offset is applied to positioning action 2 with R Point.

⑩ Even though the command of Dwell is given to the canned cycles in the form of English letters G04P_, P, which is used as data for boring, is not influenced.

12.2.3 Special Notes for Users of Canned Cycles

① RESET

ⓐ During the mode of Canned Cycles, the control device is stopped by RESET or emergent stop, boring mode and boring data are usually remembered.

ⓑ The mode of Canned Cycles can be cancelled by not only G80, but also G00, G01, G02, and G03 (G01 group).

② Single block

In case of doing canned cycles with single block, the control device stops at the end of Action 1, 2 and 6. It means that in order to make a hole, three times of operation are required. In the mode of Action 6, Stop, if the frequency of repetition remains, it stops temporarily, but in any other cases, it stops.

ⓐ Positioning the hole location in the direction of X and Y

ⓑ Positioning up to R Point

ⓒ The cycle ends and positioning at R Point (G99) or Initial Point (G98)

In case of ⓐ and ⓑ, the lamp of Temporary Stop (Red LED) lights on at the end of each, but in case of ⓒ, it lights on only when the frequency of repetition remains. Otherwise, it doesn ' t light on like the normal single block stop.

③ Temporary Stop

In case Temporary Stop is commanded between Action 3 ~ 5 under G74 and G84, the control device stops after continuously taking Action 6. If Temporary Stop is given during Action 6, it immediately stops.

④ Override

During G74 and G84 executed, feedrate override becomes 100%.

12.3 Kinds of Canned Cycles

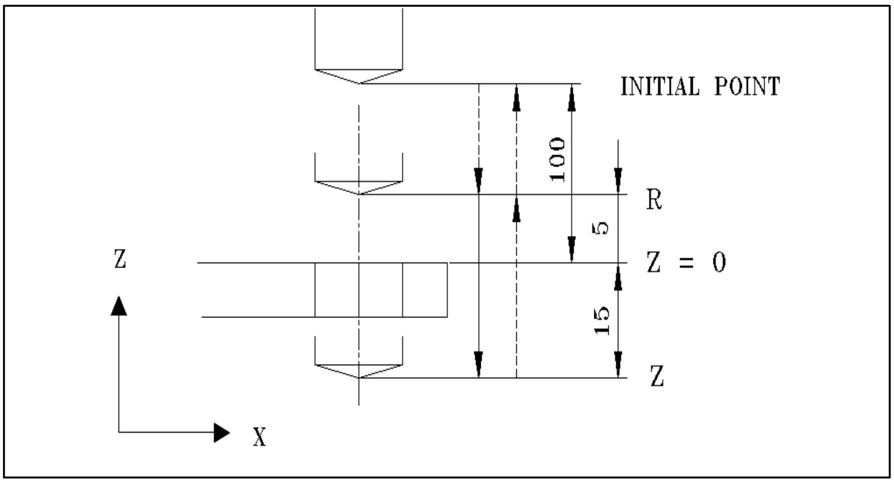
12.3.1 Drilling Cycle / Spot Drilling Cycle (G81)

G81 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ F _ K _

G81	Drill Cycle / Spot Drilling Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R point R
F _	Feedrate
K _	Frequency of Repetition

This cycle that is used for general drilling, reaming and spot boring, is used for boring a work-piece that is easy to discharge chips. In this cycle, the tool does cutting feed to the bottom of a hole and then goes out of it at rapid traverse.

Example of G81 G98 use



(G92 X0 Y0 Z100 ;)

N1 S500 M03 ;

N2 G90 G99 G81 X75 Y50 Z–15 R5 F100 ;

N3 Y–50 ;

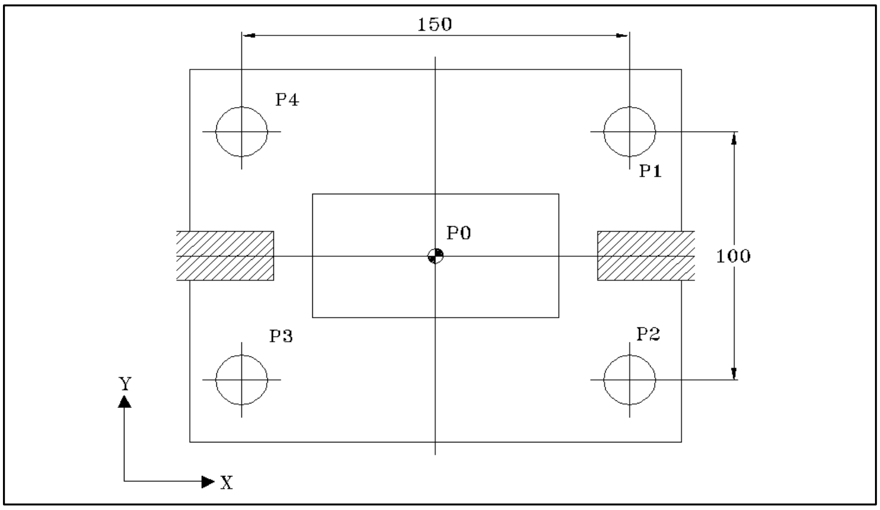
N4 X–75 ;

N5 G98 Y50 ;

N6 G80 G00 X0 Y0 M05 ;

N7 M30 ;

Example of G81 G99 use



In case there is interference with the tool at the time of Y moving

Explanation

(G92 X0 Y0 Z100 ;)
N1 S500 M03 ;
N2 G90 G98 G81 X75 Y50 Z–15 R5 F100 ; (Return to Initial Point)
N3 G99 Y– 50 ; (Return to R Point)
N4 G98 X– 75 ; (Return to Initial Point)
N5 Y50 ; (Return to Initial Point)
N6 G80 G00 X0 Y0 M05 ;
N7 M30 ;

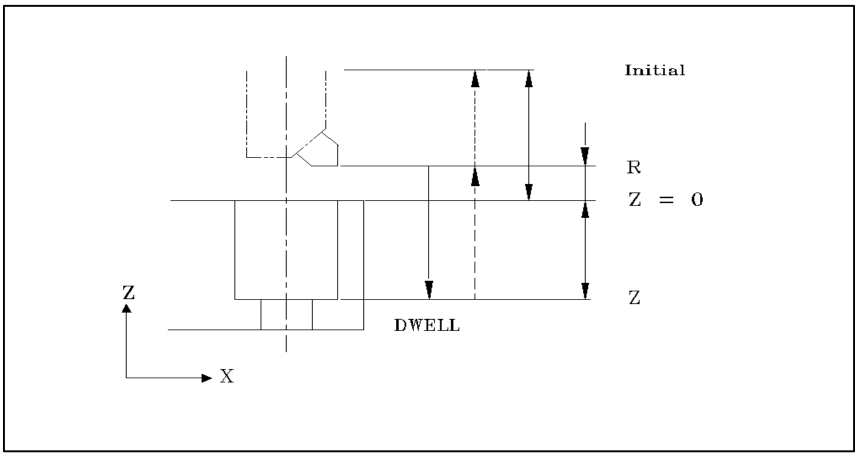
12.3.2 Drilling Dwell Cycle / Counter Boring Cycle (G82)

G82 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G82	Drilling Dwell Cycle / Counter Boring Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R point
P _	This is the command of dwell. As much as the given time, it stops progress of the program at the terminal of boring. The method of using it is same as G04. Auxiliary functions like spindle and cutting oil work normally. If P command is omitted, it is same as G81. By P500 command, it temporarily stops for 0.5 seconds.

This cycle is used for counter boring and spot boring. In order to make the bottom of a hole good, it is possible to give a command of dwell, which stops the progress of the program for particular time on the bottom of the hole.

Example of G82 use



Explanation

(G92 X0 Y0 Z100;)

N1 S600 M03;

N2 G90 G99 G82 X75 Y50 Z-50 R5 P500 F100 ;

N3 Y-50 ;

N4 X-75 ;

N5 G98 Y50 ;

N6 G80 G00 X0 Y0 M05 ;

N7 M30 ;

12.3.3 Peck Drilling Cycle (G83)

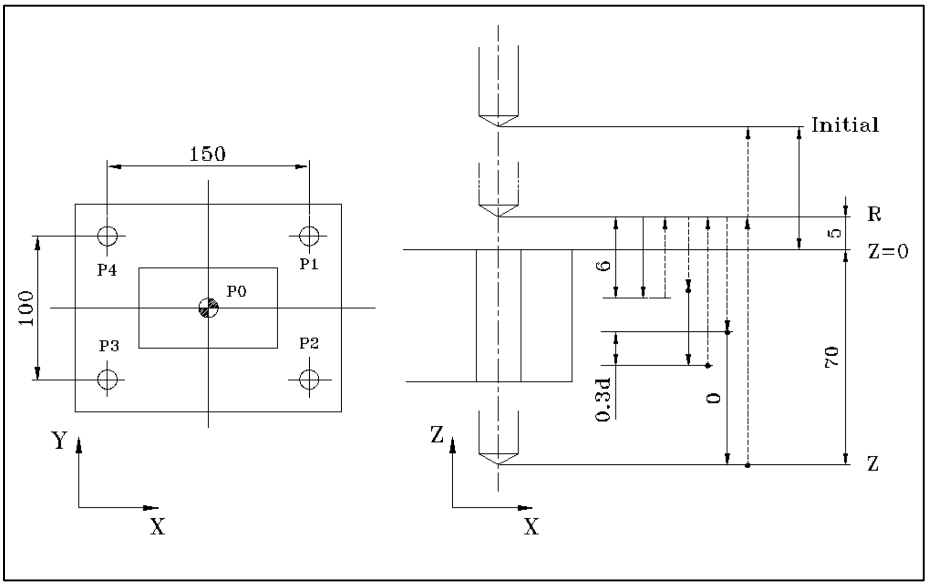
G83 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ Q _ F _ K _

G83	Peck Drilling Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R point
Q _	Use + incremental command for the amount of drilling for each time. If Q is omitted, it is same as G81 that consecutively machines from R to Z.

The hole which is small and deep is not easy to discharge out chips because it is difficult for cutting oil to go into the end of the cutting tool. Peck drill cycle, in order to solve this problem, uses intermittent cutting feed. It drills once as much as " Q ", returns to R point, moves at rapid traverse to the location higher as much as ' d ' than the location right before return, and then, cuts as much as ' Q '. It repeats this action and machines up to Z point. The surplus of peck drill cycle (d) is set with the parameter, PI 129(#3129).

It is good for making a small and deep hole or for processing heat-resistant alloy steel. It is suitable when using a material of brass.

Example of G83 use



Explanation

```
( G92 X0  Y0  Z100 ; )
N1 S500 M03 ;
N2 G90 G99 G83 X75 Y50-Z0 R5 Q6 F100 ;
N3 Y-50 ;
N4 X-75 ;
N5 G98 Y50 ;
N6 G80 G00 X0 Y0 M05 ;
N7 M30 ;
```

12.3.4 High-Speed Peck Drilling Cycle (G73)

G73 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ Q _ F _ K _

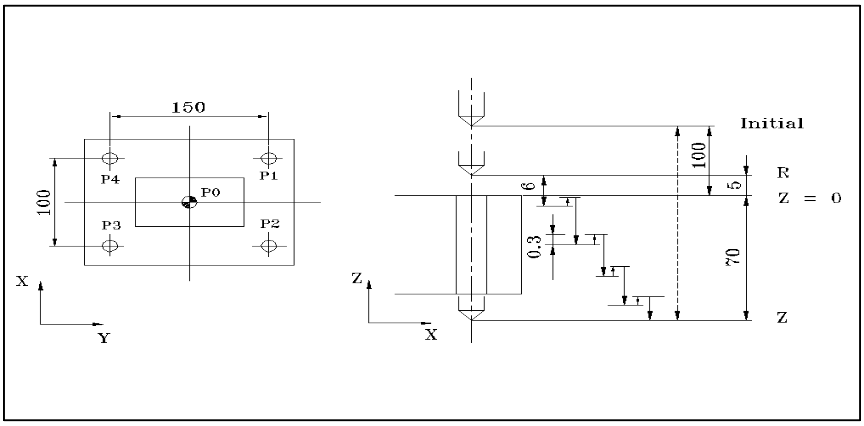
G73	High-Speed Peck Drilling Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R point
Q _	Amount of drilling for each time
F _	Feedrate
K _	Frequency of repetition

The hole which is small and deep is not easy to discharge out chips because it is difficult for cutting oil to go into the end of the cutting tool. High-speed peck drill cycle, in order to solve this problem, uses intermittent cutting feed so chips are discharged out easily without long chips.

It drills one time as much as ' Q ' and then, retreats at rapid traverse as much as ' d '. And then, it drills as much as ' Q ' and retreats at rapid traverse as much as ' d '. It repeats this process up to Z point. The surplus of peck drill cycle (d) is set with the parameter, PI 129(#3129).

Unlike peck drill cycle (G83), it doesn ' t return to ' R ' Point. Instead, it retreats as much as ' d ' and then, drills. Therefore, it can make a deep hole at high speed.

Example of G73 use



Explanation

(G92 X0 Z100;)
N1 S500 M03 ;
N2 G90 G99 G73 X75 Y50 Z-70 R5 Q6 F100 ;
N3 Y-50 ;
N4 X-75 ;
N5 G98 Y50 ;
N6 G80 G00 X0 Y0 M05 ;
N7 M30 ;

12.3.5 Tapping Cycle (G84) & RIGID Tapping Cycle (G84.2)

G84 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G84	Tapping Cycle
X _ Y _	Location data for tapping
Z _	Depth of tapping
R _	Command the coordinate of R Point
F _	Feedrate for tapping
P _	Time of dwell at the hole location
K _	Frequency of repetition

This tapping cycle makes a screw by directly revolving the spindle, drilling and counterclockwise revolving at the bottom of a hole to escape. Using a right screw tap tool, it executes tapping, which is continued until the spindle makes a clockwise revolution (M3) to reach Z Point. It returns to R point while making inverse revolutions (M4) and then, revolves clockwise again.

G84.2 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G84.2	RIGID Tapping Cycle
G84.3	Counter Rigid Tapping Cycle

RIGID tapping cycle is possibly used only with the machines which can instruct the spindle ' s position control. RIGID tap can perform tapping by synchronized control of the spindle and Z (tapping axis). In case of RIGID tapping, it reduces the time for tapping and 4-jaw independent chuck is not necessary. In addition, the start point of screw is always same.

Remarks

① If you press the button of Feed-Hold during tapping, feed of Z stops. At this time, the tap can be damaged so you should stop after finishing the current tapping work.

② Calculation formula for feedrate of tapping

$$F = n \times f$$

F	Feedrate for tapping [mm/min]
n	No. of revolutions of spindle [rpm]
f	Tap pitch [mm]

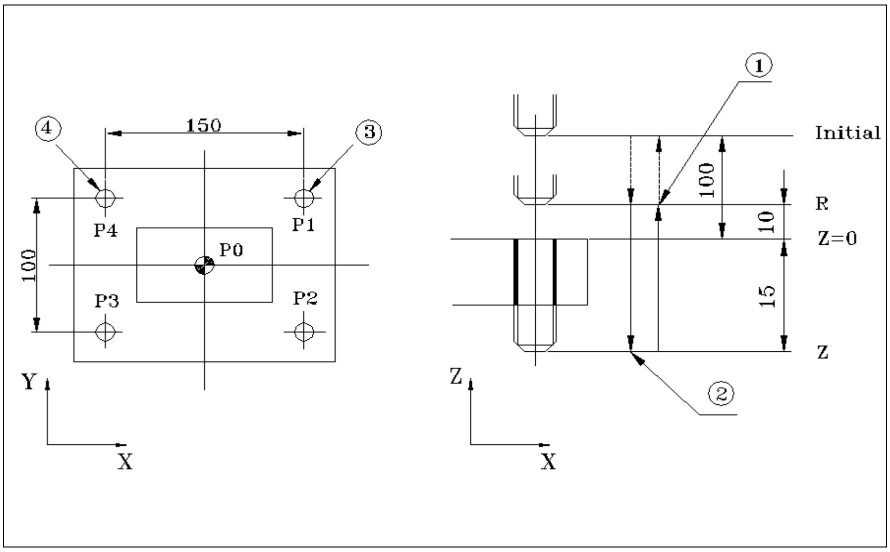
📖 Example of calculation formula

If the speed of revolution is 300rpm in case of tapping by use of M10 × P1.5, what is feedrate?

Explanation

F = n × f, SO, F = 300 × 1.5 = 450 mm.

Example of G84 use



1 : CCW revolution of spindle 2 : CW revolution of spindle

3 : Start point for machining 4 : End point for machining

Explanation

N1 S640 M03;

N2 G90 G99 G84 X75 Y50 Z-15 R10 F640; R is 10mm in case of tapping.

N3 Y–50;

N4 X–75;

N5 G98 Y50;

N6 G80 G00 X0 Y0 M05;

N7 M30;

12.3.6 Counter Tapping Cycle (G74) and Counter RIGID Tap Cycle (G84.3)

G74 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G74	Counter Tapping Cycle
X _ Y _	Location data for tapping
Z _	Depth of tapping
R _	Command the coordinate for R Point
P _	Time of dwell at the hole location
F _	Feedrate for tapping
K _	Frequency of repetition

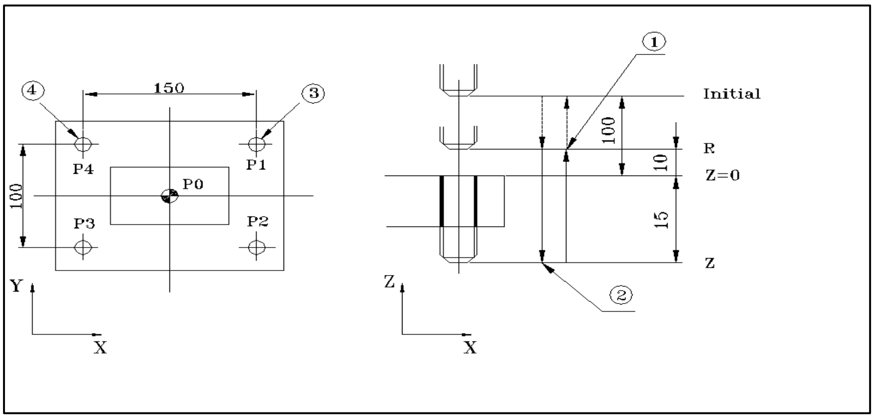
The tapping cycle is: tapping is performed by use of a left handed screw tapping tool; the spindle revolves CCW (M4) to perform tapping up to Z point; it revolves CW (M3) to return to R Point; and then, it revolves CW again. It is used for making counter screw.

G84.3 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G84.3	Counter RIGID Tapping Cycle

Counter RIGID Tap Cycle is possibly used only on the equipment which can give a command of controlling the spindle's location. RIGID tap can perform tapping by synchronized control of the spindle and Z (tapping axis). With RIGID tap cycle, tapping time is reduced and 4-jaw independent chuck is not necessary. In addition, the start point of screw is always same.

📖 Example of G74 use



1 : CCW revolution of spindle   2 : CW revolution of spindle

3 : Start point for machining   4 : End point for machining

Explanation

(G92 X0 Y0 Z100 ;)

N1 S640 M04 ;

(In case of tapping, R is 10 mm.)

N2 G90 G99 G74 X75 Y50 Z–15 R10 F640 ;

N3 Y–50 ;

N4 X–75 ;

N5 G98 Y50 ;

N6 G80 G00 X0 Y0 M05 ;

N7 M30 ;

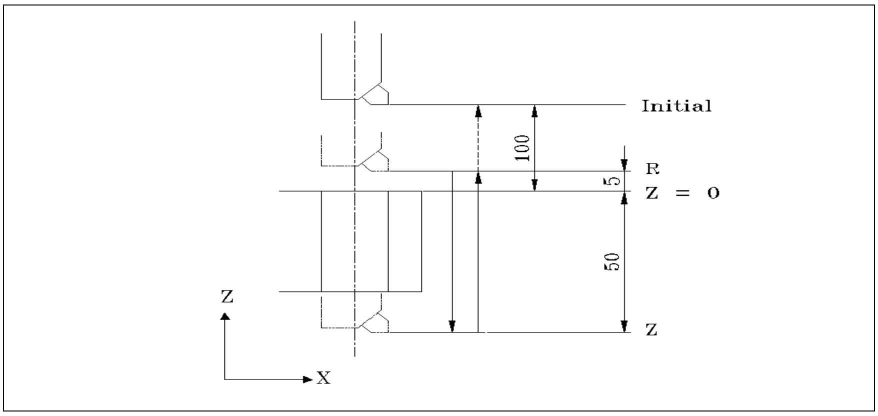
12.3.7 Boring Cycle (G85)

G85 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ F _ K _

G85	Boring Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R Point
F _	Feedrate
K _	Frequency of repetition

This cycle works on cutting while going through both the time of drilling and the time of return. It is generally applied to reamer machining.

📖 Example of G85 use



(G92 X0 Y0 Z100;)

N1 S500 M03 ;

N2 G90 G99 G85 X75 Y50 Z–50 R5 F100 ;

N3 Y–50 ;

N4 X–75 ;

N5 G98 Y50 ;

N6 G80 G00 X0 Y0 M05 ;

N7 M30 ;

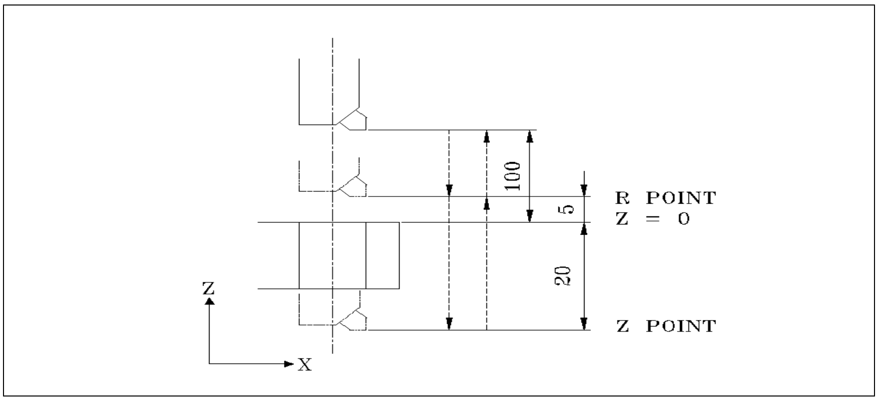
12.3.8 Boring Stop Cycle (G86)

```
G86 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ F _ K _
```

G86	Boring Stop Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R Point
F _	Feedrate
K _	Frequency of repetition

The most general boring cycle, it executes **M05 spindle stop after reaching the terminal of Z,**and returns rapidly. It is used a lot for rough boring.

📖 Example of G86 use



(G92 X0 Y0 Z100 ;)

N1 S500 M03 ;

N2 G90 G99 G86 X75 Y50 Z–50 R5.0 F100 ;

N3 Y–50 ;

N4 X–75 ;

N5 G98 Y50 ;

N6 G80 G00 X0 Y0 M05 ;

N7 M30 ;

12.3.9 Fine Boring Cycle (G76)

G76 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ Q _ F _ K _

G76	Fine Boring Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R point
Q _	Amount of shift at the bottom point of a hole

The direction of shift is set with PI 70(#3070). Generally, the amount of shift is set in the opposite direction of the boring bite before Z returns after boring up to Z terminal and doing spindle single direction orientation stop.

I _ J _ Possible to directly designate the amount of shift per axis and its direction instead of Q.Q

F _ Feedrate

K _ Frequency of repetition

Fine boring cycle executes spindle single direction orientation stop after doing cutting feed to the terminal of a hole.Under this circumstance, it moves in the opposite direction of a boring thread as much as Q (or I, J) and rapidly moves to R point.

> **Note :**
> Q, modal information, is used as the amount of drilling for G73 and G83 so you need to be careful for designating it.

12.3.10 Back Boring Cycle (G87)

G87 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ Q _ F _ K _

G87 Boring Cycle / Back Boring Cycle

X _ Y _ Location data for boring

Z _ Depth of boring

R _ Command the coordinate of R Point

Q _ Amount of shift at the bottom point of a hole

The direction of shift is set with parameter PI 70(#3070), PI 71(#3071). Generally, the amount of shift is set in the opposite direction of the boring bite before Z returns after boring up to Z terminal and spindle single direction orientation stop.

F _ Feedrate

K _ Frequency of repetition

Back boring cycle is: under the circumstance that the spindle stops in one direction (orientation), it moves in the opposite direction of the boring thread as much as Q and then, rapidly feeds to R. Unlike the general boring cycle, boring is performed in the opposite direction.

After it moves from R to the center of boring, the spindle performs cutting feed to the terminal of Z while making CW revolutions. And then, it has orientation, moves as much as Q, returns to Initial Point, and moves to the center of the tool. And then, the spindle makes a CW revolution.

Note:

Q value, modal information, is used as the amount of drilling like G73 and G83 so you need to be careful when designating it

12.3.11 Manual Boring Cycle (G88)

G88 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G88	Boring Cycle
X _ Y _	Location data for boring
Z _	Depth of boring
R _	Command the coordinate of R Point

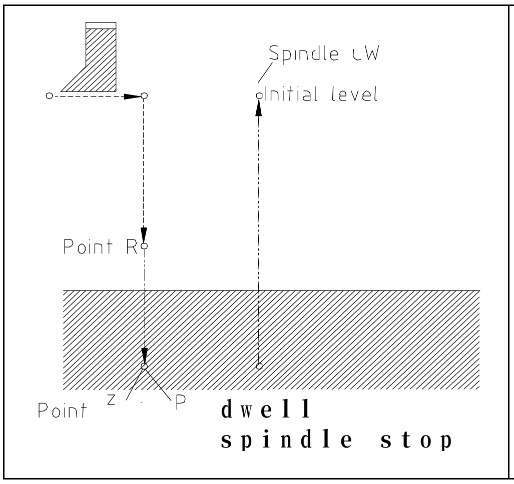
P _ This is the command of dwell. It stops the progress of the program at the terminal of boring as much as the given time. The method of using it is same as G04, and the auxiliary functions like spindle and cutting oil work normally.

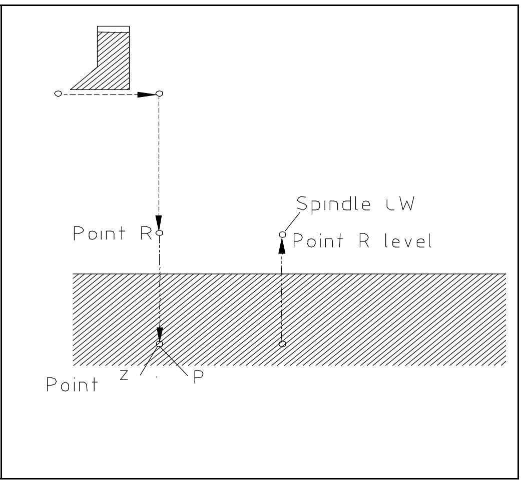
F _	Feedrate
K _	Frequency of repetition

Manual boring cycle automatically executes cutting to Z, the location of boring, dwells and then, stops the spindle. If a worker gets out of the boring location through manual operation (handle, JOG) and then, executes automatic start at a random location, it normally returns to R or Initial Point, which is G code.

This G code can execute motion by use of MPG at the manual mode if the spindle dwells at the bottom of a bore for a while and then, stops. If the key of Cycle Start is pressed in the automatic operation mode, the spindle returns to R or Initial Point at rapid traverse and then, revolves. It is used for large boring machine a lot.

G98 [Initial Point Return]	G99 [R Point Return]


Spindle CW
Initial level
Point R
Point
Z
P
dwell
spindle stop


Spindle CW
Point R
Point R level
Point
Z
P

12.3.12 Side Boring Cycle (G89)

G89 [G90 / G91] [G98 / G99] X _ Y _ Z _ R _ P _ F _ K _

G89 Side Boring Cycle

X _ Y_ Location data for boring

Z _ Depth of boring

R _ Command the coordinate of R point

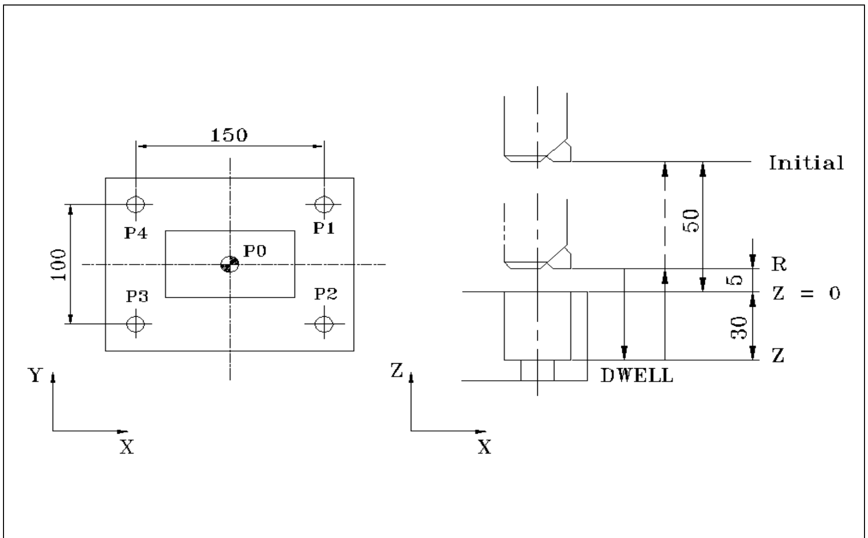
P _ This is the command of dwell, It stops the progress of the program at the terminal of boring as much as the given time. The method of using it is same as G04, and the auxiliary functions like spindle and cutting oil work normally.

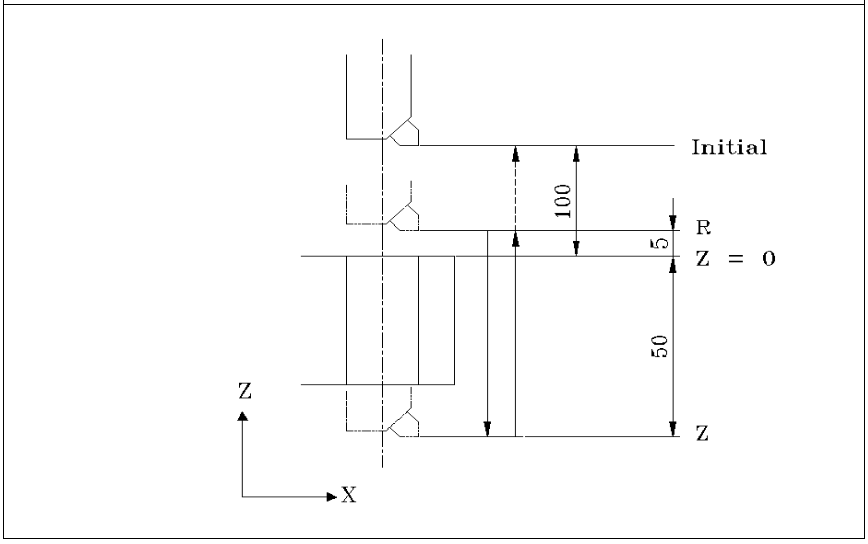
F _ Feedrate

K _ Frequency of repetition

Boring cycle G89 works on cutting, going through both the time of drilling and the time of return. It is applied for reamer machining too. It is same as G85 cycle, but it is added with the dwell function ? the tool executes the action of stop on the bottom of hole and then, goes up.

📖 Example of G89 use





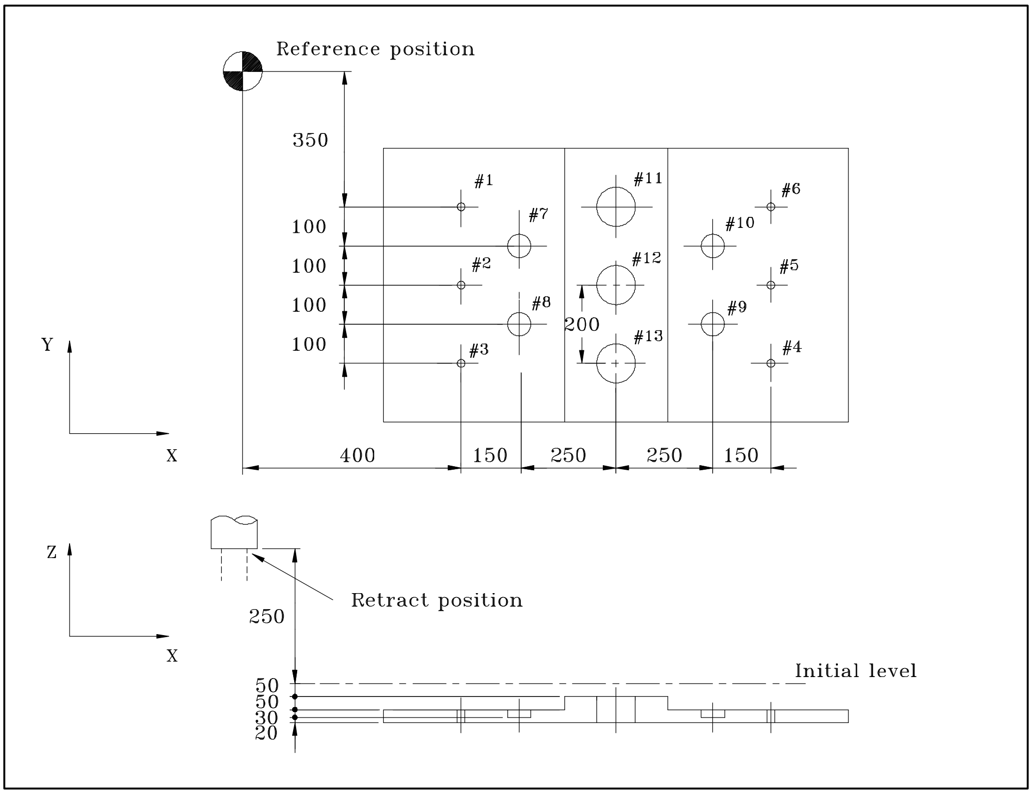
Explanation

(G92 X0 Y0 Z250 ;)
N1 S600 M03 ;
N2 G90 G99 G89 X75 Y50 Z–30 R5 P500 F100 ;
N3 Y–50 ;
N4 X–75 ;
N5 G98 Y50 ;

```
N6 G80 G00 X0 Y0 M05 ;
N7 M30
```

12.4 General Examples of Canned Cycles

🕮 Example of using tool length offset and canned cycles



1 to 6 : Drilling of a 10mm diameter hole
7 to 10 : Drilling of a 20mm diameter hole
11 to 13 : Boring of a 95mm diameter hole(depth 50 mm)

Program

```
N01 G92 X0 Y0 Z500 ;
N02 G90 G00 Z250 ;
N03 G43 Z0 H11 ;
N04 S30 M03 ;
N05 G99 G81 X400 R Y–350 Z–153 R–97 F120 ;
N06 Y–550 ;
N07 G98 Y–750;
N08 G99 X1200 ;
N09 Y–150 ;
N10 G98 Y–350 ;
N11 G00 X0 Y0 M05 ;
N12 G49 Z250 T15 M06 ;
N13 G43 Z0 H15 ;
N14 S20 M03 ;
N15 G99 G82 X550 Y–450 Z–130 R–97 P300 F120 ;
N16 G98 Y–650;
N17 G99 X1050 ;
N18 G98 Y–450 ;
N19 G00 X0 Y0 M05 ;
N20 G49 Z250
T31 M06 ;
N21 G43 Z0 H31 ;
N22 S100 M03 ;
N23 G85 G99 X800 Y–350 Z–153 R47 F50 ;
N24 G91 Y–200 K2 ;
N25 G28 X0 Y0 M05 ;
N26 G49 Z500 ;
N27 M02 ;
```

13 Tool Compensation

13.1 Tool Length Compensation

This function is to set tool length offset data with the offset memory and then, compensate them as much as the designated values when selecting a programmed tool.

In case of making a program, the tool length is not considered, but in actuality, several kinds of tools necessary for actual machining have different lengths. You can compensate the tool length by measuring each tool length, registering the tool length offset data with the compensation screen in advance, and paging each tool length address from the program whenever necessary.

You can designate the offset direction for the data inputon the compensation screen, using G43 and G44, designate the offset data set with the compensation screen, using H code, and move the location of the terminal of Z (in case of G17 plane) to ± as much as the value set on the screen.

In case of programming, using this function, you can compensate the difference between the assumed tool length value and the actual tool length value used for machining without changing the program.

13.1.1 Tool Length Compensation (G43, G44, G49)

```
{G43 / G44}   Z _ H _
G49   Z _
```

G43 : Tool Length Compensation + Direction

G44 : Tool Length Compensation - Direction

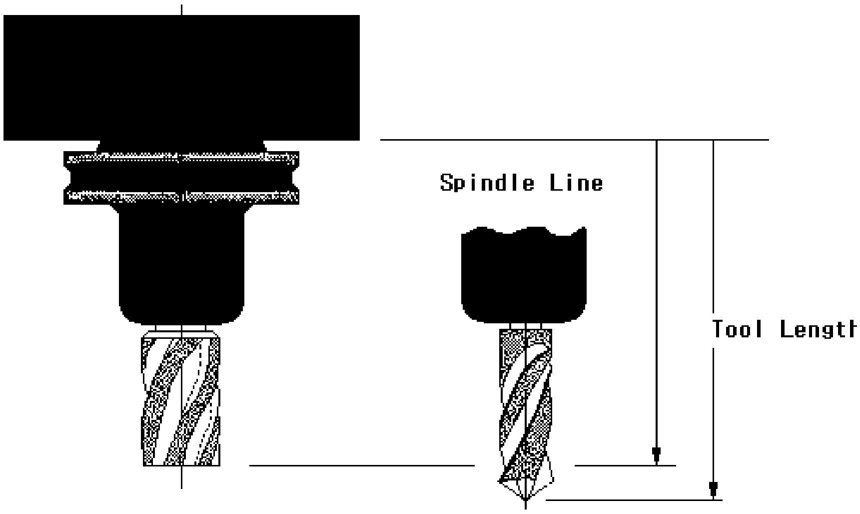
G49 : Tool Length Compensation Cancel]

Z _ : Move Z (in case of G17 plane)
Absolute Command and Incremental Command possible

H _ : Offset number with tool length compensation value remembered

In case of G43, the offset data that is designated by H code is added to the coordinate values of the terminal for Z movement that is programmed by Absolute Command or Incremental Command. In case of G44, it is subtracted from it so the coordinate value becomes the terminal.

In case the command to move Z is omitted, the length offset is applied in the + direction for G43 and ? for G44 in the next block that has the command of Z.



⏭

① Tool length compensation is applied vertically to the plane in current use. (G17/G18/G19 plane)

② G49 or H00 is used to cancel offset compensation.

③ G43, G44, G49 are modal G code, which are effective until other G code in this group appears.

④ Offset number should be designated by H code and the offset data should be set with the offset memory corresponding with this number. Offset number is available from H00 to H128. The offset corresponding to offset number 00, H00, always means 0, and it is impossible to set the offset amount corresponding with H00.

⑤ For the commands of tool length compensation and cancel, it is good to make a program like the command of moving Z. The reason is that: if it is commanded in the same method as G43 H01;, Z moves as much as the tool length (or length compensation amount) input on the compensation address 01; if it is commanded in the same method as G49;, it moves in the opposite direction as much as tool length compensation executed before G49; and if the tool length compensation value is "+", it moves down as much as the tool length from the current position so the tool can collide.

⑥ Therefore, the command of tool length compensation and cancel is given just like the command of moving Z, and it is good to designate it larger than the tool length.

▯ll Example of tool compensation

Offset data of work-piece coordinate system : -150.000 mm

Compensation amount for H1 tool length: In case 10.000 mm (= 10mm longer than the standard tool)

G54 G90 Z100. ; : Move to Z100 by G54 coordinate system

[Machine location of Z = -50.000]

[Location of Z axis work-piece coordinate system = 100.000]

G43 G90 G00 Z50. H01 ; : Move the compensation amount input on 01 of the compensation screen to 50mm of Z axis and compensate the tool length

[Tool number and tool compensation number don ' t need to be same, but if you use the same number, you can reduce any mistake in compensation while working.]

[Machine location of Z axis = -90.000(Compensate in the + direction of by 10mm)]

[Location of Z axis work-piece coordinate system = 50.000]

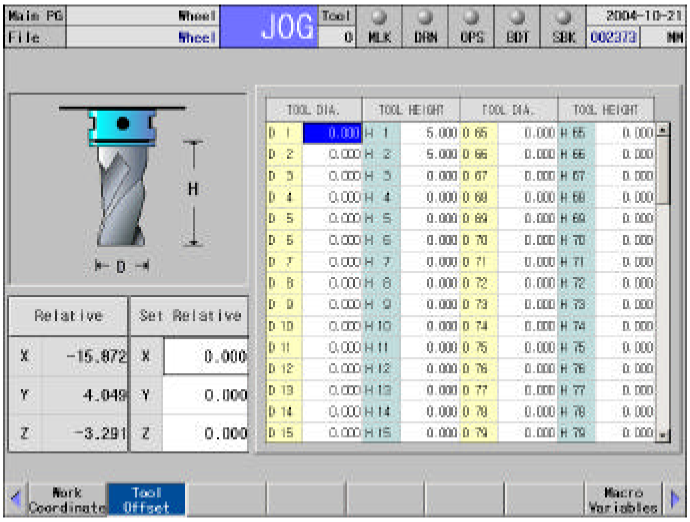
▯ll Example of tool length compensation cancel

G90 G49 G00 Z150. ; Z moves to 150.0 on the work-piece coordinate system.

(또는 G90 G43 G00 Z150. H00 ;)

13.1.2 Direct Input of Tool Length Compensation Data

Main menu ▸▸ F2 setting ▸▸ F2 offset setting screen. You can directly input the measured compensation data for each tool, using the number key on the MDI panel after moving the cursor to the relevant numbers.



13.1.3 Programmable Data Input (G10)

```
[G90 / G91]   G10   L10   P _ R _
[G90 / G91]   G10   L12   P _ R _
```

G10	Data setting
L10	H Data setting [Change of Length Offset value By Program]
L12	D Data setting [Change of Diameter Offset value By Program]
P_	Offset Register
R _	Tool Length or Tool Diameter Compensation Data
G90	Compensation data are input and in case the offset values are input already, they are replaced with the new values.
G91	Compensation data are input and in case the offset values are input already, the data are added to or subtracted from the existing offset values.

This command is used for input of tool diameter compensation data, input of tool length compensation data, and input of work-piece reference position offset data of work-piece coordinate. The compensation data can be input by program.

Generally, G10 is used a lot when an automation line or a factory for precise mass production installs a measuring equipment and automatically compensates the dimension which is changed slightly during machining. It is applied for special machining on the general production sites.

▯ II Example of tool length compensation data input

⬇

G10 G90 L10 P1 R100 ; : Input 100 for compensation number H01.

G10 L12 P1 R20 ; : Input 20 for compensation number D1.

G10 G91 L10 P1 R - 0.5 ; : Input the compensation data that are reduced by ? 0.5 for H1.

⬇

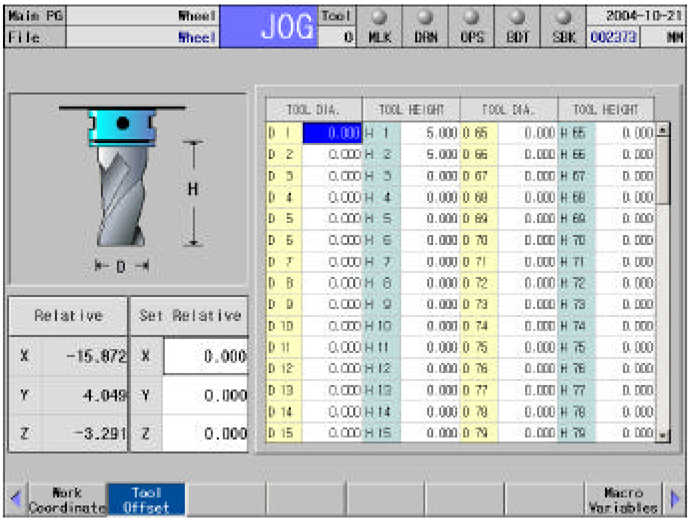
G43 G00 Z50 H02 S1500 M03 ; : Execute the spindle ' s CWt revolution while executing tool length compensation.

Input of programmable tool compensation data is done at the head of the program in general. It is done before tool compensation.

13.2 Tool Diameter Compensation

In case of making a program, the tool length is not considered, but in actuality, several kinds of tools necessary for actual machining have different lengths and diameters. The function of compensation is to register the difference with the compensation screen, to page it when machining a work-piece, and to get location compensation automatically. It is necessary to measure the difference of tool lengths and the diameters in advance, and register them with the compensation screen.

< Main menu ⏩ F2 setting ⏩ F2 offset setting screen >



13.2.1 Tool Diameter Compensation (G40, G41, G42 , G39_

```
G40   [G00 / G01]   X _ Y _
G41   [G00 / G01]   X _ Y _ D _
G42   [G00 / G01]   X _ Y _ D _
G39   {X _ Y _ / I _ J _}
```

G40	Cutter Compensation Cancel
G41	Cutter Compensation Left]
G42	Cutter Compensation Right
G39	Handle corner offset with circular interpolation

D _ Offset number with tool diameter compensation value remembered
X _Y _ / I _J _Command the vector for the next block

In case of machining by use of the side thread of a tool, the center of a tool doesn ' t meet with the program due to the tool ' s diameter so the deviation **as much as the tool radius** happens. In order to automatically compensate the deviation, it is needed to input the radius of a tool with the tool radius screen in advance and to compensate the center of the tool as much as the designated value against the programmed straight line or circular arc. This function is used a lot for end mill work that uses the side of a tool.

Main menu ⏩ F2 setting ⏩ F2 offset setting screen. Move the cursor to the number corresponding to the tool radius to be used and then, input (to be compensated as much as the radius).

In order to input the tool diameter compensation data with radius, the parameter PI 72 should be set at 1. At this time, the title of the tool diameter compensation data input is changed as shown on the below:

⏭

① Compensation data [D code]

Corresponding with [Offset Number], the number of D code that is commanded on the program, offset should be set on the tool diameter screen by use of the number key on MDI panel. D code, once it is designated, is effective until another D code is given.

The offset data corresponding with offset number 00, that is, D00, always means 0, and it is impossible to set the offset data corresponding with D00. In order to cancel offset compensation, G40 or D00 should be given.

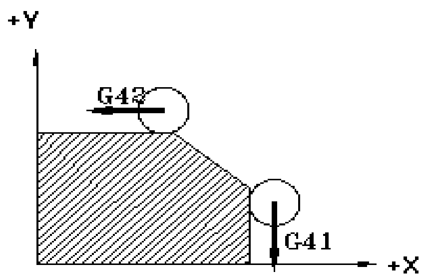
② Cancel tool diameter compensation

In case of power ON or the button of RESET on MDI panel pressed, M02 or M30 is executed and after the program is closed, the mode is changed into Tool Diameter Compensation Cancel. In G40 mode, the size of offset is always 0, and the central passage of the tool is same as the route of the program.

③ The program should be always closed under G40 mode. If it is closed with G41/G42, the program is closed in the position that is far away as much as the compensation data.

④ Tool diameter compensation is applied only in the designated plane.

G code	Description
G40	Cancel tool thread-end compensation
G41	Based on the direction of tool progress, the tool progresses in the left direction of the work-piece. [Tool compensation is done in the left direction of the work-piece.]
G42	Based on the direction of tool progress, the tool progresses in the right direction of the work-piece. [Tool compensation is done in the right direction of the work-piece.]



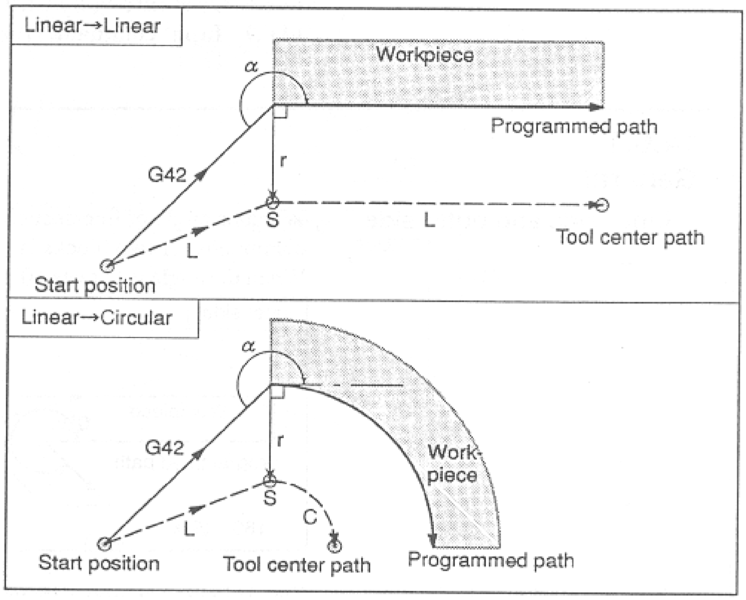
13.2.2 Tool Diameter Compensation C type (G41,G42)

(1) Start ? Up mode

- Start-Up mode is that under the mode of Tool Diameter Cancel, G41/G42 is given and the tool diameter compensation starts.
- Start-Up mode is that G41 or G42 is given, or after that, the block for axis feed is designated first.
- In the block of Start-Up, the command of axis feed should be given by more than tool radius.
- In the mode of Start-Up or Start-Up Cancel, the circular arc command [G02/G03] is not executed and if it is executed, an alarm happens.

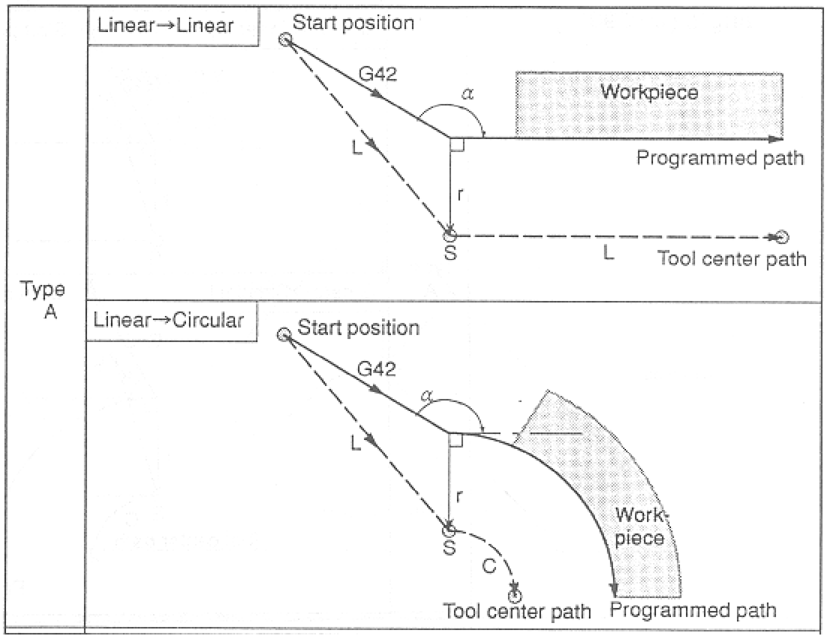
Meaning of letter in figure	
S	Location where it stops by Single Block signal (conforms to the vertical location of the current block)
SS	Location where it stops by Single Block signal
L	The center of the tool moves along the straight line.
C	The center of the tool moves along the arc.
R	Radius of the tool
○	Center of the tool

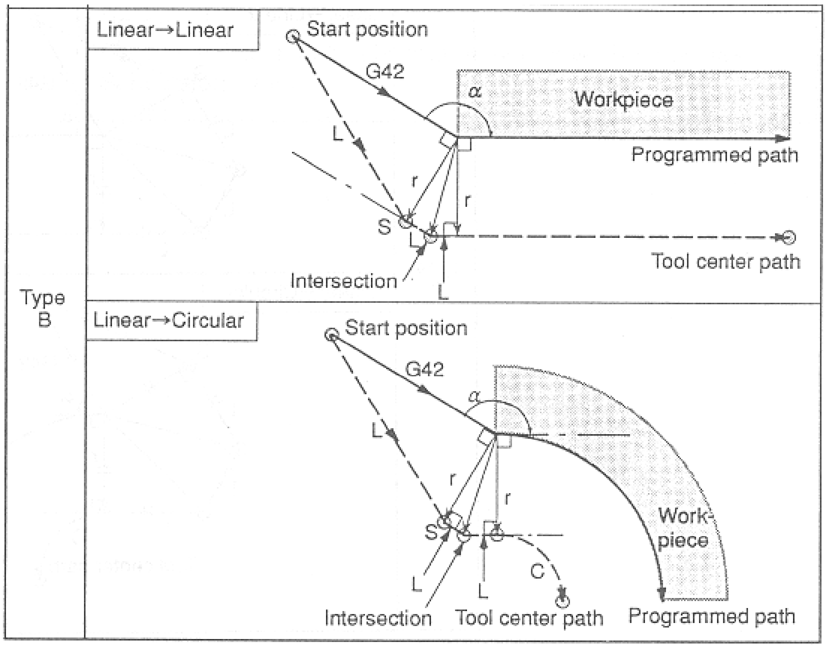
- Path of the tool starting right compensation (G42) for the corner having ofα >= 180°


Linear→Linear
Workpiece
α
Programmed path
r
G42
S
L
Tool center path
L
Start position
Linear→Circular
α
r
G42
Work-
piece
S
C
L
Start position
Tool center path
Programmed path

- Path of the tool starting right compensation (G42) for the corner having 90° <= α <= 180°

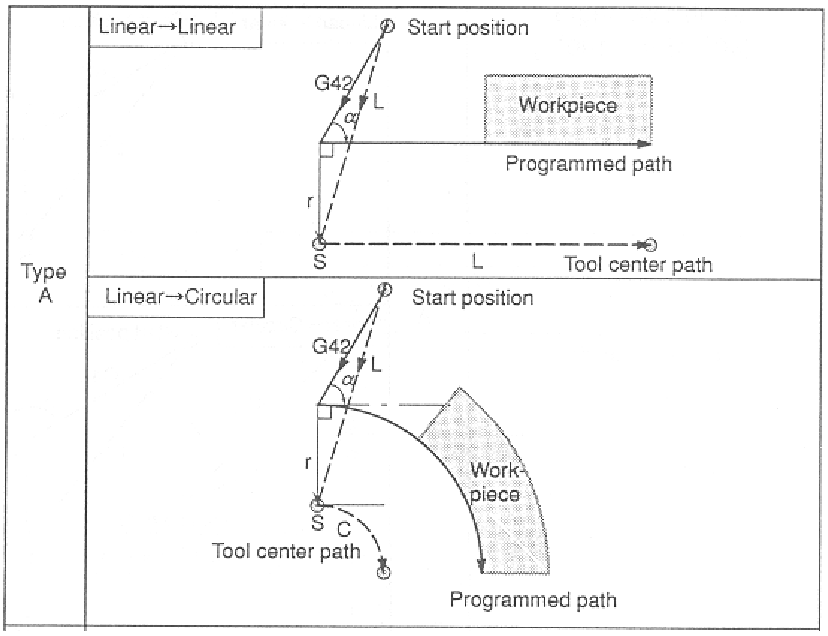
According to the parameter (PI128(#3128)), it is possible to change A type (direct) or B type (detour).

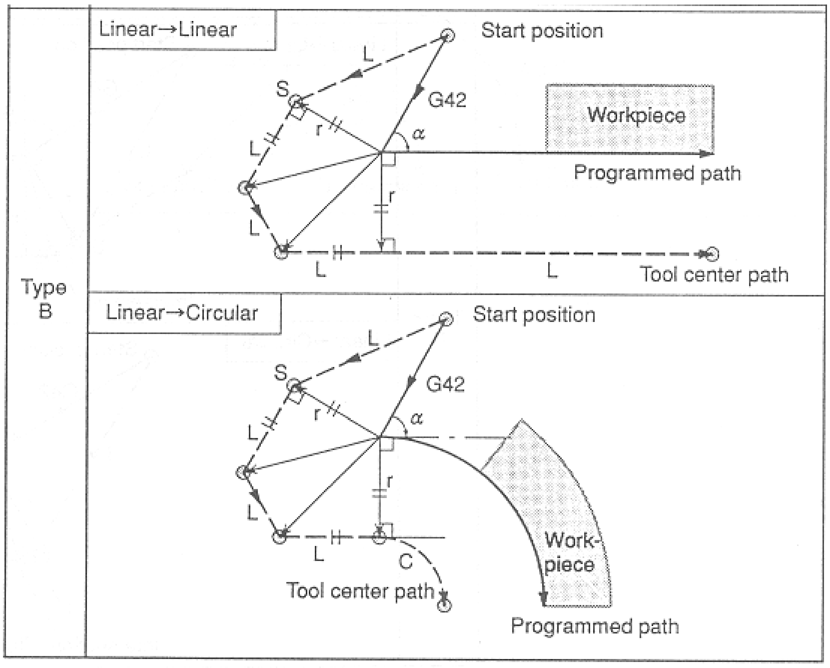



Type
B
Linear→Linear
Start position
G42
α
Workpiece
L
Programmed path
r
r
S
L
Tool center path
Intersection
L
Linear→Circular
Start position
G42
α
L
r
r
Work-
piece
S
C
L
L
Intersection
Tool center path
Programmed path

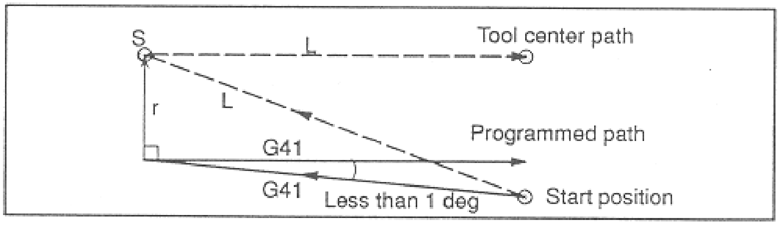
● Path of the tool starting right compensation (G42) for the corner having α < 90°

According to the parameter (PI128(#3128)), it is possible to change A type (direct) or B type (detour).

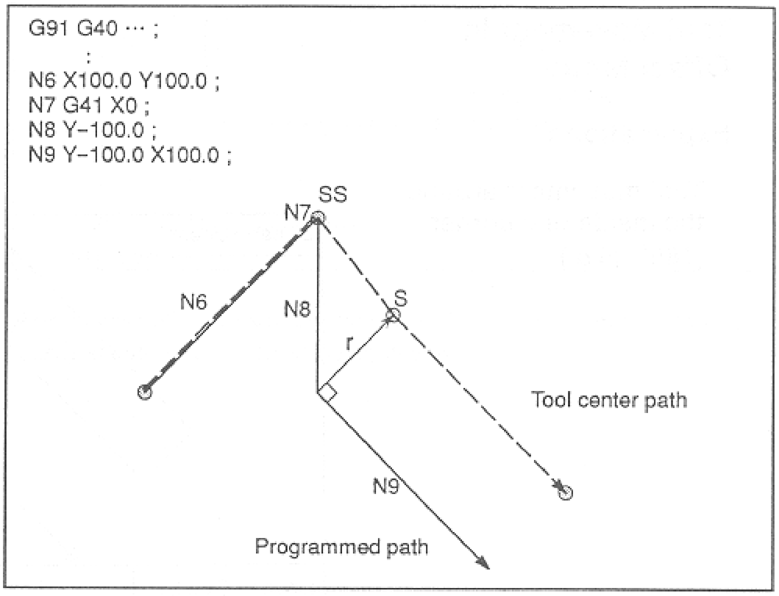



Type B
Linear→Linear
Start position
L
S
G42
r
α
L
L
r
Workpiece
Programmed path
L
L
Tool center path
Linear→Circular
Start position
L
S
G42
r
α
L
L
r
L
C
Work-
piece
Tool center path
Programmed path

- Path of the tool starting left compensation (G41) for the corner having 1° > α



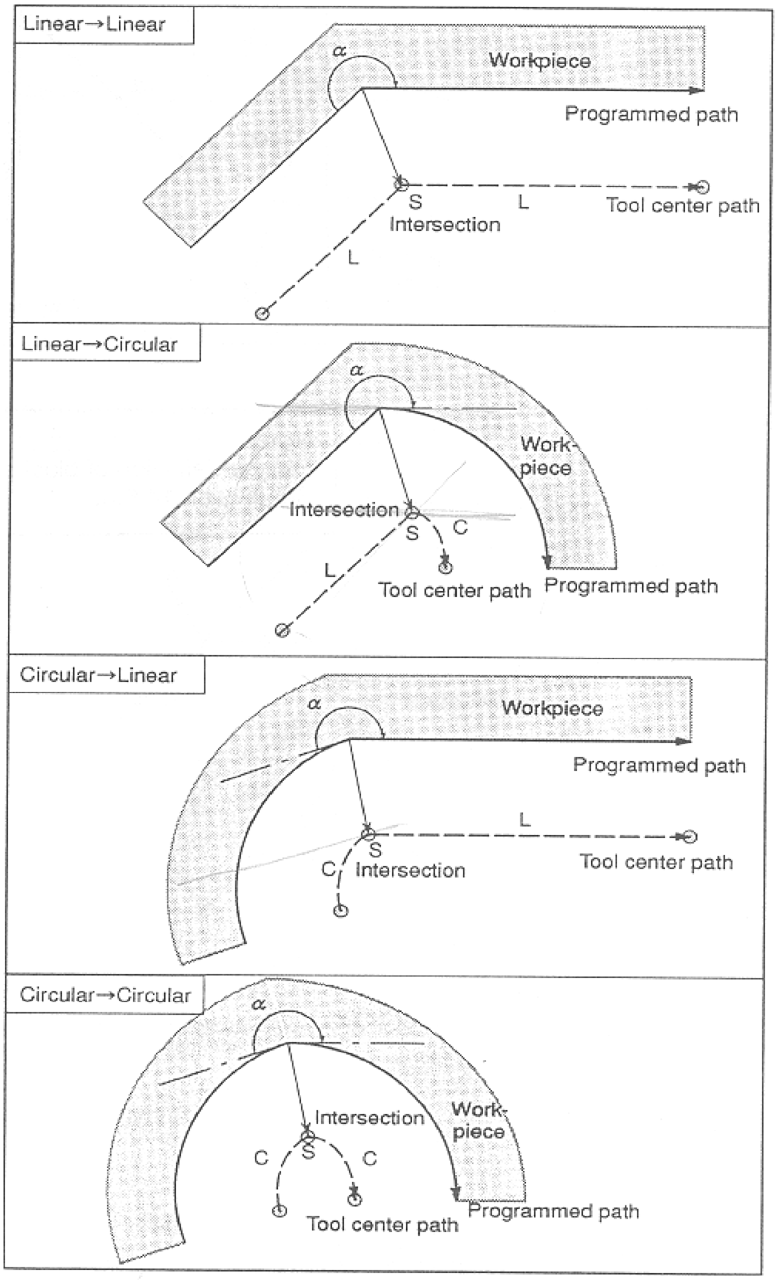
- Movement of the tool in case there is no axis feed data in the diameter compensation start block (Start-Up)



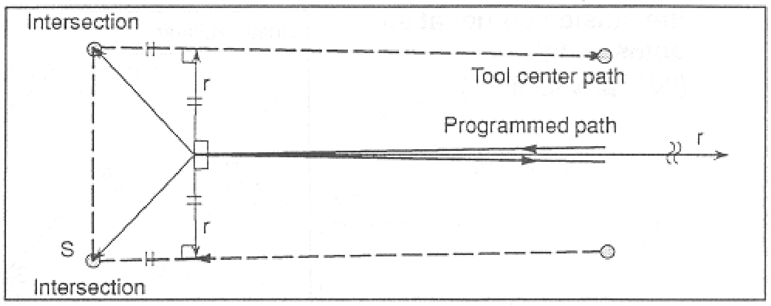
(2) **Offset Mode**

: Tool-centered movement mode until G40 appears after Start-Up mode

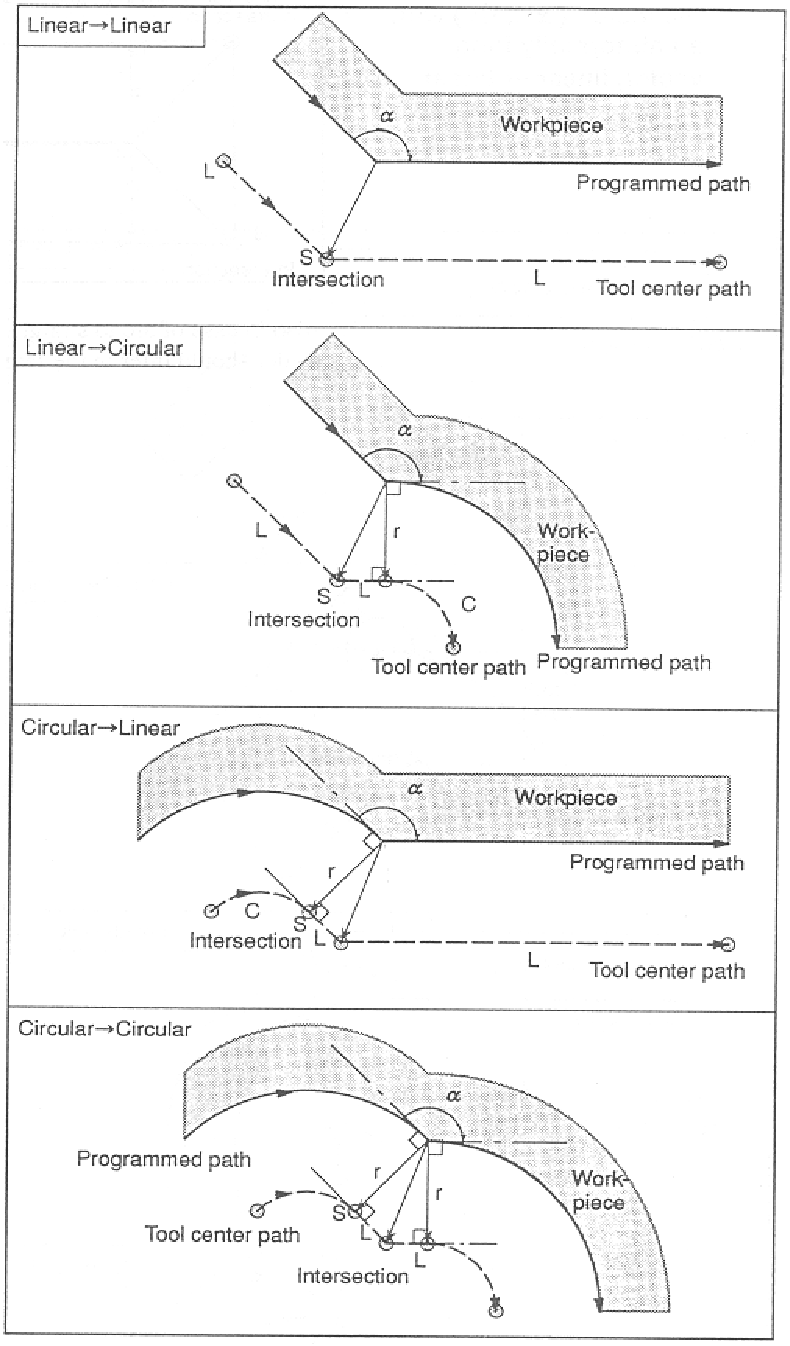
- Path of the internal tool for the corer having 180° <= α



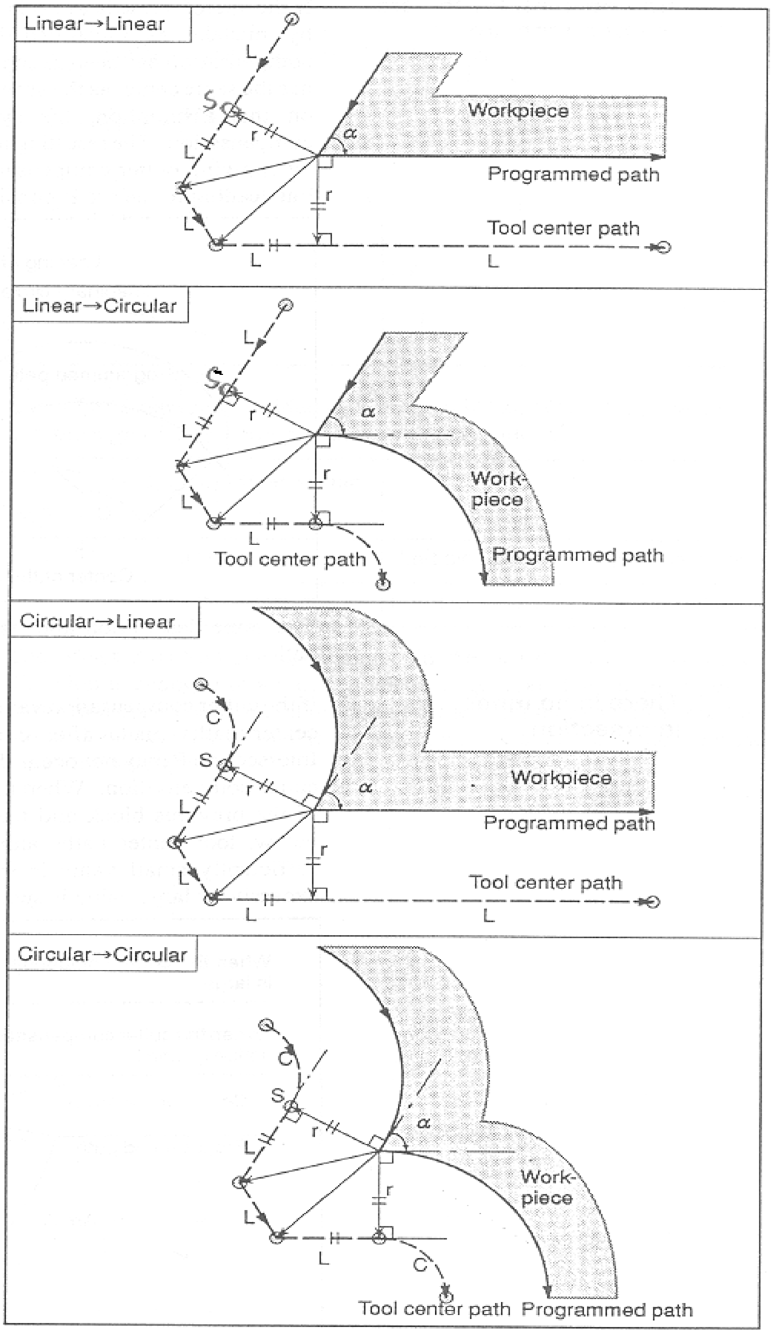
- Path of the tool for the section having less than 1 degree of radian between two straight lines



- Path of the external tool for the section having 90° <= α < 180°

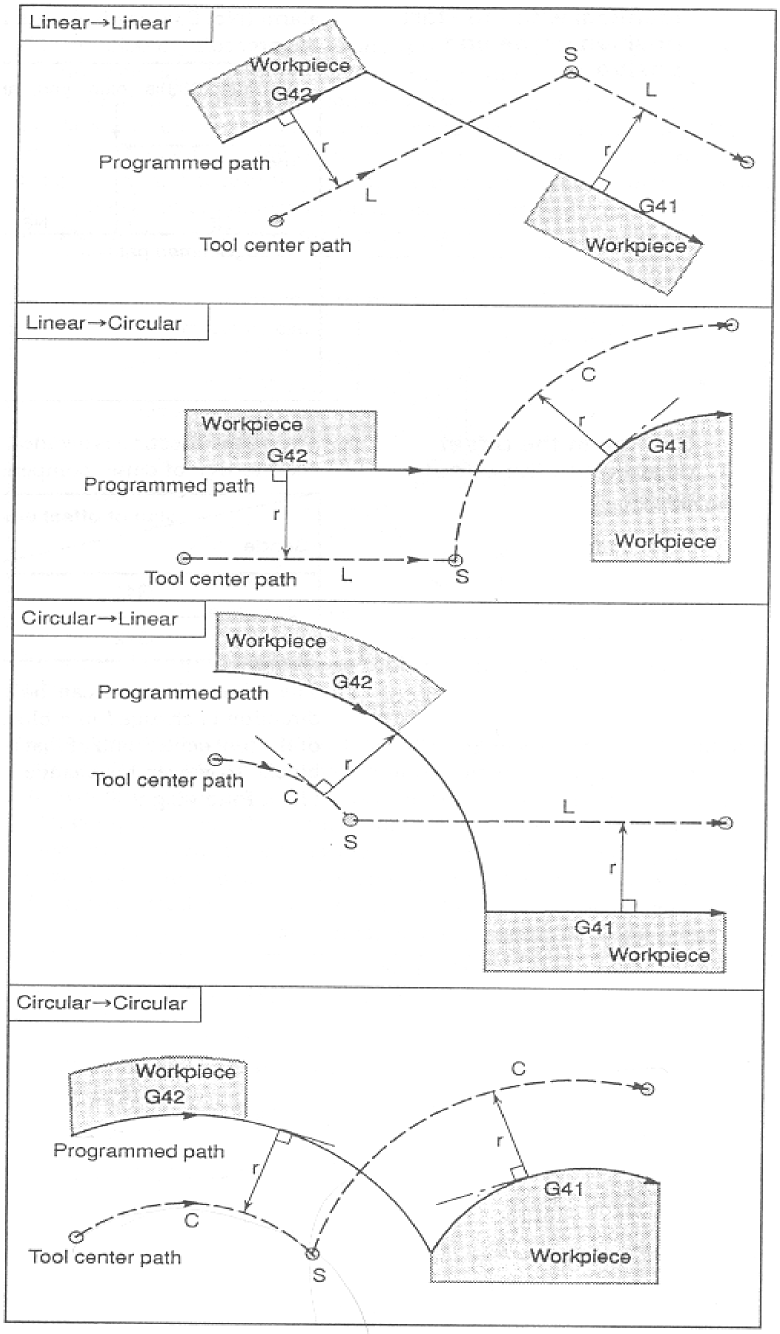


- Path of the external tool for the section having α < 90°

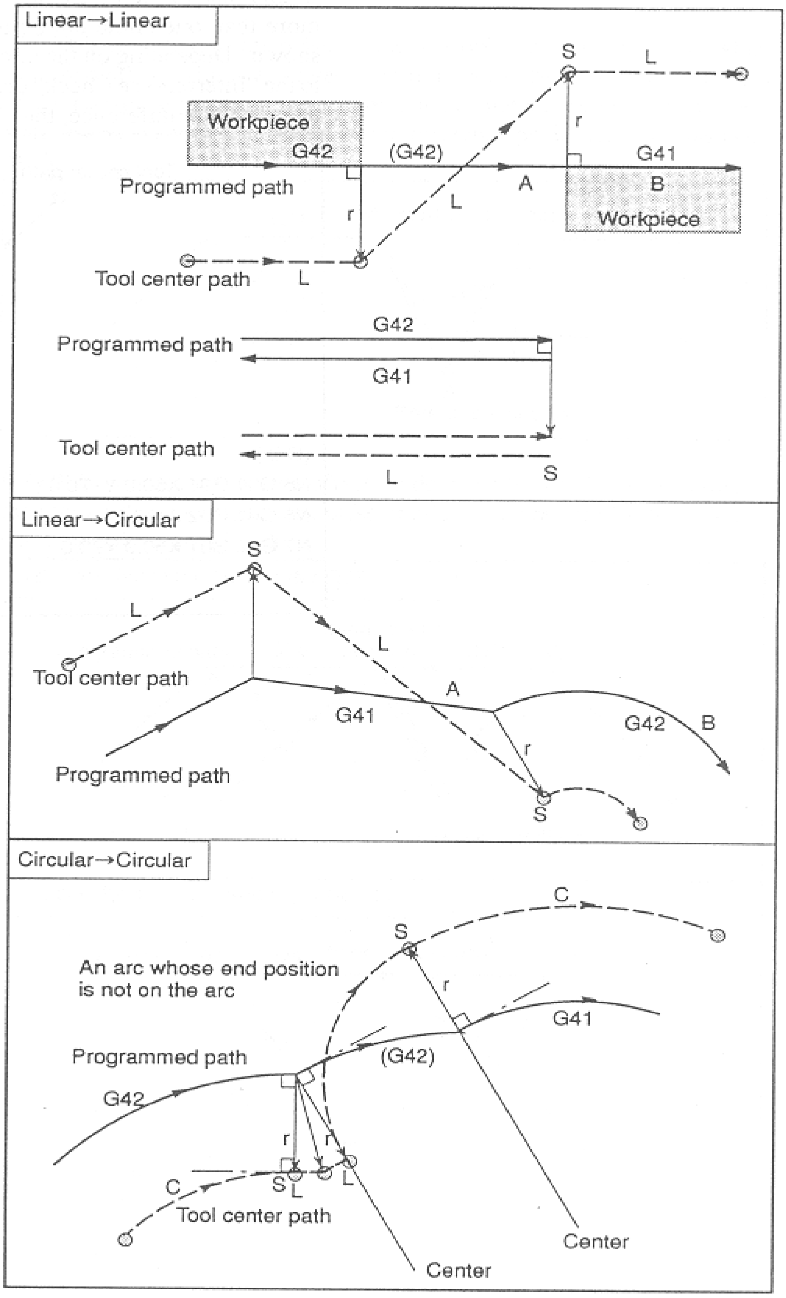


(3) **Path of the tool for the section where G41/G42 is changed during Offset mode**

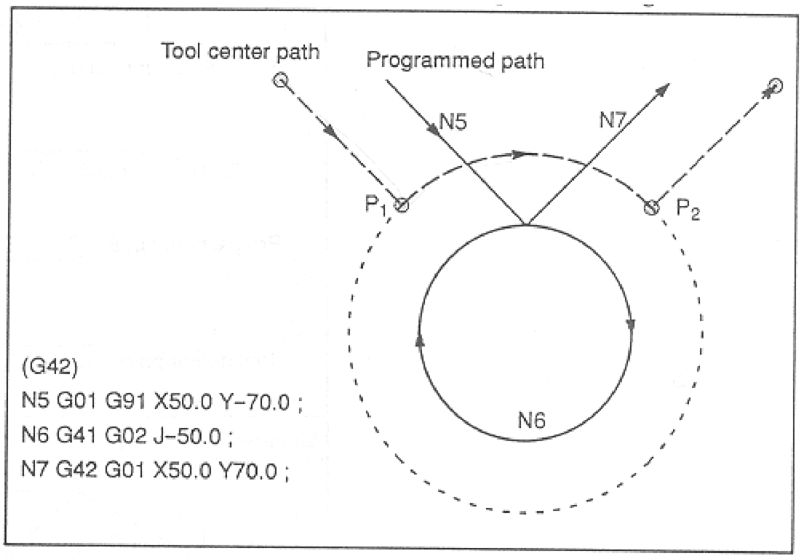
- Central path of the tool in case there is a node



- In case there is no node in the central path of the tool



- The path of tool center is longer than the length of arc.

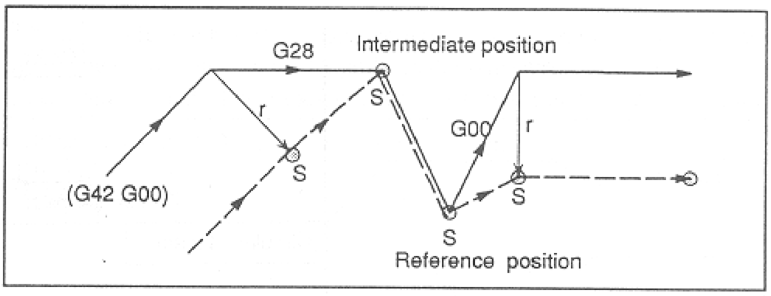


- In case tool diameter compensation is temporarily cancelled

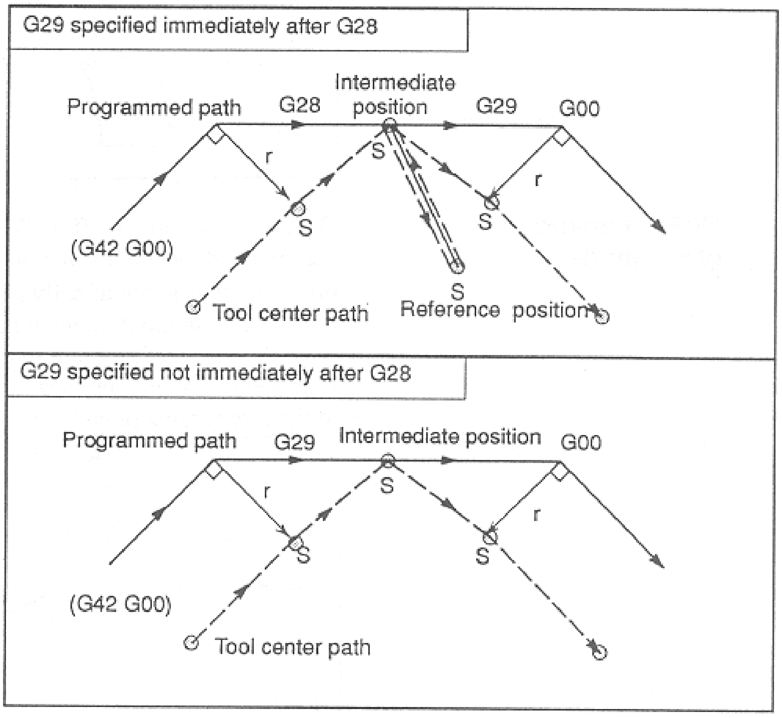
In case there is a command related with machine coordinate system like G28, G29, G53 during diameter compensation

In case there is a command related with coordinate setting like G92, G54~G59, G52 during diameter compensation

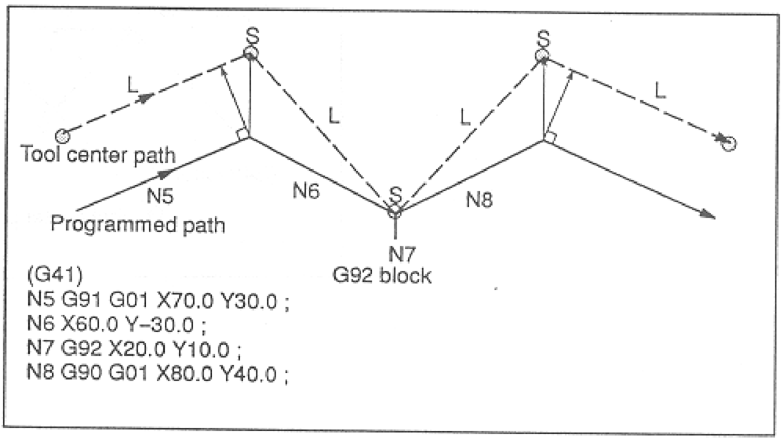
① In case there is a command of G28 during diameter compensation



② In case there is a command of G29 or G28 during diameter compensation

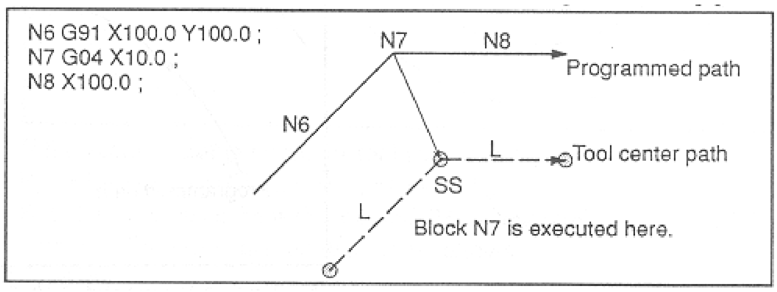


③ In case there is a command of G92 during diameter compensation



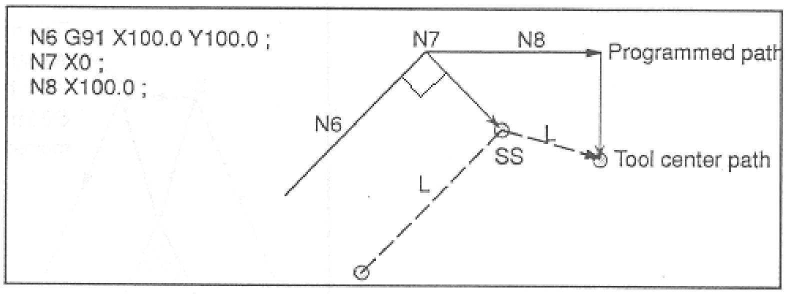
● Central path of the tool in case there is no block for axis feed during diameter compensation

① In case there is a block not related with the axis feed during diameter compensation



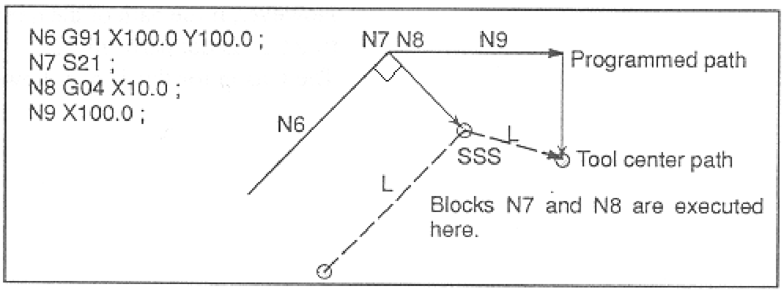
② In case there is a block for axis feed during diameter compensation, but the feed amount is 0

(Located vertically to N6 block, Single block stop)



③ In case there are two blocks not related with the axis feed during diameter compensation

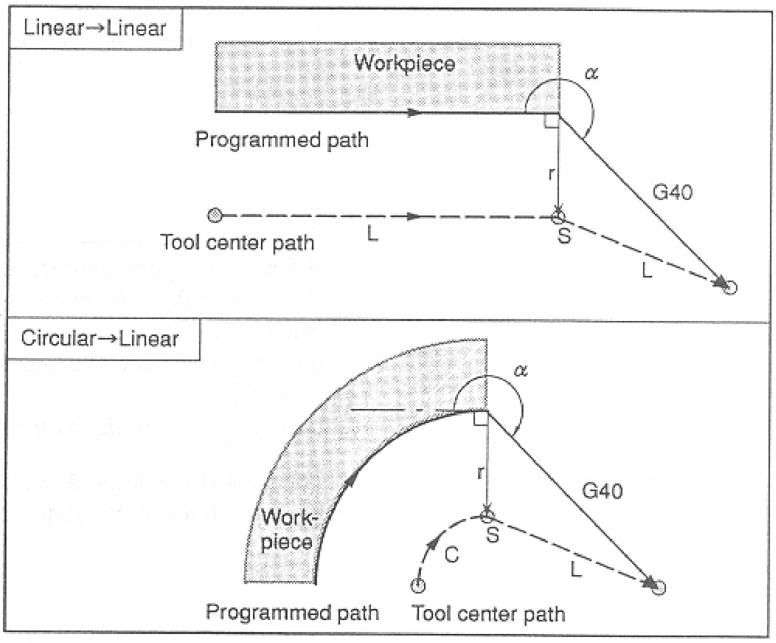
(Located vertically to N6 block, Single block stop)


N6 G91 X100.0 Y100.0 ;
N7 S21 ;
N8 G04 X10.0 ;
N9 X100.0 ;
N7 N8
N9
Programmed path
N6
SSS
L
L
Tool center path
Blocks N7 and N8 are executed here.

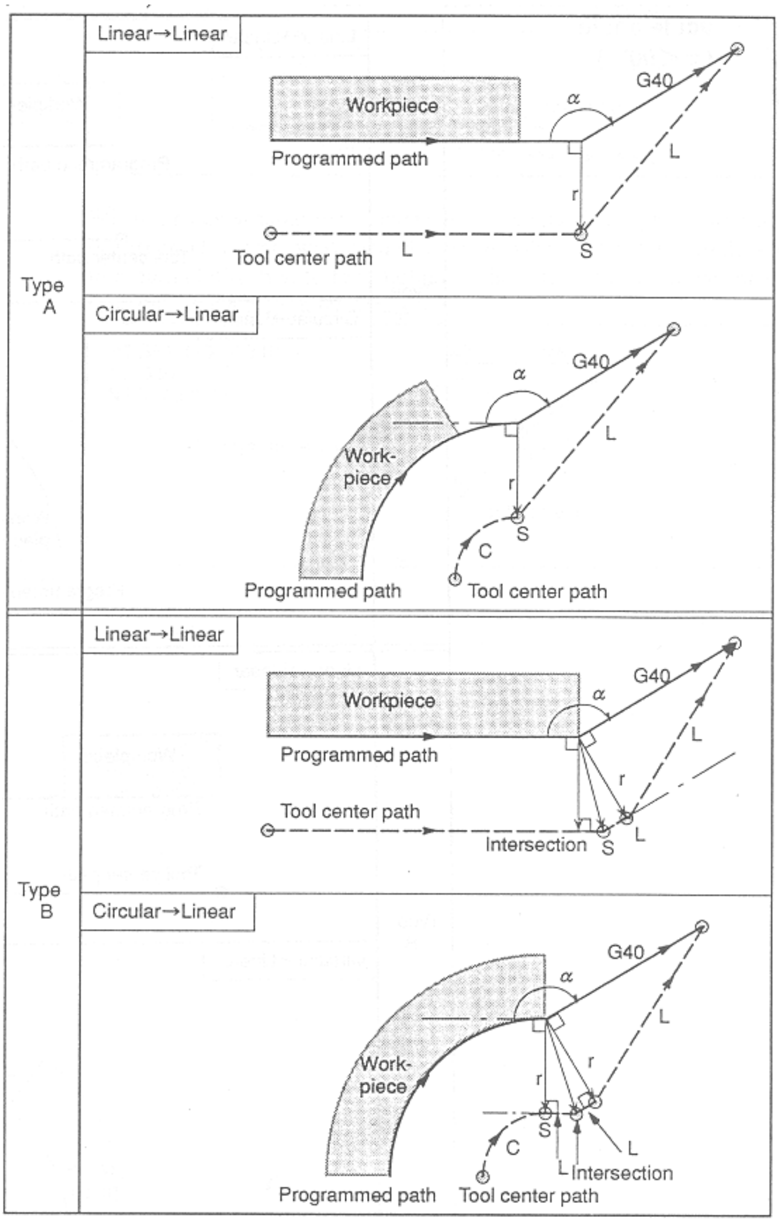
(4) Offset Cancel Mode

: In case the next block has G40 command or has a command to cancel offset, the block is called as Offset Cancel Mode.

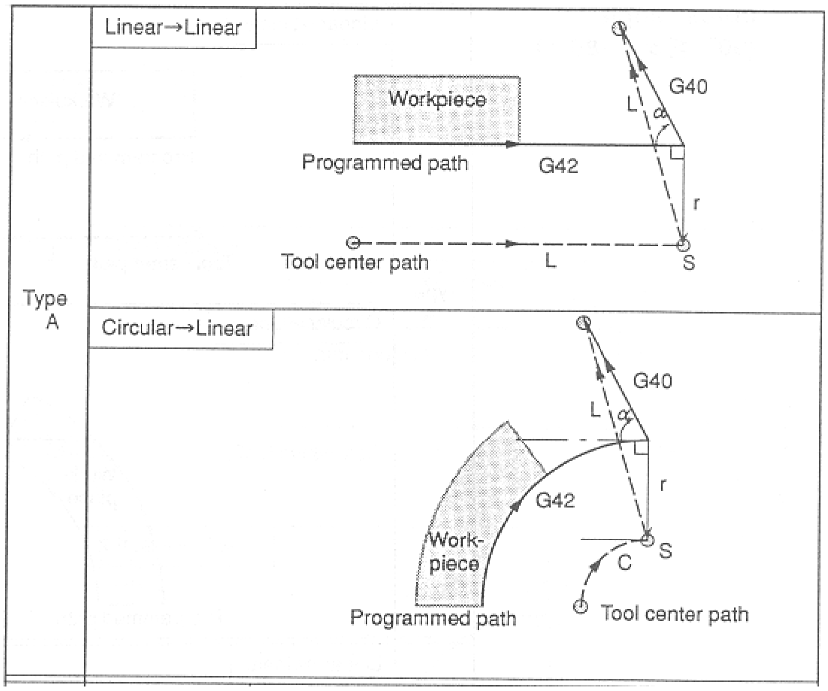
- Path of the internal tool for the corner having 180° <= α (G40)

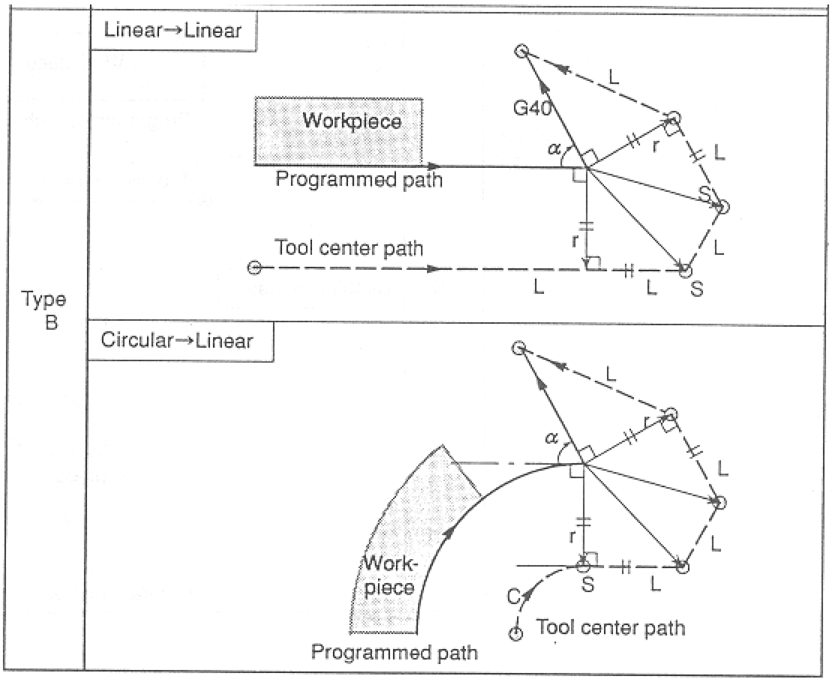


- Path of the external tool for the corner having 90° <= α < 180° (G40)

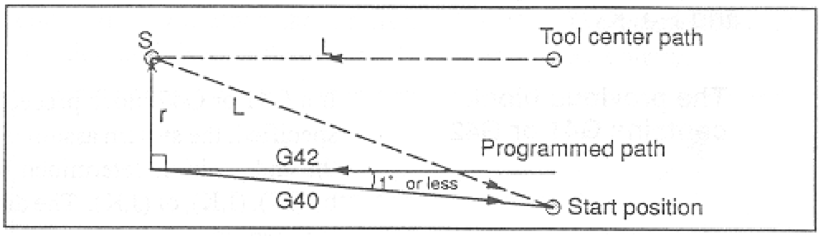


- Path of the external tool for the corner having $\alpha < 90°$ (G40)

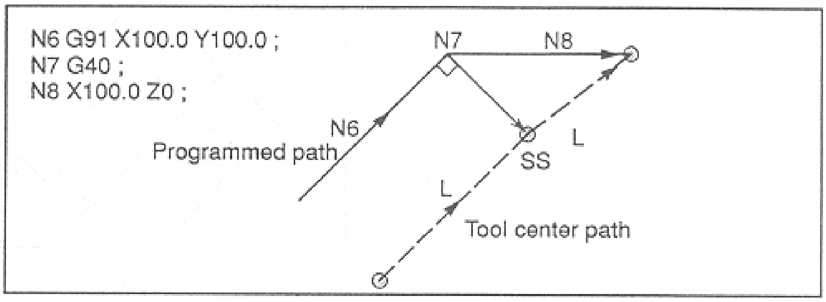



Type B
Linear→Linear
Workpiece
G40
α
r
L
S
Programmed path
Tool center path
Circular→Linear
Work-
piece
C
Programmed path
Tool center path

- Path of the tool for the section having radian of less than 1 degree between two straight lines (G40)



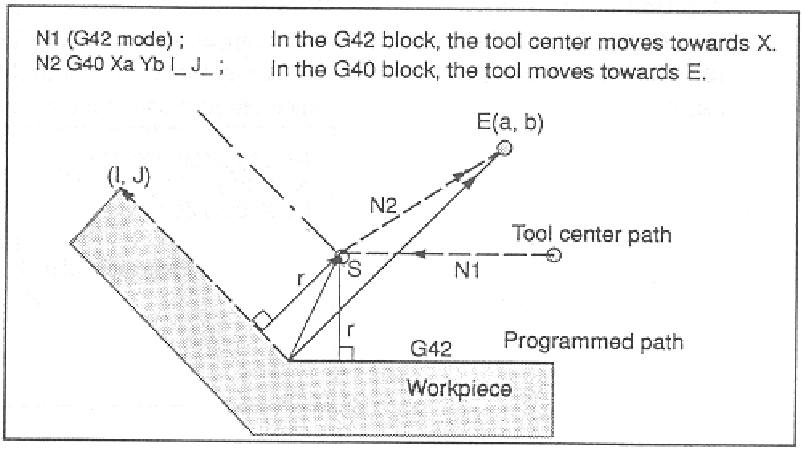
- In case there is no command of axis feed block (G40)



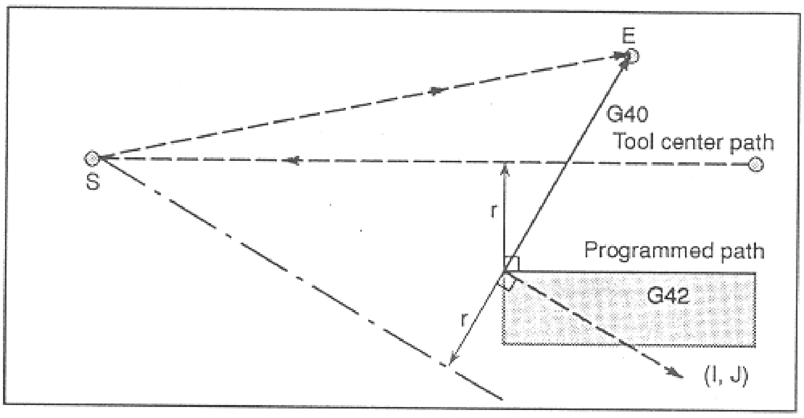
(5) Handling G40 block including I_, J_, K_

In case there is G40 command block including I_,J_,K_ during G41/G42 offset mode, you can calculate the end of the current block feed, thinking as if G40 block has virtual axis feed. Vector direction by (I,J), (I,K) or (J,K) is considered.

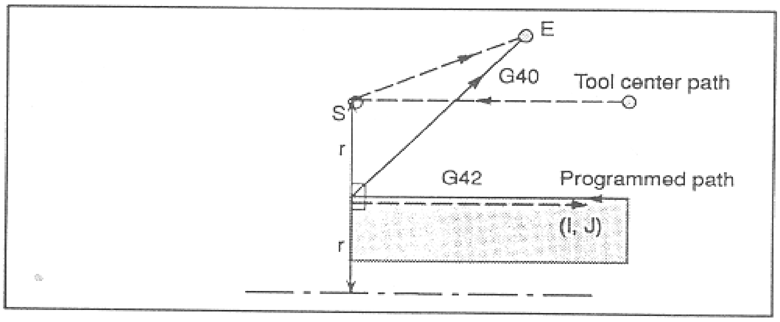
- Creation of tool path by use of (I,J)



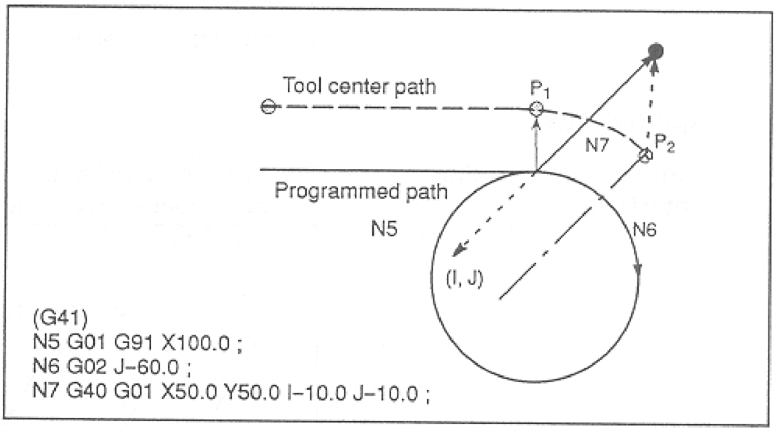
- In case there is a node



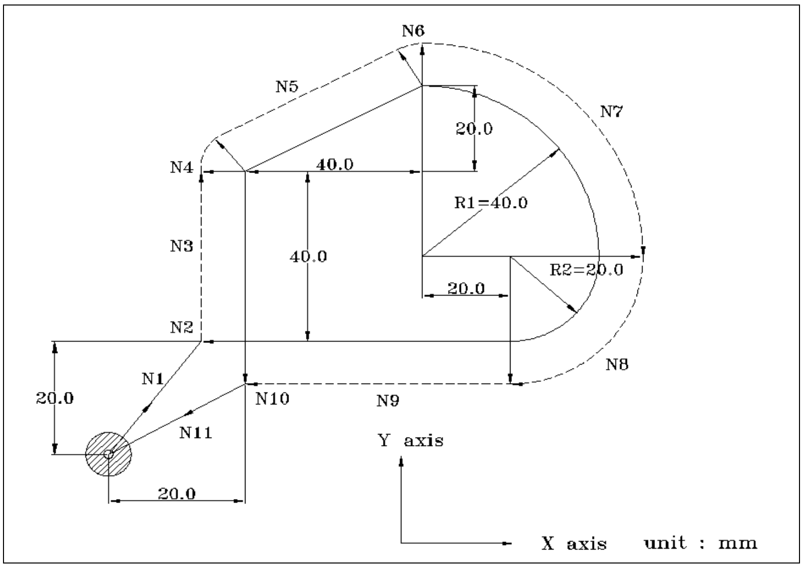
- In case there is no node



- In case the path of the tool center is longer than the length of arc, it moves to P2 point from P1 point.



ll Example of tool diameter compensation



N1 G91 G17 G00 G41 X20. Y20. D08 ; (D08 tool offset number)

N2 G01 Z ? 25. F100 ; (The value of tool radius is input on the relevant number)

N3 Y40. F250 ;

N4 G39 X40. Y20.; (Path of arc type compensation)

N5 X40 Y20. ;

N6 G39 X40. Y-20.; (Path of arc type compensation)

N7 G02 X40. Y? 40. R40.0 ;

N8 X ? 20. Y ? 20. R20 ;

N9 G01 X ? 60. ;

N10 G00 Z25. ;

N11 G40 X ? 20. Y ? 20.;

N12 M30 ;

13.3 Tool Offset (G45~G48)

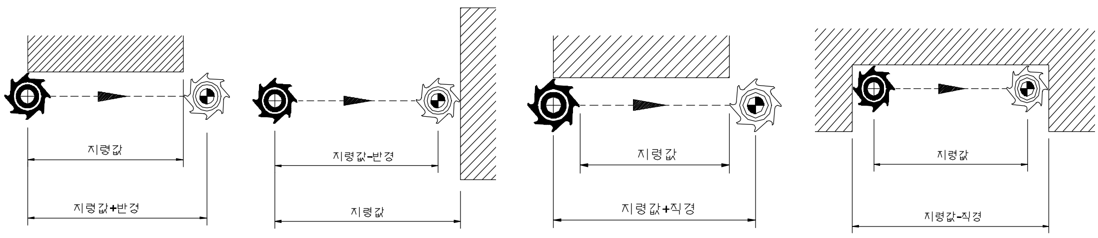
G45	[G01 / G02 / G03]	X _ Y _ D _
G46	[G02 / G03]	X _ Y _ D _
G47	[G02 / G03]	X _ Y _ D _
G48	[G02 / G03]	X _ Y _ D _

G45	Tool Offset Increase
G46	Tool Offset Decrease
G47	Tool Offset Double Increase
G48	Tool Offset Double Decrease

D	Too Radius Compensation Number
X, Y	Command for Coordinate of X and Y on G17 Plane

Main menu ⏩ F2 setting ⏩ F2 offset setting. You can input the tool radius by use of the number key on MDI panel after moving the cursor to the relevant number on the compensation screen. If you runs the program, using G45~G48, the designated moving distance of the axis increases, decreases, double increases or double decreases as much as the tool radius set with the offset memory. It is one-shot code.

G45 compensation G46 compensation G47 compensation G48 compensation



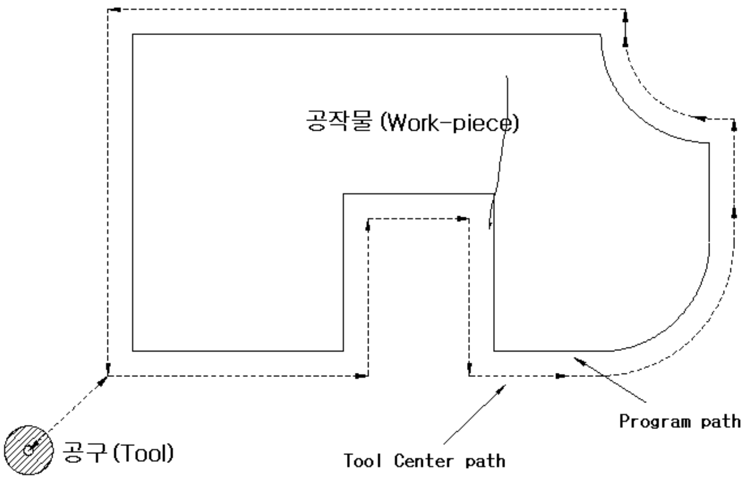
① In case that corresponding with the number of D code designated on the program [Offset Number], you use offset for circular interpolation G02 or G03 on the MDI panel, I or J can be given according to the setting with the parameter. And in case of 1/3 or 3/4 circle based on the designation, the offset can compensated by G45~G48 command. However, rotation of coordinate or parallel operation is not carried out. It is not used together with tool diameter compensation like G40, G41, G42.

② Ranges of offset data possible for setting

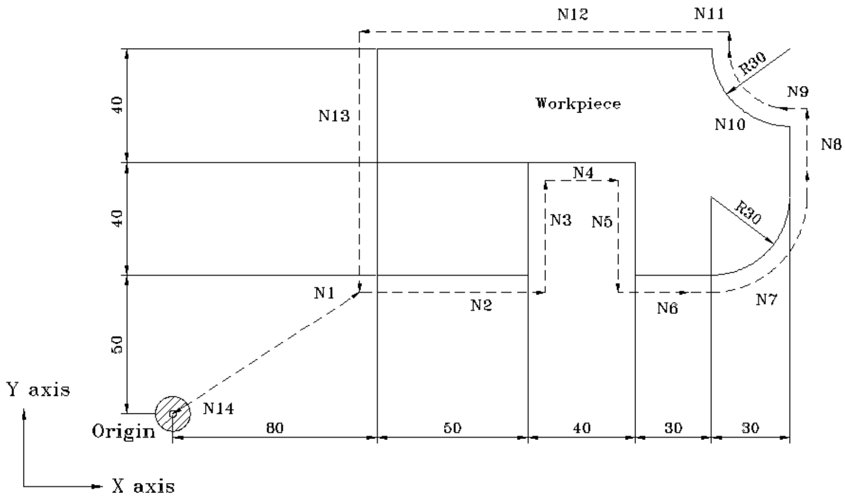
	Metric G21	Inch G20
Offset Value	0 ~ ± 999.999 mm	0 ~ ± 99.999 inch
	0 ~ ± 999.999 deg	0 ~ ± 999.999 deg

③ Tool offset is effective in the sub-axis. Even under the absolute command of G90, it is increased or decreased in the moving direction from the terminal of the previous block to the location designated by a certain block G45~G48.

Ⅱ Example of tool offset



Ⅱ Example of tool offset



N1 G91 G46 G00 X80 Y50 D01 ;
N2 G47 G01 X50 F120 ;
N3 Y40;
N4 G48 X40 ;
N5 Y-40 ;
N6 G45 X30 ;
N7 G45 G03 X30 Y30 J30 ;
N8 G45 G01 Y20 ;
N9 G46 X0 ; : The amount of decrease is 0 and it moves by the set offset data 10 in the direction of -X .
N10 G46 G02 X-30 Y30 J30 ;
N11 G45 G01 Y0 ; : The amount of increase is 0 and it moves by the set offset data 10 in the direction of +Y .
N12 G47 X-120 ;
N13 G47 Y-80 ;
N14 G46 G00 X-80 Y-50 ;

14 Custom Macro

If you use macro program, you can do several machining works having similar shapes by changing some numbers as shown on the below: For repetition of the same action, sub-program is effective, but macro program can use variable, calculation command, conditional branch, etc. so it is widely used.

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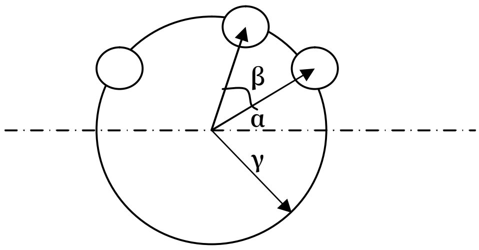
Macro program can be paged with a simple command like sub-program.

Ex) Macro program for boring on a cylinder



Required data : - Radius of the cylinder where a hole is located (γ)
- Angle where a hole starts (α)
- Angle of arc where a hole will be arranged (β)
- Number of holes

G65 P_p R_ γ _A_ α _B_ β K_k;

p : Macro program number
γ : Radius
α : Start angle
β : Angle between circle
k : Number of circles

14.1 Custom Macro Command

14.1.1 Simple paging (G65)

G65 P _ L _ <Designate factors>

G65	Macro paging
P _	Program Number
L _	Frequency of repetition

The biggest advantages of custom macro are:

- Variables can be used,
- Operation among variables is possible,
- It is possible to set the actual value to a variable through macro,
- Variables are classified into local variables, common variables and system variables according to the variable number,
- Variables always have the values in the form of real number, and
- The program number should be 4-digits.

(1) How to designate factors I

- It is possible to assign to all the addresses except for G, L, N. P, and O. .(A_ B_ C_ ··· Z_)
- In general, the order of designating alphabet doesn ’ t matter, but I, J, and K should be designated in order.

B___ A___ D___ I___ K___ : Normal

B___ A___ D___J___ I___ : Abnormal

- Address based on the method 1 of assigning factors (except for G, L, N, P, O)

어드레스	커스텀 매크로 본체의 변수
A	#1
B	#2
C	#3
D	#7
E	#8
F	#9
H	#11
I	#4
J	#5
K	#6
M	#13
Q	#17
R	#18
S	#19
T	#20
U	#21
V	#22
W	#23
X	#24
Y	#25
Z	#26

(2) How to designate factors II

A_ B_ C_ I_ J_ K_ I_ J_ K_ …

- It is possible to designate factors with address A, B, C. It is possible to designate the factor consisting of a set of address I, K, J up to maximum 10 sets.
- In case several variables are designated with the same address, the order of decision should be designated.
- It is possible to omit address that doesn ’ t need designation.

- The address that is designated by the method II and the numbers of the variables that are used in the macro correspond as follows:

어드레스	커스텀 매크로 본체의 변수
A	#1
B	#2
C	#3
I1	#4
J1	#5
K1	#6
I2	#7
J2	#8
K2	#9
I3	#10
J3	#11
K3	#12
I4	#13
J4	#14
K4	#15
I5	#16
J5	#17
K5	#18
I6	#19
J6	#20
K6	#21
I7	#22
J7	#23
K7	#24
I8	#25
J8	#26
K8	#27
I9	#28
J9	#29
K9	#30
I10	#31
J10	#32
K10	#33

1~10 next to I, J, K means the order of the designated sets, which are not commanded in actuality.

(3) Mixture of type I and II

Even though there are factors of the designation method I and II in the block factor of G65, an alarm doesn't happen. If I factor and II factor corresponding to the same factor are designated by both methods, the designation by type I is effective.

```
예)
    G65   A1.0   B2.0   I3.0   I 4.0   D5.0   P1000;

변수 :   #1   =   1.0 (A)
         #2   =   2.0(B)
         #3
         #4   =   3.0(I1)
         #5
         #6
         #7   =   5.0(D)
```

14.1.2 Macro modal paging (G66 /G67)

G66 P _ L _ <Designate factor>

G67

G66	Macro modal paging
G67	Macro modal paging cancel
P _	Program number
L _	Frequency of repetition

Factor designation is same as simple paging (G65).

Note:

① In the G66 block, G66 should be commanded prior to all the factors.
② In G66 block, macro program paging should not be performed.
③ Macro modal paging (G66) is possible only in the main program.
④ G66 and G67 should be always in the same program.
⑤ During macro paging, execute the block having the command of moving and then, page the designated macro.

Example

Drill Cycle : Execute drill cycle at each point of positioning

```
G66 P9082 R(R Point) Z(Z Point) X(dwell);
X __ ;
… ;
G67;
O9082(Macro program)
G00 Z#18;
G01 Z#26;
G04 X#24;
G00 Z-[ROUND[#18] + ROUND[#26]];
M99;
```

14.1.3 Macro paging by G code

(1) It is possible to set G code that pages macro program 9010.nc ~ 9019.nc with parameter PI 85 ~ PI 94 (#3085~3094).

PI 0085	0.0		9010.nc Calling G Code
PI 0086	0.0		9011.nc Calling G Code
PI 0087	0.0		9012.nc Calling G Code
PI 0088	0.0		9013.nc Calling G Code
PI 0089	0.0		9014.nc Calling G Code
PI 0090	0.0		9015.nc Calling G Code
PI 0091	0.0		9016.nc Calling G Code
PI 0092	0.0		9017.nc Calling G Code
PI 0093	0.0		9018.nc Calling G Code
PI 0094	0.0		9019.nc Calling G Code

Instead of N__ G65 P___ < Factor designation>;, N__ GXX < Factor designation >; can give a command. (G code XX and the relevant macro program are set with parameter, PI 85 ~ PI 94 (#3085 ~ 3094)

(2) It is possible to use maximum 10 G codes out of G01~G255 (except for G00) for paging macro program (9010~9019). G code is available up to one decimal place

Notes:

① It is impossible to use G65, G66, G67.
② it is impossible to use in the paged macro program.

14.1.4 Macro paging by M code

(1) M code that pages macro program 9020.nc ~ 9029.nc can be set with parameter PI 95 ~ PI 104 (#3095~3104).

PI 0095	6		9020.nc Calling M Code
PI 0096	10		9021.nc Calling M Code
PI 0097	55		9022.nc Calling M Code
PI 0098	56		9023.nc Calling M Code
PI 0099	57		9024.nc Calling M Code
PI 0100	58		9025.nc Calling M Code
PI 0101	59		9026.nc Calling M Code
PI 0102	0		9027.nc Calling M Code
PI 0103	0		9028.nc Calling M Code
PI 0104	0		9029.nc Calling M Code

Instead of N__ G65 P____ < Factor designation>;, N__ Mxx < Factor designation >; can give a command. (M code XX and the relevant macro program are set with parameter, PI 95 ~ PI 104(#3095~3104)

(2) There is no signal of M FIN and M code command (same as M98 and M99).

(3) It is possible to use 10 M codes out of M01~M97.
(Impossible to use M02 and M30)

- In case this Mxx is commanded during sub-program paging by G code, M code or T code, it is handled as general M code.
- These M codes are different from general M codes, which should be commanded at the head (or next to N_) of a block.

14.1.5 Sub-program paging by M code

(1) Possible to page sub-program by M code set with the parameter (PI 95~104)

N__ G__ X__ Y__ Mxx; instead of N__ G__ X__ Y__ M98 P____;

> **Notes:**
> In case this Mxx is commanded during sub-program paging by G code, M code or T code, it is handled as general M code.

14.1.6 Sub-program paging by T code

(1) Possible to page sub-program by T code set with the parameter, PI 105(#3105)

N__ G__ X__ Y__ Txx; instead of N__ G__ X__ Y__ M98 P9000;

14.1.7 Difference between M98 and G65

(1) G65 can make factor designation.

(2) M98 carries out other M code, P or L in the same block, and then, is branched into sub-program. G65 is used as a branch function.

(3) In case there is O, N, P, L in the block, M98 stops by single stop. G65 doesn't stop.

(4) G65 can change the level of local variable, but M98 can't. That is, event though #i is defined before G65 is command, #i in the program paged by G65 is a different variable from the previous #i. In case #i is defined before M98, #i is the same variable even though sub-program is paged by M98.

(5) It is possible to page G65 four times, including G66. With M98, it is possible to page up to 8 stages including G65 and G66.

14.1.8 Multiple paging (G66)

G66 P _ L _ <Factor designation>

(1) Multiple paging

- It is possible to page macro program during macro program same as sub-program. (modal paging is possible only in the main).
- Possible to page up to maximum 4 stages

(2) Multiple modal paging

- In the multiple modal paging, <u>the designated macro is paged whenever motion is executed</u>. In case the macro, modal, is designated multiple, <u>the next macro is paged whenever motion of the first macro is executed.</u> Macro is paged in order as per designated.
- In the program paged by macro modal, macro modal paging (G66) is not supported.

Ex) (Main program)

....

G66 P9100; (Setting for macro modal paging. Paging is made after the next block for axis feed)

Z1000;	(1-1)
G66 P9200	
Z15000;	(1-2)
G67;	P9200 cancelled
G67;	P9100 cancelled
Z-25000;	(1-3)
O9100;	
X5000;	(2-1)
M99;	
O9200;	
Z6000;	(3-1)
Z7000;	(3-2)
M99;	

(3) Custom macro level and local variable

When a macro is paged by G65, G66 or G code (which pages macro program), the macro level increases by one (level 0 ~ level 4). As a result, the level of local variable increases by 1 too.

Main program Macro level 1 Macro level 2 Macro level 3 Macro level 4
<Figure>< Figure >< Figure >< Figure >< Figure >

Local variable
(Level 0) (Level 1) (Level 2) (Level 3) (Level 4)
< Figure >< Figure >< Figure >< Figure >< Figure >

- There are local variables of #1 ~ #33 in the main program (level 0).
- If macro (level 1) is paged by G65, the local variable for the main program is stored and the new local variable #1 ~ #33(level 1) for the macro program of level 1 is prepared.
- Whenever macro is paged (level 2, 3, 4), the local variables (level 1, 2, 3) are stored and the new local variables of the next level is prepared.
- Returned to each macro by M99, the local variables (level 0, 1, 2, 3) stored at level 2 and 3 are restored with the values preserved.

14.2 Making out the custom macro main body

14.2.1 Type of macro program main body

- It is the same as sub-program.
- It is possible to edit freely up to O0001 - O8999
- It is possible to delete/edit up to O9000 - O9999, depending on the setting with parameter PA 2.
- It is possible to use variables and give the commands of operation and control in the custom macro program.
- Designation of real values corresponding with the variables is made by macro paging command. (G65 A_ B_)

14.2.2 How to use variables

- Variables give flexibility and universality to the macro program by designating macro as variables instead of giving the values directly, and giving the variable values whenever paging a macro.
- It is possible to use several variables and the variables are identified by their numbers.

(1) Expression of variable

#i (i= 1, 2, 3, ···)

#[Expression]

(2) Use of variables

- The numbers following the address can be substituted with variables.

= Possible to program with address # 1, address-#1

Ex) F#103 ----- Same as F15 at the time of #103= 100

Z-#110 ------ Same as Z-250 at the time of #110 = 250

- How to replace the variable number with variable

In case of #100 = 105 and #105=-500, "##100" is not used and instead, "#[#100]" is used.

- Address /, :, O, N are not used for variable.

n(n= 1, 2, .. 9) of Optional Block Skip /n can ’ t be used as variable.

- Any values that exceed the designated maximum value can ’ t be given to address.

In case of #140 = 1000, G#140 is over the designated maximum value.

- In [Expression], all the variables are operated in the type of double.

(3) Undecided variable

- The value of the undecided value is 0.

<Table >

<Table >

- Expression and setting of variable formula

It is possible to set variable value at MDI mode.

14.2.3 Kinds of variables

Variables are classified into local variable, common variable and system variable, depending on the variable number.

Item	No.	Meaning	Feature
Local Variable	#1 ~ #33	Used as local in the macro. Local variable #i used in the macro that is paged at a certain time and #i used in the macro that is paged at any other time than the above are different.	Local variable is used for exchange of factors (/, O, N not allowed) Local variable is 0 in the initial status.
	#34 ~#99	Local variable	“
Common Variable	#100 ~ #199 #200 ~ #699	Common variable #i in the main program and #i in the paged sub (macro)-program are common. Common variable #i as operation result can be used in other macros(#[#i]). User can freely use this variable.	#100 ~ #199 : Cleared 0 with power OFF/ON. #200 ~ #699 : Not cleared even with power OFF

System Variable	#1600~#1663 : Lathe tool thread end type #1664~#1727 : Lathe tool thread R #1728~#1791 : Lathe tool X axis offset #1792~#1855 : Lathe tool Y axis offset #1856~#1919 : Lathe tool Z axis offset #2112~#2239 : Tool diameter compensation offset #2240~#2367 : Tool diameter compensation offset #2377~#2385 : G54 offset (X, Y, Z...) #2386~#2394 : G55 offset (X, Y, Z...) #2395~#2403 : G56 offset (X, Y, Z...) #2404~#2412 : G57 offset (X, Y, Z...) #2413~#2421 : G58 offset (X, Y, Z...) #2422~#2430 : G59 offset (X, Y, Z...) #6718 ~ : 32 G code modals #4379 ~ #4382 :Servo tracking error #7000 ~#7034: UI (G115~G118) #7500 ~#7534: UO (F105~F108) #8000 ~#8031 : Second reference position by axis #8032 ~#8063 : Third reference position by axis #8064 ~#8095 : Fourth reference position by axis #4018 : TPG related macro 관련 매크로 (Refer to the back for the details 뒷장에 설명 있음)	Its use in the system is fixed. Macro variable connected with the parameter	

#4018 : Macro variable related with TPG mode	Macro variable that informs whether it is under TPG or not is added, which is not to handle macro program while TPG is being done. (TPG in case #4018 is 1, and machining mode in case #4018 is 0.) Example of use) It is used to omit if it is TPG mode in case of interface between PLC and F map in order to select a needle at QT (9010.nc)

- Classification of system variable System

(1) System variable for input of interface signals (32bit) : #7000 ~ #7031 , #7032 ~#7035

- Input Signal

System Variable	Point	Interface input signal	G map
#7000	1	2^{0} UI 0 00	**G115.0**
#7001	1	2^{1} UI 0 01	**G115.1**
#7002	1	2^{2} UI 0 02	**G115.2**
#7003	1	2^{3} UI 0 03	**G115.3**
#7004	1	2^{4} UI 0 04	**G115.4**
#7005	1	2^{5} UI 0 05	**G115.5**
#7006	1	2^{6} UI 0 06	**G115.6**
#7007	1	2^{7} UI 0 07	**G115.7**
#7008	1	2^{8} UI 0 08	**G115.8**
#7009	1	2^{9} UI 0 09	**G115.9**
#7010	1	2^{10} UI0 10	**G115.10**
#7011	1	2^{11} UI 0 11	**G115.11**
#7012	1	2^{12} UI 0 12	**G115.12**
#7013	1	2^{13} UI 0 13	**G115.13**
#7014	1	2^{14} UI 0 14	**G115.14**
#7015	1	2^{15} UI 0 15	**G115.15**
#7016	1	2^{16} UI 0 16	**G115.16**
#7017	1	2^{17} UI 0 17	**G115.17**
#7018	1	2^{18} UI 0 18	**G115.18**
#7019	1	2^{19} UI 0 19	**G115.19**
#7020	1	2^{20} UI 0 20	**G115.20**
#7021	1	2^{21} UI 0 21	**G115.21**
#7022	1	2^{22} UI 0 22	**G115.22**
#7023	1	2^{23} UI 0 23	**G115.23**
#7024	1	2^{24} UI 0 24	**G115.24**
#7025	1	2^{25} UI 0 25	**G115.25**
#7026	1	2^{26} UI 0 26	**G115.26**
#7027	1	2^{27} UI 0 27	**G115.27**
#7028	1	2^{28} UI 0 28	**G115.28**
#7029	1	2^{29} UI 0 29	**G115.29**
#7030	1	2^{30} UI 0 30	**G115.30**
#7031	1	2^{31} UI 0 31	**G115.31**
#7032	32	UI0 0 ? UI0 31	**G115**
#7033	32	UI1 0 ? UI1 31	**G116**
#7034	32	UI2 0 ? UI2 31	**G117**
#7035	32	UI3 0 ? UI3 31	**G118**

$$\#7032 = \sum_{i=0}^{31} \#[7000 + i] * 2^{i}$$

Value of variable	Input Signal
1	Contact closed(HIGH)
0	Contact open(LOW)

(2) System variable for output of interface signal (32bit) : #7500 #7531, #7532 ~ #7535

- Output Signal

System Variable	Point	Interface input signal	F map
#7500	1	2^{0} UO 000	F105.0
#7501	1	2^{1} UO 001	F105.1
#7502	1	2^{2} UO 002	F105.2
#7503	1	2^{3} UO 003	F105.3
#7504	1	2^{4} UO 004	F105.4
#7505	1	2^{5} UO 005	F105.5
#7506	1	2^{6} UO 006	F105.6
#7507	1	2^{7} UO 007	F105.7
#7508	1	2^{8} UO 008	F105.8
#7509	1	2^{9} UO 009	F105.9
#7510	1	2^{10} UO 010	F105.10
#7511	1	2^{11} UO 011	F105.11
#7512	1	2^{12} UO 012	F105.12
#7513	1	2^{13} UO 013	F105.13
#7514	1	2^{14} UO 014	F105.14
#7515	1	2^{15} UO 015	F105.15
#7516	1	2^{16} UO 016	F105.16
#7517	1	2^{17} UO 017	F105.17
#7518	1	2^{18} UO 018	F105.18
#7519	1	2^{19} UO 019	F105.19
#7520	1	2^{20} UO 020	F105.20
#7521	1	2^{21} UO 021	F105.21
#7522	1	2^{22} UO 022	F105.22
#7523	1	2^{23} UO 023	F105.23
#7524	1	2^{24} UO 024	F105.24
#7525	1	2^{25} UO 025	F105.25
#7526	1	2^{26} UO 026	F105.26
#7527	1	2^{27} UO 027	F105.27
#7528	1	2^{28} UO 028	F105.28
#7529	1	2^{29} UO 029	F105.29
#7530	1	2^{30} UO 030	F105.30
#7531	1	2^{31} UO 031	F105.31
#7532	32	UO0 0 ? UO0 31	F105
#7533	32	UO1 0 ? UO1 31	F106
#7534	32	UO2 0 ? UO2 31	F107
#7535	32	UO3 0 ? UO3 31	F108

$$\#8032 = \sum_{i=0}^{31} \#[8000 + i] * 2^{i}$$

When uo[100+I] is low, Vi = 0,
When uo[100+I] is high, Vi = 1,

(3) Tool offset data: #2001~ #2901

- Shift data of work-piece coordinate

Axis	Shift
X	#2368
Y	#2369
Z	#2370

- 2^{nd}, 3^{rd}, 4^{th} reference position

Axis	2nd	3rd	4th
X	#8000	#8032	#8064
Y	#8001	#8033	#8065
Z	#8002	#8034	#8066

- **In case of lathe**

- Tool offset data

Div	No	Geometric offset	Wear offset
X	1 ~ 64	#1728 ~ #1791	#1920 ~ #1983
Z	1 ~ 64	#1856 ~ #1919	#2048 ~ #2111
R	1 ~ 64	#1664 ~ #1727	
T	1 ~ 64	#1600 ~ #1663	
Y	1 ~ 64	#1792 ~ #1855	#1984 ~ #2047

- **In case of milling**

- Tool diameter compensation offset data: #2112~#2239

D No	Variables
1	#2112
2	#2113
..	..
127	#2238
128	#2239

- Tool length compensation offset data: #2240~#2367

H No	Variables
1	#2240
2	#2241
..	..
127	#2366
128	#2367

- Work-piece coordinate offset

Axis	Coordinate	No
X	G54	#2377
	G55	#2386
	G56	#2395
	G57	#2404
	G58	#2413
	G59	#2422
Y	G54	#2378
	G55	#2387
	G56	#2396
	G57	#2405
	G58	#2414
	G59	#2423
Z	G54	#2379
	G55	#2388
	G56	#2397
	G57	#2406
	G58	#2415
	G59	#2424
4th ~ 9th	G54	#2380 ~
	G55	#2389~
	G56	#2398~
	G57	#2407~
	G58	#2416~
	G59	#2425~

(4) Single Block stop: #3083

If you put the values in the below table into the system variable #3083, single block stop doesn’t happen in the block after that.

#3083	Single Block Stop
1	Don’t control
0	Control

(5) Target quantity for machining and quantity of machining (- value input forbidden)

#6101 : Quantity of machining (possible to initialize into 0)
#2431 : Target quantity for machining

(6) Information on modal in use

System	Modal information
#6718	G code group 1
#6719	G code group 2
#6720	G code group 3
..	..
#6749	G code group 32
#4882	Command D
#4883	Command H
#4721	Command F
#6716	Block No
#4792	Command S

(7) Coordinate Information

System Variable	Coordinate Information	Remarks
#6205 #6206 #6207 #6208	Machine coordinate of X after block feed is completed Machine coordinate of Y after block feed is completed Machine coordinate of Z after block feed is completed Machine coordinate of 4 axes after block feed is completed	Tool diameter compensation and tool length compensation data are considered (machine coordinate system)
#4083 #4084 #4085 #4086	Absolute coordinate of X on the current work-piece coordinate system Absolute coordinate of Y on the current work-piece coordinate system Absolute coordinate of Z on the current work-piece coordinate system Absolute coordinate of 4 axes on the current work-piece coordinate system	Tool diameter compensation and tool length compensation data are considered. Absolute coordinate on work-piece coordinate
#6319 #6320 #6321	Coordinate of X on the machine coordinate where skip signal happens Coordinate of Y on the machine	Location where skip signal becomes ON in G31 block. Machine coordinate

#6322	coordinate where skip signal happens Coordinate of Z on the machine coordinate where skip signal happens Coordinate of 4 axes on the machine coordinate where skip signal happens	
#4379 #4380 #4381 #4382	Deviation of servo position of X Deviation of servo position of Y Deviation of servo position of Z Deviation of servo position of 4 axes	

Note:

① If skip signal doesn't happen while G31 is carried out, the coordinate of the block end becomes signal position.

② Don't substitute arbitrary value for system variable.

▯II Example of macro paging

G65 P9300 X (Middle point) Y (Middle point) Z (Middle point)

Macro program main body

```
O9300
#1 = #5001;
#2 = #5002;
#3 = #5003;
G00 X#24 Y#25;
G04; (dwell in order to read #5201)
G91 X[Xp - #5021] Y[Yp - #5022] Z[Zp - #5023];
..
X#24 Y#25 Z#26;
X#1 Y#2;
Z#3;
M99
```

14.2.4 Operation command

(1) Constant of variable and substitution
#i = #j

(2) Deriving operation
#i =#j + #k
#i =#j - #k
#i =#j OR #k
#i =#j XOR #k

(3) Multiplicative operation
#i =#j * #k
#i =#j / #k
#i =#j AND #k

(4) Function
#i = SIN[#j]
#i = COS[#j]
#i = TAN[#j]
#i = ATAN[#j]
#i = SQRT[#j]
#i = ABS[#j]
#i = ROUND[#j]
#i = AND[#j]
#i = OR[#j]
#i = FIX[#j]
#i = FUP[#j]

● Method of using ROUND
① When it is used for operation command or conditional expression for IF and WHILE, decimals are rounded off.
#1 = ROUND[1.2345]; → #1 = 1.0
IF [#1 LE ROUND[#2]] GOTO 10; → #2 is 3.567, ROUND[#2] becomes 4.0.
② When it is used during the command of address, ROUND is done in the minimum setting

unit for the address.

When G01 X[ROUND[#1]]; → #1 is 1.4567, if the minimum setting unit for X is 0.001, this block becomes G01 X1.456;.

```
Ex) N1 #1 = 1.2345;
    N2 #2 = 2.3456;
    N3 G91 G01 X#1 F100;              (Act as  X1.235)
    N4 X#2;                           (Act as X2.346)
    N5 X-[#1 + #2];                   (1.2345 + 2.3456 = 3.5801so act
                                      as X3.580)
    N5 X-[ROUND[#1]+ROUND[#2]];       (1 + 2 = 3 so act as X3.)
```

(5) Combination of operations

Operation is done in the order of variable, multiplicative operation and deriving operation.

#i=#j + #k*SIN[#1];

(6) Change in operation order by [ ]

Using [ ], you can decide the order of operation.

[ ] includes [ ] of a variable so it is possible up to 10 times.

#i = SIN[[[#j + #k] * #l + #m] * #n];

14.2.5 Control command

(1) Branch

```
IF  [  <probability>  ]
GOTO n
```

<Probability> : EQ, NE, GT, LT, GE, LE

n : Sequence number to find in case probability is true

Using variable or [<expression>] instead of n, it is impossible to make the next block variable.

(2) Repetition

```
WHILE    [<probability>]
DOm
(m = 1, 2, 3…)
∫

END m
```

When <probability> exists, it repeats from the next block of Dom to the block of END m

When <probability> doesn ’ t exist, it executes END m block.

WHILE [<probability>] is same as IF so it can be omitted. When omitted, it repeats from DOm to ENDm without limitation.

WHILE [<probability>] Dom and ENDm are used in pair, and by the identification number m, each opposite can be identified.

Ex)

```
#120 = 1;
    N1 WHILE [ #120 LE 10] DO 1;
    N2 WHILE [#30 EQ 1 ] DO 2;
    .
.
N3 END 2;
    .
    #120 = #120 +1;
N33 END 1;
```

Repeat 10 times

(Note) Notes for making repetition program

① Should command DO m at the head and END m at the back

```
END 1
.
DO 1          (Impossible)
```

② DO m and END m should correspond one-to-one in the same program.

```
DO 1
.
DO 1
.
END 1                    (Impossible)
DO 1
.
END 1
.
END 1                    (Impossible)
```

③ Possible to use the same identification number several times DO 1

```
.
END 1
.
DO 1
.
END 1                    (Possible)
```

④ Multiplicity of DO is triple.

```
DO 1
.
DO 2
.
DO 3
.
END 3
.
END 2
.
END 1
```

⑤ The range of DO is possible in turn. DO 1

.

DO 2

.

END 1

.

END 2 (Impossible)

⑥ It can branch out of the range of DO.

DO 1

.

GOTO 9000

.

END 1

.

N9000 (Possible)

⑦ It can ' t branch in from the range of DO.

GOTO 9000

.

DO 1

.

N9000

.

END 1 (Impossible)

DO 1

.

N9000

.

END 1

.

GOTO 9000 (Impossible)

⑧ Possible to page custom macro main body or sub-program within the range of DO. Multiplicity of DO is up to triple in the custom macro main body or sub-program.

DO 1

.

G65 … (Possible)

```
.
G66 ⋯          (Possible)
.
G67            (Possible)
.
END 1
.
DO 1
.
M98 ⋯          (Possible)
.
END 1
```

14.2.6 Macro description and CNC description

Item	Description
Macro description	Operation command Control command (Block including GOTO, DO, END) Macro paging command (Block including macro paging by G65, G66, G67, G code)
Sameness between Macro description and CNC description	Block paging sub-program (Including M98, M, T) M99 block not including any other command address than O, N, P, L

Difference between macro description and CNC description	In the macro program, the normal single block stop doesn’t happen. During Nose R compensation, macro description is not regarded as the block which doesn’t have axis feed. Time of execution is different. - In case there is macro description next to the block where buffering doesn’t work like M code and G31 block, it is carried out after the previous block ends. - In case there is macro description next to the block where buffering is carried out. 1) When thread end R compensation mode is not in Once the current block starts to work, the next macro description is carried out immediately. It is carried out until the next CNC description comes out. 2) When thread end R compensation mode is in

14.3 Registering custom macro main body

It is same as the method of registering sub-program (9000.NC ~ 9999.NC).
It is possible to make a program by use of macro variable in general programs.

14.4 Restrictions

(1) MDI mode operation

Command of macro paging is possible.

Command of sub-program paging is impossible.

(2) Single Block

In any other blocks than those for macro paging command, operation command and control command, single block stop is possible even during macro performance. In the blocks of macro paging command (G65, G66, G67), operation command and control command, single block stop is possible by parameter setting.

(3) Optional Block Skip

In case the operation expression includes /, it is not regarded as optional block skip.

(4) Operation in EDIT mode

You can protect macro main body of 9000 ~ 9999 or sub-program from being deleted or edited, using parameter (PA2).

(5) RESET

In case it is cleared by RESET, the local variable (#1~#33) and the common variable (#100~#199) are cleared to be 0. However, sometimes it can ' t be cleared by parameter (PIU 74).

The system variable is not cleared.

Paging status of custom macro main body and sub-program and status of DO are all cleared to return to the main program.

(6) Indication of program restart page

T code, M code that is used for paging sub-program by T code, and T code are not indicated same as M98.

(7) Feed Hold

If you instruct feed hold while executing macro description, it stops after the macro description is completed.

(8) Other matters

- Available variables

#0, #1 ~ #33 (Possible to deliver factors), #34 ~ #99,
#100~#149, #150~#199,
#500~#699,
System variable

- Available variable values

Maximum : $\pm 10^{47}$
Minimum : $\pm 10^{29}$

- Constant available in <Express>

Maximum : ±99999999.999
Minimum : ±0.0000000001
Decimal is possible.

- Multiplicity for macro paging

Maximum 4

- Multiplicity of [ ]

Maximum 10 folds

- Multiplicity of sub-program paging

8 folds including macro paging

14.5 Example of program

14.5.1 Macro Program PLANE DRILL

```
G40 G49 G80 ;
G28 G91 Z0. ;
G28 X0.Y0. ;
G90 G92 X150. Y150. Z200.;
Z50.
G0 X0 Y0

#100 = 15.    ; X DRILL NUMBER
#102 = 10.    ; X DISTANCE
#103 = 0;     ; X COUNT

#111 = 5.     ;Y LINE COUNT
#112 = 10.    ; Y DISTANCE
#113 = 0;     ; Y COUNT

#200 = 0.     ; START X POS
#201 = 0.;     START Y POS

G90 X[#200] Y[#201]

N10
G90 X[#200] Y[#201 + #112 * #113]
N20
;G81   X_Y_   Z-15. R2. F200.
G90 X[#200 + #102 * #103] Y[#201 + #112 * #113]
G1 Z-10. F100.
G0 Z5.
#103 = #103 + 1;
IF [#103 LT #100-1] GOTO 20

#103 = 0;
```

```
#113 = #113 + 1;
IF [#113 LT #1111] GOTON10

G91 G0 Y50.
X50.

#102 = #102 + 1;
G49 G00 Z200. M05;
M02;
```

14.5.2 Circular Arc Machining (Divide by 1 degree)

```
(Macro Program)
G40 G49 G80 ;
G28 G91 Z0. ;
G28 X0.Y0. ;
G90 G92 X150. Y150. Z200. ;
Z50.
G0 X0 Y0

#100 = 0. ; Start angle
#101 = 50.; Arc radius

N20
G1 X[SIN[#100] * #101] Y[COS[#100] * #101]
#100 = #100 + 1;
IF [#100 LE 360.0 ] GOTO 20

G91 G0 Y50.
X50.

#102 = 1;   count

#102 = #102 + 1;
G49 G00 Z200. M05;
M02;
```

14.5.3 Examples of WHILE- ENDm

```
(Macro Program CIRCLE DRILL)
G40 G49 G80 ;
G28 G91 Z0. ;
G28 X0.Y0. ;
G90 G92 X150. Y150. Z200.;
G0 Z10.
G0 X0 Y0
G0 X0 Y0

#100 = 0.  ; START ANGLE
#101 = 50 ; RADIUS
#102 = 45. ; BETWEEN ANGLE

N150 WHILE [#100 LE [360.-#102]] DO 210

N200 WHILE [#101 GE 10.] DO 220
;G98 G81 X[SIN[#100] * #101] Y[COS[#100] * #101] Z-15. R2. F200.
G0 X[SIN[#100] * #101] Y[COS[#100] * #101]
G1 Z-20. F100.
G0 Z10.
#101 = #101- 10.; (INCREASE)
END 220

#100 = #100 + #102  ; (RADIUS DECREASE)
#101 = 50.           ;RADIUS

END 210

G91 G0 Y50.
X50.
G90 G49 G00 Z200. M05;
M02;
```

15 Special Functions

15.1 High Speed Cutting

15.1.1 High Speed Cutting Mode (G10.3)

```
G10.3
G11.3
```

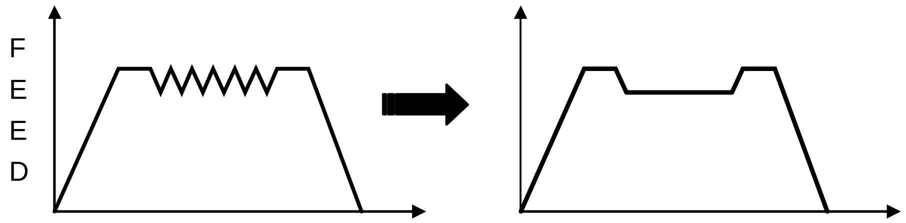
G10.3	High Speed Cutting Mode ON
G11.3	High Speed Cutting Mode OFF

15.1.2 High speed cutting by use of preceding control and optimal algorithm

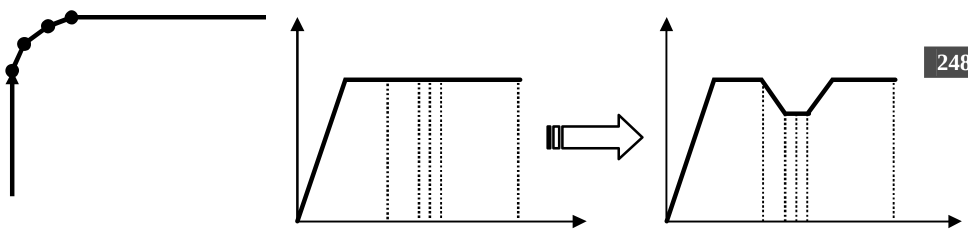
The preceding control algorithm and feed forward control function compensate dynamics of a machine and minimize form error of cutting. By using these functions, high-speed and high-precision aperture control can be obtained, which satisfy the minimum form error and maximum cutting speed required at the time of high-speed cutting very small consecutive blocks. Radius errors at the time of high-speed corner cutting and high-speed arc cutting have been greatly improved. It is used for the machines requiring high-speed machining and high-precision control.

15.1.3 Features of high speed cutting

- Without additional hardware, high-speed and high-precision cutting is possible.
- It uses high-speed interpreter, Pentium 133MHz (475 blocks per second). In case of a small block of 1 mm, the cutting speed of maximum 475mm/sec is possible.
- Maximum 100-block look ahead algorithm has been applied.
- Preceding feed forward algorithm has been applied.
- Using the algorithm of acceleration/deceleration before interpolation, it has minimized form errors in cutting.
- The high-speed algorithm in consideration of dynamics of a machine has been applied. It has a smoothing interpolation function for smooth free curved line cutting having big radius of curvature In the high-speed cutting mode, linear block (G01) and circular arc interpolation (G02/G03) are possibly handled.
- During high-speed cutting, it is possible to set and change the work-piece coordinate.
- During high-speed cutting, it is possible to compensate tool diameter and length.
- During high-speed cutting, it is possible to adjust feed override.
- Acceleration/deceleration in consideration of allowable speed change at the corer and of allowable acceleration/deceleration for block are possible.
- It smoothly handles speed profile and improves cutting speed.
- A small straight line which has little difference in speed between blocks automatically resets allowable accelerating speed for the corner and then, reduces cutting load.
- Feed Hold and Single Block are possible.
- During high-speed cutting, it is possible to change the mode.
- During high-speed cutting, interlock application by axis is possible.
- During high-speed cutting, it is possible to use TPS function.
- During high-speed cutting, it is possible to use manual ABS On/Off function.
- During high-speed cutting, it is possible to change the mode into manual operation and then, to do manual operation.
- It is possible to check high-speed cutting under machine lock status.



Speed Profile Smoothing Function



Corner Speed Control of Continious Small Blocks

15.1.4 Parameter Setting for High-Speed Cutting

< Parameter ->Cutting 1->Setting for high-speed function >

프로그램 | 사용자 | 가공 1 | 가공 2 | 시스템 | 매크로 | 축 | I/O 설정 | 특수기능 | HMI

NO.	Value	Unit	Comment
			High-Speed Machining Function Setting
PM 0680	1		High-Speed Machining Type Selection (0:Type 1, 1: Type 2)
PM 0701	0	msec	Accel/Decel. Time before High-Speed Machining Interpolation
PM 0704	0	msec	Accel/Decel. Time after High-Speed Machining Interpolation(Type
PM 0710	99999.0	mm/min	Max. Cutting Feed of High-Speed Machining
PM 0711	100.0	mm/min	Min. Cutting Feed of High-Speed Machining
PM 0720	0	mm/min	High-Speed Machining Vector Error Allowance
PM 0721	0.0100	mm	High-Speed Machining Shape Error Allowance (Type 1)
PM 0725	3		High-Speed Machining Prior Interpolation Factor
PM 0729	0		R Speed Limiting Function Usage Status (Type 2) (0:No Use 1:Us
PM 0730	0.000	mm	R Speed Limiting Function Max. Block Length (Type 2)
PM 0731	0.000	mm	R Speed Limiting Function Standard Radius (Type 2)
PM 0732	0.0	mm/min	R Speed Limiting Function Standard Machining Speed (Type 2)
PM 0735	0		High-Speed Machining Speed Selection (0:Command Feed 1:Max
PM 0740	0		G00 Block during High-Speed Machining(0:High-Speed,1:Normal)

(1) Select high-speed type (Parameter PM 680)

: High-Speed Cutting Type 1: It reads only the block that corresponds to the minimum distance which a machine can be decelerated, and then, it performs high-speed cutting. It can handle straight line blocks only.

High-Speed Cutting Type 2: It reads maximum 100 blocks in advance and performs high-speed cutting. It can handle straight line block and arc block. Generally type 1 is not used and type 2 is usually used.

(2) Setting for time constant for acceleration/deceleration before high-speed cutting interpolation (Parameter PM 701)

: Input the time that feedrate reaches the standard speed under the status of stop. The standard speed is 1000mm/min. Generally, a value between 25~ 45 is set.

(3) Setting for time constant for acceleration/deceleration after high-speed cutting interpolation (Type 2) (Parameter PM 704)

: This function is to apply acceleration/deceleration to the speed profile that is completely handled by the high-speed controller. With this function, you can get the effect of Jerk compensation, together with smooth cutting result. However, compared with the nominal value, cutting shape is distorted in proportion to time constant so you should be careful. It is set at 0 in general.

(4) **Maximum and minimum speed for high-speed cutting (Parameter PM 710,711)**

: This function is to set the maximum cutting speed and the minimum cutting speed at the time of high-speed cutting.

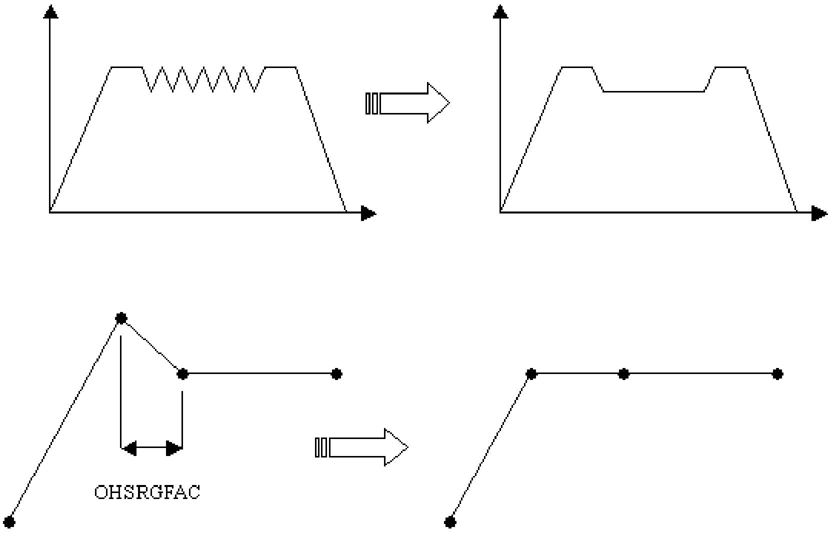
(5) **Setting for allowable vector error for high-speed cutting (Parameter PM 720)**

: It means changes in the allowable speed at the corners of a block. If it is set at 0, it is automatically calculated and set.

(6) **Preceding interpolation factor for high-speed cutting (Parameter PM 725)**

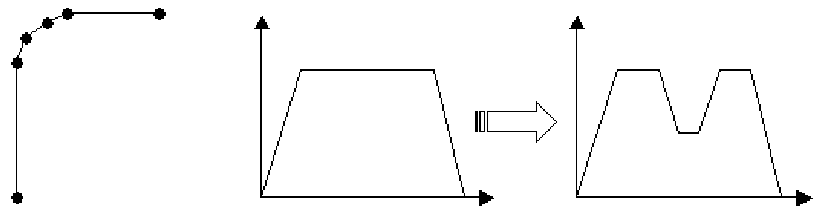
: Preceding interpolation factor for high-speed cutting has big influence on cutting illumination. Large value improves illumination while it decreases cutting speed. In high-speed cutting type 1, the forecasted distance becomes larger as much as double of the set value. In the high-speed cutting type 2, as shown on the below figure, it means the amount of smoothing the speed profile.

In case acceleration or deceleration is carried out under the set sampling, it is handled for smoothing and the set value means the number of samplings. In general, it is set at 3.



(7) **Whether to use the function of speed limit by radius of curvature (R) at the time of high-speed cutting (Type 2) (PM 729)**

: This is a function to limit the feedrate to acceleration on a small curved line consisting of small blocks at the time of high-speed cutting, and then, to reduce cutting load. Here allowable acceleration is calculated from OHSRLMTR and OHSRLMTFO, which is applied in small linear block and arc block.



(8) Maximum block length with speed limit function applied by radius of curvature (R) at the time of OHSRLMTR and OHSRLMTF high-speed cutting (Type 2) (PM 730)

: The speed limit function is applied to consecutive small blocks having the length less than the set value.

(9) Handling rapid traverse blocks (G00) during high-speed cutting (PM 740)

: If it is set at 0, rapid traverse block is handled with high-speed cutting, and if it is set at 1, it is handled with general cutting.

15.2 Scaling Function (G50,G51)

Scaling function is to reduce or enlarge the size of the programmed shape and then, to program in case cutting of the same shape is repeated. It can be applied variously by designating the whole magnifications or different magnification of each axis. The tool offset data is excluded from the subject for scaling.

15.2.1 Increase/decrease by same magnification

```
G50
G51 X _ Y _ Z _ P _
```

G50	Scaling Cancel
G51	Scaling On
X_ Y_ Z_	The center coordinate value for scaling command is should be absolute G90. If X_Y_Z_ are omitted, the current location where G51 is commanded becomes the center of scaling.
P_	It is possible to give a command in the form of real number to address that designates magnification of scaling. The unit of magnification is always 1/1000.

(Double increase = P2000, 0.5 decrease = P500)

For the motion command after this command, scaling is performed by magnification of P that is designated based on the center of scaling. The scaling mode can be cancelled by G50.

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① In case P is not commanded, the value set with the parameter, PI 108~116 (#3108~3116) becomes the basic magnification.

② Scaling is not applied to tool diameter compensation, tool length compensation, and tool offset data.

③ If X, Y, Z are omitted, the center of G51 is the current coordinate of the time point where G51 is commanded.

④ In case of milling, if there is no G code command like G50, G51, G code set with parameter, PI 149(#3149) becomes the default modal.

15.2.2 Increase/decrease by independent magnification

```
G50
G51 X _ Y _ Z _ I _ J _ K _
```

G50	Scaling Cancel
G51	Scaling On

X_ Y_ Z_ The center coordinate values of the scaling command should be absolute G90.

If X_ Y_ Z_ are omitted, the point where G51 is commanded becomes the center of scaling.

I_ J_ K_ Scaling magnification for each axis is designated in the relation of I = X axis, J = Y axis, K = Z axis. It is always 1/1000.

Example: In case the command of circular arc radius is absolute G90:

	Explanation
G90 G00 X0. Y100.	: Absolute coordinate is moved to X0.Y100. at rapid traverse.
G51 X0. Y0. Z0. I2000. J1000.	: X ' s feed double increases, and Y ' s feed one-time increases.
G02 X100. Y0. R100. F500.	: The end point of arc is X200. Y0.

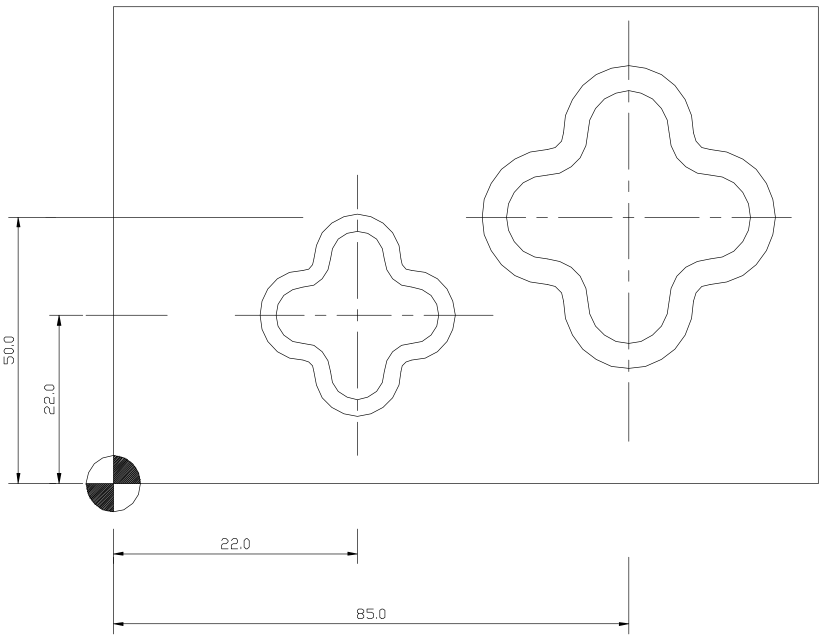
① In case P or I, J, K are not commanded, the value set with PI 108~116 (#3108~3116) becomes the scale factor.

② Scaling is not applied to tool diameter compensation, tool length compensation, and tool offset data.

③ If X, Y, Z are omitted, the center of G51 is the current coordinate of the time point where G51 is commanded.

④ In case of milling, if there is no G code command like G50 and G51, G code set with parameter, PI 149(#3149) becomes the default modal.

Example of scaling program



G00 G90 X0. Y0. ;	
Z2. ;	
G52 X22. Y22. ;	(Local coordinate system setting)
G51 X0. Y0. I1000. J1000. ;	(Scaling magnification of 1 time)
M98 P1234 ;	(Sub - program paging)
Z2. ;	
G52 X85. Y50. ;	(Local coordinate system setting)
G51 X0. Y0. I1500. J1500.;	(Scaling magnification of 1.5 times)
M98 P1234 ;	(Sub - program paging)
G50 ;	
G52 X0. Y0. ;	(Local coordinate system cancel)

Notes:

① G51 command is given as single block and is cancelled by G50.

② Coordinates are indicated as scaled values.

③ Scaling is not applied in the manual operation.

④ Scaling is not applied in the following feed action in case canned cycles feed to Z.

- Drilling amount Q of G73/G83 and return relief
- Fine boring (G76)
- Shift of X and Y axes in back boring (G87)

⑤ G27, G28, G29, G30, G92 should be commanded in G50 mode.

⑥ 5-decimal places of the scaling results are ignored.

⑦ In case scaling and rotation are commanded together, scaling is applied first and then, rotation is applied. Therefore, the center of rotation is influenced by scaling.

⑧ The center of scaling that is commanded in G51 block is not applied during incremental command (G91) after G51 is commanded (the current point that G51 is commanded is the center), but after absolute command (G90) is given, it is applied as the center.

15.3 Mirror Image (G50,G51)

G50 G51 X _ Y _ Z _ I -_ J -_ K -_

G50 Mirror Image Cancel

G51 Mirror Image Setting

X_ Y_ Z_ The center coordinate value of mirror image should be absolute G69.
If X_ Y_ Z_ are omitted, the point where G51 is commanded becomes the center of the mirror image.

I_ J_ K_ You should designate in negative number by use of ' - ' .

The mirror image magnification of each axis should be designated in the relation of I = X axis, J = Y axis, and K = Z axis. It is always 1/000.

ex) I-1000 : Mirror image of X (one time)

J-1500 : Mirror image of Y (1.5 times)

K-2000 : Mirror image of Z (2 times)

P _ In case the mirror image magnification for each axis is same, you should use P_ command instead of I_ J_ K_ like scaling command.

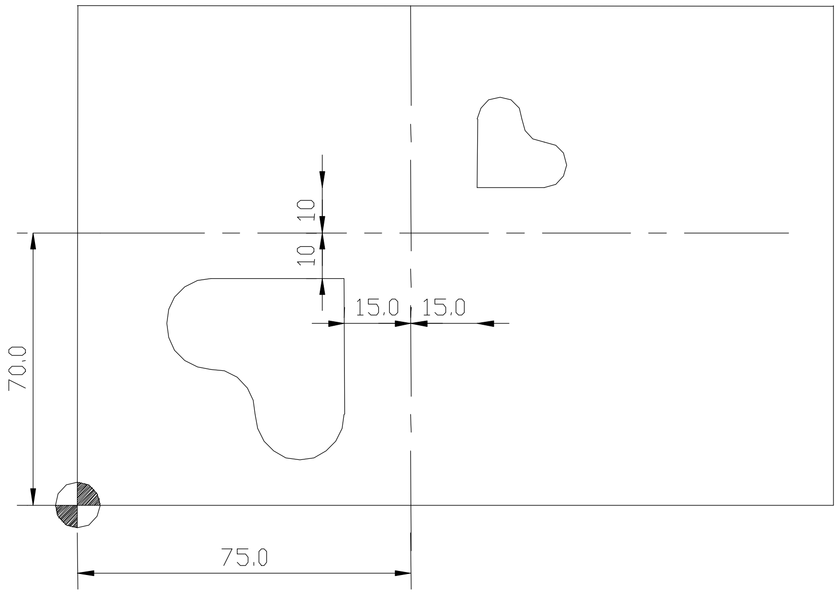
You should designate in a negative number by use of ' - ' .

(2 times increase of mirror image = P-2000,

1/2 decrease of mirror image = P-500)

Scaling is the function to use the programmed forms by reduction or enlargement. Mirror image function is made by adding the mirror function to the scaling function. Therefore, it is possible to reduce/enlarge the programmed forms and at the same time, it is possible to symmetrically move. If you designate I_ J_ K_(or P_) in positive numbers when using mirror image G51, scaling function gets to work, and if you designate in negative numbers, mirror image function works.

II Example of mirror image program



G54 G91 XZ0. ;	
G54 G90 G00 X0. Y0. ;	
G52 G90 X90. Y80. ;	(Local coordinate system 1 setting)
G00 X0. Y0. ;	
M98 P1234 ;	(Sub-program paging)
Z2 ;	
G52 X60. Y80. ;	(Local coordinate system 2 setting)
G51 X0. Y0. I-1500. J-1500. ;	(1.5 times increase of scale)
	(Mirror image for X and Y ON)
M98 P1234 ;	(Sub-program paging)
G50 ;	
G52 X60. Y60. ;	(Local coordinate system cancel)

15.4 Coordinate Rotation

15.4.1 Coordinate Rotation (G68, G69)

```
G17 G68 X _ Y _ R _
G18 G68 Z _ X _ R _
G19 G68 Y _ Z _ R _
G69
```

G68	Coordinate System Rotation
G69	Coordinate System Rotation Cancel
G17 X_ Y_	Coordinate values of G17 plane and rotation center
G18 Z_ X_	Coordinate values of G18 plane and rotation center
G19 Y_ Z_	Coordinate values of G19 plane and rotation center
R _	Rotation angle

With the function of coordinate rotation, you can rotate a shape designated by a program at a random angle by instructing the rotation center and rotation angle of the shape so you don ' t need to calculate the coordinates with trigonometrical function. This function can be applied by making the shape in a sub-program in case the work-piece is a shape pattern that the shape seems to rotate.

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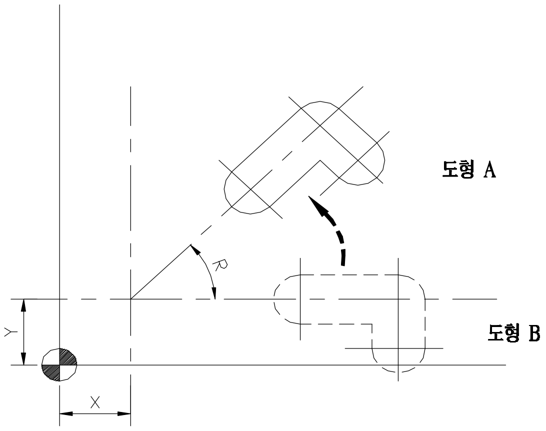
① If the command of center point is omitted, the current position (work-piece coordinate system) where G68 is commanded becomes the center of rotation.

② In case there is a incremental command (G91) block next to G68 command block, the center of rotation by G68 gets ignored until the absolute command block (G90) is commanded. That is, the current position where G68 is commanded becomes the center of rotation.

③ In case R command is omitted, the value set with parameter PI 117(#3117) becomes the rotation angle.

④ G69 can be commanded in the same block together with other commands.

⑤ Tool diameter compensation data, tool length and tool offset are carried out after the coordinate of the commanded program rotates.



R : It is the angle command for coordinate rotation. Counterclockwise is + direction.

The command method is same as the angle command of polar coordinate.

Note:

① In order for the rotation function to be always applied without G68 command, parameter PI 150(#3150) should be set at 1.
② In case R command has decimal point, it is angle unit.
③ G68 command should be given under G00, G01 mode.
④ G27, G28, G29, G30 , G92 should be given under G69 mode.

15.4.2 Relations with other functions

(1) Tool diameter compensation C type

During tool diameter compensation, G68 and G69 command can be given. The plane for coordinate rotation should be the plane that tool diameter compensation is carried out.

▯ll Example

```
N1 G92 X0 Y0   G69 G01
N2 G42 G90 X1000. Y1000. F100 D01
N3 G68 R-30.0                          (Counterclockwise 30.0 degree)
N4 G91 X2000.0
N5 G3 Y1000.0 J500.
N6 G1 X-2000.
N7 Y-1000.
N8 G69 G40 G90 X0 Y0 M30
```

(2) Coordinate rotation under scaling mode

In case the command of coordinate rotation is given under scaling mode (G51), scale is applied to the center of the coordinate (a, b), but scale is not applied to rotation angle R.

In case there is a command of axis feed, scale is applied first and then, rotation is applied.

▯ll Example

```
G51... (Scaling mode)
G68... (Coordinate rotation ON)
...
G69... (Coordinate rotation OFF)
G50... (Scaling mode cancel)
```

> **Note:**
> The command of coordinate rotation should not be given under the status of tool diameter compensation in the scaling mode. The command of coordinate rotation should be always given prior to tool diameter compensation (G41/G42).

Ⅱ Example

(G40 mode)
G51 ... (Scaling mode)
G68... (Coordinate rotation ON)
.
.
G41.. (Tool diameter compensation left)

(3) Repetition command (sub-program paging)

It is possible to page sub-program, changing the rotation angle.

Ⅱ Example

G92 X0 Y0 G69 G17
G01 F200 H01
M98 P2100
M98 P2200 L7
G00 G90 X0 Y0
M30

O2200
G68 X0 Y0 G91 R45.0 (O rotation angle increases by 45 degrees and the rotation center is same as the existing rotation center.)
G90 M98 P2100
M99

O2100
G90 G1 G42 X0 Y-10.0
X4.142
X7.071 Y-7.071

G40
M99

■ Example of coordinate rotation

Explanation

N1 G92 X-50 Y-50 G69 G17 ;

N2 G68 X70 Y30 R60 ; (Rotation center : X 70. Y 30.)

N3 G90 G01 X0 Y0 F200 (G91 X50 Y50) ; (Rotation angle 60°)

N4 G91 X100 ;

N5 G01 Y100 ;

N6 X-100 ;

N7 Y-100 ;

N8 G69 G90 X-50 Y-50 M30 ;

15.5 Forbidden Zone Set and Cancel

This function is to prevent a tool ' s sudden separation and to obtain safe machining by setting a particular machining zone and moving zone of a tool.

15.5.1 H/W Limit

This is to set the maximum stored stroke of a machine, using H/W Limit switch. Outside of this zone is set as the forbidden zone. If this zone is invaded, Over Travel alarm happens and the power of servo motor becomes cut.

How to cancel this zone is: set Mode Select switch on OP panel at JOG, put Axis Select key on the relevant axis, and press O. T. Release key, together with Axis Feed key. Then, it will be moved to a safe zone.

15.5.2 S/W Limit

You should page the system screen, move the cursor to the item corresponding to Soft Limit of each axis, and directly input by use of the number key on MDI panel. The outside of the set zone can be set as forbidden zone. If this zone is invaded, S/W Limit alarm happens and the power of servo motor is not cut.

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① Soft limit check is possible after completion of reference position return. Soft Limit check parameter is input with the combination of X:1, Y:2, Z:4.
② PM 3378~3409 (#23378~23409) sets by axis whether to use Soft Limit function or not.
③ Soft Limit zone is set by axis with parameter PM 3410~3473 (#23410~#23473). The order of input is minimum/maximum per axis.

15.5.3 Programmable S/W Limit set and cancel (G22, G23)

G22 X _ Y _ Z _ I _ J _ K _ **G23**

G22 Stored Stroke Check Function On

G23 Stored Stroke Check Function Off

X_ Y_ Z_ As shown on the below figure, it inputs machine coordinate values with the distance from the first reference position to A. The coordinate of apex close to the first reference position becomes A point.

I_ J_ K_ As shown on the below figure, it inputs machine coordinate values with the distance from the first reference position to B. The coordinate of apex diagonal to A point becomes B point.

Range for the command Like I value > X value , J value > Y value, K value > Z value, I_ J_ K_ value should be always larger than X_ Y_ Z _.

If this zone is invaded while a tool feeds under G22, the error message, " G22 related Limit Over " appears on the screen and the feed stops. In this case, you can cancel the mode with G23 or get into the manual mode to move the tool in the opposite direction and get out of the zone.

⏭

① G22 and G23, modal G code, are continuously effective once they are given. If G23 is commanded, regardless of the designated forbidden zone for feed, it works.

② In order to use G22 feed forbidden zone, parameter PM 3474 (#23474) should be set at 0. If it is set at 1, G22 feed forbidden zone is ignored.

③ In the section set with G22, parameter PM 3475 (#23475) sets inside or outside as the forbidden zone.

④ Feed command is given in the form of G94 F500 G01 IP.

⑤ With the initial power ON, the lathe group uses G95 (feed per revolution) while milling uses G94 (feed per minute).

⏭

① It is impossible to know machine coordinate under the condition that reference position return is not made. Therefore, the forbidden zone should be set as a single block and its execution is possible after return to reference position.

② when the inside is forbidden zone, if a tool invades this zone, an alarm happens at the time of next feed. If the outside is forbidden zone, an alarm immediately happens. In case an alarm of the forbidden zone happens, you should feed the axis in the opposite direction.

③ When using several tools, the tools have different compensation data and the machine coordinates are different so setting for each tools is required.

⏸ Example of forbidden zone setting

<Figure

⏭

G30 G91 Z0

G22 X-120 Y-100 I-190 J-150 (Command the coordinate of A-a (-120,-100) and the coordinate of A-b (-190,-150) as machine coordinate values)

⬇ (Zone for machining program)

G23 (Forbidden zone cancel)

⬇

G30 G91 Z0

G22 X-110 Y-90 I-200 J-160 (Command the coordinate of B-a’ and the coordinate of B-b’ as machine coordinate values)

⬇ (Machining program zone)

G23 (Forbidden zone cancel)

15.6 Stitch Function

15.6.1 Stitch Function Mode

G63.1 G63.2

G63.1 Stitch Function Mode ON
G63.2 Stitch Function Mode OFF

15.6.2 Explanation of Stitch Function

This function makes axis feed of table get started at the time of point when a needle is at a particular height, following its up and down movement. It prevents the needle from being broken and it is possible to use stitch function by use of very short block. In the stitch function, the command block feeds at rapid traverse. The time of acceleration/deceleration of rapid traverse and the speed of rapid traverse are changed automatically into 8-folded values set with the parameter, depending on the length of the command block.

15.6.3 Screen of parameter setting related with stitch function

< Parameter-> Special functions -> Stitch control >
<Figure>

If parameter PM7308 is set at 0, general rapid feedrate and acceleration/deceleration time constant are applied. If it is set at 1, rapid feedrate and acceleration/deceleration time constant are changed, depending on the length of the block. The feedrate and acceleration/deceleration time constant, which are influenced by the length of each block, are decided as shown on the below, depending on the values set with OPPG0X(Y)F and OPPG0X(Y)T
<Table>
<Figure>
<Figure>
<Figure>
<Figure>

PM 7321 ~ 7327 : Standard length of each stage for rapid feed block

PM 7328 ~ 7335 : Feed of each stage of rapid feed block for X

PM 7336 ~ 7343 : Acceleration and deceleration time of each stage of rapid feed block for X

PM 7344 ~ 7351 : Feed of each stage of rapid feed block for Y

PM 7352 ~ 7359 : Acceleration and deceleration time of each stage of rapid feed block for Y

PM 7374 ~ 7381 : Setting for output time of PPF signal before X feed

PM 7382 ~ 7389 : Setting for output time of PPF signal before Y feed completion

PM 7391 : Setting for the time of delay that PPF becomes low.

PM 7392: Setting for the time of delay that the next block is progressed when PPFS is high.

PM 7393: Positioning at nibbling (stitch) mode and setting for the time of delay between positioning and PPF output

PM 7394 : Setting for the time of delay between NPFIN and the next block at nibbling (stitch)

PM 7395 : Setting for the time of delay that PPF signal becomes low by PPE signal at nibbling (stitch) mode

PM 7408 : Maximum length of rapid feed block at nibbling (stitch) mode